

A journey into Alexandrov geometry:
curvature bounded below

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Preface

This book is similar to our “Invitation to Alexandrov geometry” written jointly with Stephanie Alexander [6]. We try to demonstrate the beauty and power of Alexandrov geometry by reaching interesting applications and theorems with minimal preparation. This time it is about curvature bounded below.

This subject is more technical and requires more preparation, so we had to skip several proofs. Namely, we do not prove the existence part of generalized Picard’s theorem (5.3) and Perelman’s theorem about conic neighborhoods (9.2); up to the last lecture, the rest is nearly rigorous. Several important statements are left as exercises (they are marked with **Exercise!**). Solutions to all of them are given at the end of the book.

In Lecture 1, we discuss necessary preliminaries and fix notation.

Lecture 2 introduces the main object of our study — spaces with curvature bounded below in the sense of Alexandrov.

In Lecture 3, we formulate and prove the globalization theorem that the local Alexandrov condition implies the global one. To simplify the presentation, we treat only the compact case; it captures the main ideas of the proof with fewer technicalities.

In Lecture 4, we develop calculus — tangent space and space of directions, differential, and gradient.

Lecture 5 introduces gradient flow, which will be used further as the main technical tool.

Lecture 6 proves the line splitting theorem. We then introduce and study linear subspaces of tangent spaces.

In Lecture 7, we introduce linear dimension and volume. Next, we prove the Bishop–Gromov inequality and the right-inverse theorem, introduce distance charts, and show that all reasonable types of dimension are the same for Alexandrov spaces.

In Lecture 8, we show that a lower curvature bound survives under Gromov–Hausdorff limits and prove Gromov’s selection theorem. Further, we present Perelman’s construction of strictly concave functions

and apply it together with Gromov's selection theorem to prove the homotopy finiteness theorem. This proof illustrates the main source of applications of Alexandrov geometry.

In Lecture 9, we introduce the boundary of finite-dimensional Alexandrov spaces and prove the doubling theorem.

In Lecture 10, we show that quotients of Alexandrov spaces by isometric group actions are Alexandrov spaces. This gives another source of applications of Alexandrov geometry; several examples are given.

Finally, in Lecture 11 we briefly discuss convex surfaces in Euclidean space; this subject is the main precursor to modern Alexandrov geometry.

Let us list some available texts on Alexandrov spaces with curvature bounded below:

- ◇ The first introduction to Alexandrov geometry of all dimensions was given in the original paper by Yuriy Burago, Michael Gromov, and Grigory Perelman [20] and its extension [87] written by Perelman.
- ◇ A brief and reader-friendly introduction was written by Katsumi Shiohama [114, Sections 1–8].
- ◇ Another reader-friendly introduction, written by Dmitri Burago, Yuriy Burago, and Sergei Ivanov [19, Chapter 10].
- ◇ Survey by Conrad Plaut [105].
- ◇ Survey by the third author [96].
- ◇ Our book, written jointly with Stephanie Alexander [7].

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Lecture 1

Preliminaries

A Prerequisites

The main prerequisite: the reader should know and love elementary geometry, including convexity.

We also assume that the reader is familiar with the following topics in metric geometry:

- ◊ Compactness and proper metric spaces; recall that a metric space is proper if all its closed balls (of finite radius) are compact.
- ◊ Complete metric spaces and completion.
- ◊ Curves, semicontinuity of length, and rectifiability.
- ◊ Hausdorff and Gromov–Hausdorff convergence. These are discussed briefly in 1J–1N. The definitions are there, but some prior familiarity with these concepts will be very helpful.

All these topics are treated in detail in the classic book of Dmitri Burago, Yurii Burago, and Sergei Ivanov [19]; see also a short book of the third author [102]. Occasionally, we use the Baire category theorem and Rademacher’s theorem, but these could be used as black boxes.

We use some topology. An introductory text in algebraic topology should be sufficient most of the time. For some examples, we apply more advanced results, but these are used as black boxes.

Since most of the applications come from Riemannian geometry, it is beneficial to be familiar with the Toponogov comparison theorem and related topics. The classical book by Jeff Cheeger and David Ebin [23] contains more than one needs.

B Notations

The distance between two points x and y in a metric space $\mathcal{X}$ will be denoted by $|x - y| = |x - y|_{\mathcal{X}}$, or $\text{dist}_x y = \text{dist}_x(y)_{\mathcal{X}}$. Furthermore, we define the distance function to a subset $A \subset \mathcal{X}$ as

$$\text{dist}_A x = \text{dist}_A(x)_{\mathcal{X}} := \inf \{ |a - x|_{\mathcal{X}} : a \in A \}.$$

We may use the subscript $\mathcal{X}$ to emphasize that the distance is taken in the space $\mathcal{X}$.

Given radius $r \in [0, \infty]$ and center $x \in \mathcal{X}$, the sets

$$B(x, r) = \{ y \in \mathcal{X} : |x - y| < r \},$$

$$\bar{B}[x, r] = \{ y \in \mathcal{X} : |x - y| \leq r \}$$

are called, respectively, the open and the closed balls. The notations $B(x, r)_{\mathcal{X}}$ and $\bar{B}[x, r]_{\mathcal{X}}$ might be used if we need to emphasize that these balls are taken in the metric space $\mathcal{X}$.

The diameter of $\mathcal{X}$ is defined as

$$\text{diam } \mathcal{X} = \sup \{ |x - y| : x, y \in \mathcal{X} \}.$$

The radius of $\mathcal{X}$ is defined as

$$\text{rad } \mathcal{X} = \inf \{ R > 0 : B(x, R) = \mathcal{X} \text{ for some } x \in \mathcal{X} \}.$$

We will denote by $\mathbb{S}^m$, $\mathbb{E}^m$, and $\mathbb{H}^m$ the m -dimensional sphere (with angle metric), the Euclidean space, and the Lobachevsky space, respectively. More generally, $\mathbb{M}^m(\kappa)$ will denote the model m -space of curvature κ ; that is,

- ◇ if $\kappa > 0$, then $\mathbb{M}^m(\kappa)$ is the m -sphere of radius $\frac{1}{\sqrt{\kappa}}$, so $\mathbb{S}^m = \mathbb{M}^m(1)$
- ◇ $\mathbb{M}^m(0) = \mathbb{E}^m$,
- ◇ if $\kappa < 0$, then $\mathbb{M}^m(\kappa)$ is the Lobachevsky m -space $\mathbb{H}^m$ rescaled by the factor $\frac{1}{\sqrt{-\kappa}}$; in particular $\mathbb{M}^m(-1) = \mathbb{H}^m$.

C Nets and packing

Let S be a subset of a metric space $\mathcal{X}$. Recall that a set $Z \subset \mathcal{X}$ is called an ε -net of S if for any point $s \in S$, there is a point $z \in Z$ such that $|s - z| \leq \varepsilon$.

The ε -pack of $\mathcal{X}$ (or packing number) is the maximal number (possibly infinite) of points in $\mathcal{X}$ at distance $> \varepsilon$ from each other; it is denoted by $\text{pack}_{\varepsilon} \mathcal{X}$. If $n = \text{pack}_{\varepsilon} \mathcal{X} < \infty$, then a set $\{x_1, x_2, \dots, x_n\}$ in $\mathcal{X}$ such that $|x_i - x_j| > \varepsilon$ is called a maximal ε -packing in $\mathcal{X}$.

We will use the following characterizations of compact sets.

1.1. Exercise! *Let X be a closed subset of a complete metric space.*

(a) *Show that X is compact if and only if it admits a finite ε -net for any $\varepsilon > 0$.*

(b) *Show that X is compact if and only if it admits a compact ε -net for any $\varepsilon > 0$.*

1.2. Exercise! *Show that any maximal ε -packing $x_1, \dots, x_n$ is an ε -net. Conclude that a complete metric space $\mathcal{X}$ is compact if and only if $\text{pack}_\varepsilon \mathcal{X} < \infty$ for any $\varepsilon > 0$.*

D Length spaces

Let $\mathcal{X}$ be a metric space. If for any $\varepsilon > 0$ and any pair of points $x, y \in \mathcal{X}$, there is a path α connecting x to y such that

$$\text{length } \alpha < |x - y| + \varepsilon,$$

then $\mathcal{X}$ is called a length space and the metric on $\mathcal{X}$ is called a length metric.

1.3. Exercise. *Let $\mathcal{X}$ be a complete length space. Show that for any compact subset $K \subset \mathcal{X}$ there is a compact path-connected subset $K' \subset \mathcal{X}$ that contains K .*

Induced length metric. From the definition of length, it follows that if $\alpha: [0, 1] \rightarrow \mathcal{X}$ is a path from x to y (that is, $\alpha(0) = x$ and $\alpha(1) = y$), then

$$\text{length } \alpha \geq |x - y|.$$

Set

$$\|x - y\| = \inf_{\alpha} \{ \text{length } \alpha \},$$

where the greatest lower bound is taken for paths α from x to y . It is straightforward to check that $(x, y) \mapsto \|x - y\|$ is an ∞ -metric; that is, $(x, y) \mapsto \|x - y\|$ is a metric in the extended positive reals $[0, \infty]$. The metric $\|* - *\|$ is called the induced length metric.

1.4. Exercise. *Suppose $(\mathcal{X}, \|* - *\|)$ is a complete metric space. Show that $(\mathcal{X}, \|* - *\|)$ is complete; that is, any Cauchy sequence of points in $(\mathcal{X}, \|* - *\|)$ converges in $(\mathcal{X}, \|* - *\|)$.*

Let A be a subset of a metric space $\mathcal{X}$. Given two points $x, y \in A$, consider the value

$$|x - y|_A = \inf_{\alpha} \{ \text{length } \alpha \},$$

where the greatest lower bound is taken for all paths α from x to y in A . In other words, $|\ast - \ast|_A$ denotes the induced length metric on the subspace A . (The notation $|\ast - \ast|_A$ conflicts with the previously defined notation for distance $|x - y|_{\mathcal{X}}$ in a metric space $\mathcal{X}$. However, most of the time we will work with ambient length spaces where the meaning will be unambiguous.)

E Geodesics

Let $\mathcal{X}$ be a metric space, and let $\mathbb{I}$ be a real interval. A distance-preserving map $\gamma: \mathbb{I} \rightarrow \mathcal{X}$ is called a geodesic¹; in other words, $\gamma: \mathbb{I} \rightarrow \mathcal{X}$ is a geodesic if

$$|\gamma(s) - \gamma(t)| = |s - t|$$

for any pair $s, t \in \mathbb{I}$.

If $\gamma: [a, b] \rightarrow \mathcal{X}$ is a geodesic such that $p = \gamma(a)$, $q = \gamma(b)$, then we say that γ is a geodesic from p to q . In this case, the image of γ is denoted by $[pq]$, and, with abuse of notation, we also call it a geodesic. We may write $[pq]_{\mathcal{X}}$ to emphasize that the geodesic $[pq]$ lies in the space $\mathcal{X}$.

In general, a geodesic from p to q need not exist, and if it exists, it need not be unique; for example, any meridian is a geodesic between poles on the sphere. However, once we write $[pq]$ we assume that we have chosen such a geodesic.

A geodesic path is a geodesic with constant-speed reparameterization by the unit interval $[0, 1]$.

A metric space is called geodesic if any pair of its points can be joined by a geodesic. A metric space $\mathcal{X}$ is called ℓ -geodesic if any two points $x, y \in \mathcal{X}$ such that $|x - y| < \ell$ can be connected by a geodesic. For instance, any geodesic space is ∞ -geodesic.

Evidently, any geodesic space is a length space.

1.5. Exercise. *Show that any proper length space is geodesic.*

F Menger's lemma

Given two points x, y in a metric space $\mathcal{X}$, a point $z \in \mathcal{X}$ is called a midpoint between them if

$$|x - z| = \frac{1}{2} \cdot |x - y| \quad \text{and} \quad |y - z| = \frac{1}{2} \cdot |x - y|.$$

¹Other texts may refer to geodesics in our sense of the word as *shortest path* or *minimizing geodesic*. Also, our meaning of the term *geodesic* is closely related to, but different from, the term used in Riemannian geometry.

If $\mathcal{X}$ is geodesic, then for any two points there is a midpoint between them. The following lemma says that the converse holds for complete metric spaces.

1.6. Menger's lemma. *Let $\mathcal{X}$ be a complete metric space. Assume that for any pair of points $x, y \in \mathcal{X}$, there is a midpoint z . Then $\mathcal{X}$ is a geodesic space.*

This lemma is due to Karl Menger [78, Section 6].

Proof. Choose $x, y \in \mathcal{X}$; set $\gamma(0) = x$ and $\gamma(1) = y$.

$$x = \overset{\circ}{\gamma(0)} \quad \overset{\circ}{\gamma(\frac{1}{4})} \quad \overset{\circ}{\gamma(\frac{1}{2})} \quad \overset{\circ}{\gamma(\frac{3}{4})} \quad \overset{\circ}{\gamma(1)} = y$$

Let $\gamma(\frac{1}{2})$ be a midpoint between $\gamma(0)$ and $\gamma(1)$. Further, let $\gamma(\frac{1}{4})$ and $\gamma(\frac{3}{4})$ be midpoints between the pairs $(\gamma(0), \gamma(\frac{1}{2}))$ and $(\gamma(\frac{1}{2}), \gamma(1))$ respectively. Applying the above procedure recursively, on the n -th step we define $\gamma(\frac{k}{2^n})$, for every odd integer k such that $0 < \frac{k}{2^n} < 1$, as a midpoint of the already defined $\gamma(\frac{k-1}{2^n})$ and $\gamma(\frac{k+1}{2^n})$.

This way we define $\gamma(t)$ for all dyadic rationals t in $[0, 1]$. Moreover, the map γ has Lipschitz constant $|x - y|$. Since $\mathcal{X}$ is complete, γ can be extended continuously to $[0, 1]$ with the same Lipschitz constant. Therefore,

$$\text{length } \gamma \leq |x - y|.$$

Hence γ is a geodesic path from x to y . □

1.7. Exercise. *Let $\mathcal{X}$ be a complete metric space. Assume that for any pair of points $x, y \in \mathcal{X}$, there is an almost midpoint; that is, given $\varepsilon > 0$, there is a point z such that*

$$|x - z| < \frac{1}{2} \cdot |x - y| + \varepsilon \quad \text{and} \quad |y - z| < \frac{1}{2} \cdot |x - y| + \varepsilon.$$

Show that $\mathcal{X}$ is a length space. Conversely, if $\mathcal{X}$ is a length space, then for any two points there exist almost midpoints between them.

G Triangles and model triangles

Triangles. Given a triple of points p, q, r in a metric space $\mathcal{X}$, a choice of geodesics $([qr], [rp], [pq])$ will be called a triangle and denoted by $[pqr] = [pqr]_{\mathcal{X}}$.

Given a triple of points $p, q, r \in \mathcal{X}$, there may be no triangle $[pqr]$ simply because one of the pairs of these points cannot be joined by a geodesic. Also, many different triangles with these vertices may exist,

any of which can be denoted by $[pqr]$. If we write $[pqr]$, it means that we have chosen such a triangle.

Model triangles. Given three points p, q, r in a metric space $\mathcal{X}$, let us define the model triangle $[\tilde{p}\tilde{q}\tilde{r}]$ (briefly, $[\tilde{p}\tilde{q}\tilde{r}] = \tilde{\Delta}(pqr)_{\mathbb{E}^2}$) to be a triangle in the Euclidean plane $\mathbb{E}^2$ with the same sides; that is,

$$|\tilde{p} - \tilde{q}|_{\mathbb{E}^2} = |p - q|_{\mathcal{X}}, \quad |\tilde{q} - \tilde{r}|_{\mathbb{E}^2} = |q - r|_{\mathcal{X}}, \quad |\tilde{r} - \tilde{p}|_{\mathbb{E}^2} = |r - p|_{\mathcal{X}}.$$

In the same way, we can define the hyperbolic and the spherical model triangles $\tilde{\Delta}(pqr)_{\mathbb{H}^2}$, $\tilde{\Delta}(pqr)_{\mathbb{S}^2}$ in the Lobachevsky plane $\mathbb{H}^2$ and the unit sphere $\mathbb{S}^2$. In the latter case, the model triangle is said to be defined if in addition

$$|p - q| + |q - r| + |r - p| < 2 \cdot \pi.$$

Then the model triangle again exists and is unique up to an isometry of $\mathbb{S}^2$.

Model angles. If $[\tilde{p}\tilde{q}\tilde{r}] = \tilde{\Delta}(pqr)_{\mathbb{E}^2}$ and $|p - q|, |p - r| > 0$, the angle measure of $[\tilde{p}\tilde{q}\tilde{r}]$ at $\tilde{p}$ will be called the model angle of the triple p, q, r and will be denoted by $\tilde{\angle}(p_r^q)_{\mathbb{E}^2}$.

For example, if $|p - q| = |q - r| = |r - p|$, then $\tilde{\angle}(p_r^q)_{\mathbb{E}^2} = \frac{\pi}{3}$ regardless of the existence and relative position of geodesics $[pq]$ and $[pr]$.

In the same way we define $\tilde{\angle}(p_r^q)_{\mathbb{M}^2(\kappa)}$; in particular, $\tilde{\angle}(p_r^q)_{\mathbb{H}^2}$ and $\tilde{\angle}(p_r^q)_{\mathbb{S}^2}$. We may use the notation $\tilde{\angle}(p_r^q)$ if it is evident which of the model spaces is meant.

1.8. Exercise! Show that for any triple of points p, q , and r , the function

$$\kappa \mapsto \tilde{\angle}(p_r^q)_{\mathbb{M}^2(\kappa)}$$

is nondecreasing in the interval of definition.

H Hinges and their angle measure

Hinges. Let $p, x, y \in \mathcal{X}$ be a triple of points such that p is distinct from x and y . A pair of geodesics $([px], [py])$ will be called a hinge and will be denoted by $[p_y^x] = ([px], [py])$.

Angles. The angle measure of a hinge $[p_y^x]$ is defined as the following limit

$$\angle[p_y^x] = \lim_{\bar{x}, \bar{y} \rightarrow p} \tilde{\angle}(p_{\bar{y}}^{\bar{x}}),$$

where $\bar{x} \in]px]$ and $\bar{y} \in]py]$. The angle is only defined if this limit exists.

If $\angle[p_y^x]$ is defined, then

$$0 \leq \angle[p_y^x] \leq \pi.$$

1.9. Exercise. Suppose that in the above definition, one uses spherical or hyperbolic model angles instead of Euclidean ones. Show that it does not change the value $\angle[p_y^x]$.

1.10. Exercise. Give an example of a hinge $[p_y^x]$ in a metric space with undefined angle measure $\angle[p_y^x]$.

I Triangle inequality for angles

1.11. Proposition. Consider three geodesics $[px_1]$, $[px_2]$, and $[px_3]$ in a metric space. Suppose all the angle measures $\alpha_{i,j} = \angle[p_{x_j}^{x_i}]$ are defined. Then

$$\alpha_{1,3} \leq \alpha_{1,2} + \alpha_{2,3}.$$

Proof. We can assume that $\alpha_{1,3} > 0$. Furthermore, since $\alpha_{1,3} \leq \pi$, we can assume that $\alpha_{1,2} + \alpha_{2,3} < \pi$.

Denote by γ_i the unit-speed parametrization of $[px_i]$ from p to x_i . Given small $\varepsilon > 0$, for all sufficiently small $t, \tau, s \in [0, \infty)$ we have

$$\begin{aligned} & \sqrt{t^2 + \tau^2 - 2 \cdot t \cdot \tau \cdot \cos(\alpha_{1,3} - \varepsilon)} \leq |\gamma_1(t) - \gamma_3(\tau)| \leq \\ & \leq |\gamma_1(t) - \gamma_2(s)| + |\gamma_2(s) - \gamma_3(\tau)| < \\ & < \sqrt{t^2 + s^2 - 2 \cdot t \cdot s \cdot \cos(\alpha_{1,2} + \varepsilon)} + \\ & + \sqrt{s^2 + \tau^2 - 2 \cdot s \cdot \tau \cdot \cos(\alpha_{2,3} + \varepsilon)} \leq \end{aligned}$$

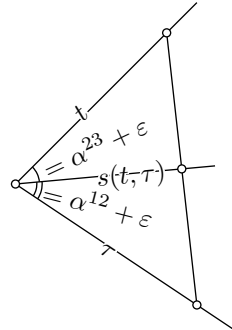
Below we define $s(t, \tau)$ so that for $s = s(t, \tau)$, this chain of inequalities can be continued as

$$\leq \sqrt{t^2 + \tau^2 - 2 \cdot t \cdot \tau \cdot \cos(\alpha_{1,2} + \alpha_{2,3} + 2 \cdot \varepsilon)}.$$

Thus

$$\alpha_{1,3} \leq \alpha_{1,2} + \alpha_{2,3} + 3 \cdot \varepsilon.$$

Hence the result follows.



To define $s(t, \tau)$, consider three half-lines $\tilde{\gamma}_1, \tilde{\gamma}_2, \tilde{\gamma}_3$ in a Euclidean plane starting at one point, such that $\angle(\tilde{\gamma}_1, \tilde{\gamma}_2) = \alpha_{1,2} + \varepsilon$, $\angle(\tilde{\gamma}_2, \tilde{\gamma}_3) = \alpha_{2,3} + \varepsilon$, and $\angle(\tilde{\gamma}_1, \tilde{\gamma}_3) = \alpha_{1,2} + \alpha_{2,3} + 2 \cdot \varepsilon$. We parametrize each half-line by the distance from the starting point. Given two positive numbers t, τ , let $s = s(t, \tau)$ be the number such that $\tilde{\gamma}_2(s) \in [\tilde{\gamma}_1(t), \tilde{\gamma}_3(\tau)]$. Clearly, $s \leq \max\{t, \tau\}$, so t, τ, s may be taken sufficiently small. $\square$

1.12. Exercise! *Prove that the sum of adjacent angles is at least π .*

More precisely, suppose two hinges $[p_z^x]$ and $[p_z^y]$ are adjacent; that is, they share side $[pz]$, and the union of two sides $[px]$ and $[py]$ forms a geodesic $[xy]$. Show that

$$\angle[p_z^x] + \angle[p_z^y] \geq \pi$$

whenever each angle on the left-hand side is defined.

Give an example showing that the inequality can be strict.

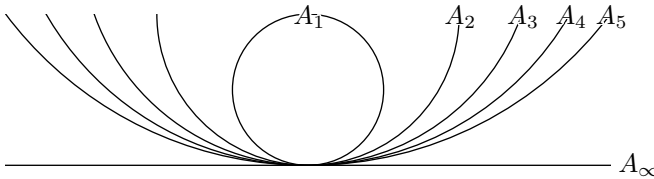
1.13. Exercise. *Let γ be the unit-speed parametrization of $[qx]$ from q to x . Assume that the angle measure $\varphi = \angle[q_x^p]$ is defined. Show that*

$$|p - \gamma(t)| \leq |q - p| - t \cdot \cos \varphi + o(t).$$

J Hausdorff convergence

1.14. Definition. *Let $A_1, A_2, \dots$ be a sequence of closed sets in a metric space $\mathcal{X}$. We say that the sequence A_n converges to a closed set A_∞ in the sense of Hausdorff if, for any $x \in \mathcal{X}$, we have $\text{dist}_{A_n} x \rightarrow \text{dist}_{A_\infty} x$ as $n \rightarrow \infty$.*

For example, suppose $\mathcal{X}$ is the Euclidean plane and A_n is the circle with radius n and center at the point $(0, n)$. Then A_n converges to the x -axis as $n \rightarrow \infty$.



Further, consider the sequence of one-point sets $B_n = \{(n, 0)\}$ in the Euclidean plane. It converges to the empty set; indeed, for any point x we have $\text{dist}_{B_n}(x) \rightarrow \infty$ as $n \rightarrow \infty$ and $\text{dist}_\emptyset(x) = \infty$ for any x .

The following exercise is an extension of the so-called Blaschke selection theorem to our version of Hausdorff convergence.

1.15. Exercise. *Show that any sequence of closed sets in a proper metric space has a convergent subsequence in the sense of Hausdorff.*

1.16. Exercise. *Construct a metric space with a sequence of points $\{x_0, x_1, \dots\}$ such that $|x_0 - x_n| = 1$ for any $n > 0$ and the two-point sets $\{x_0, x_n\}$ converge to the one-point set $\{x_0\}$ in the sense of Hausdorff.*

K Hausdorff metric

1.17. Definition. *Let A and B be two nonempty compact subsets of a metric space $\mathcal{X}$. Then the Hausdorff distance between A and B is defined as*

$$|A - B|_{\text{Haus } \mathcal{X}} := \sup \{ |\text{dist}_A x - \text{dist}_B x| : x \in \mathcal{X} \}.$$

The following observation gives a useful reformulation of the definition:

1.18. Observation. *Suppose A and B are two nonempty compact subsets of a metric space $\mathcal{X}$. Then $|A - B|_{\text{Haus } \mathcal{X}} < R$ if and only if B lies in the R -neighborhood of A , and A lies in the R -neighborhood of B .*

According to the following exercise, Hausdorff convergence of nonempty compact subsets is the convergence in the Hausdorff metric.

1.19. Exercise. *Let $A_1, A_2, \dots$, and A_∞ be compact nonempty sets in a proper metric space $\mathcal{X}$ such that all A_i are contained in a compact subset of $\mathcal{X}$.*

Show that $|A_n - A_\infty|_{\text{Haus } \mathcal{X}} \rightarrow 0$ as $n \rightarrow \infty$ if and only if $A_n \rightarrow A_\infty$ in the sense of Hausdorff.

Does the statement remain true without the assumption that all A_i are contained in a compact subset of $\mathcal{X}$?

L Gromov–Hausdorff convergence

Let $\mathcal{X}_1, \mathcal{X}_2, \dots$, and $\mathcal{X}_\infty$ be a sequence of proper metric spaces. Suppose that there is a metric on the disjoint union

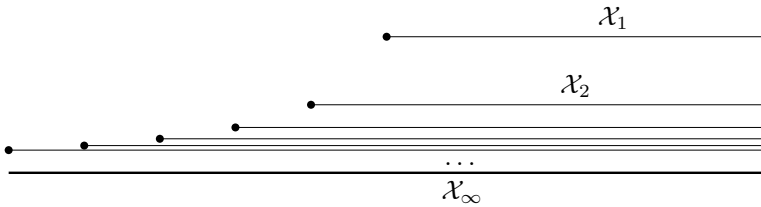
$$\mathbf{X} = \bigsqcup_{n \in \mathbb{N} \cup \{\infty\}} \mathcal{X}_n$$

that satisfies the following property:

1.20. Property. *The restriction of the metric on each $\mathcal{X}_n$ and $\mathcal{X}_\infty$ coincides with its original metric, the space $\mathbf{X}$ is proper, and $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ as subsets of $\mathbf{X}$ in the sense of Hausdorff.*

In this case we say that the metric on $\mathbf{X}$ defines a convergence $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ in the sense of Gromov–Hausdorff. The metric on $\bigsqcup \mathcal{X}_n$ makes it possible to talk about limits of sequences $x_n \in \mathcal{X}_n$ as $n \rightarrow \infty$, as well as weak limits of a sequence of Borel measures μ_n on $\mathcal{X}_n$ and so on.

The limit space is not uniquely defined by the sequence. For example, if each space $\mathcal{X}_n$ in the sequence is isometric to the half-line, then its limit might be isometric to the half-line or the whole line. The first convergence is evident and the second could be guessed from the picture.



Note that any sequence of spaces has an empty space as its limit in some Gromov–Hausdorff convergence. As we will see further, if the limit is nonempty and compact, then it is unique up to isometry.

Pointed convergence. The isometry class of the limit can be fixed by marking a point p_n in each space $\mathcal{X}_n$. We say that $(\mathcal{X}_n, p_n)$ converges to $(\mathcal{X}_\infty, p_\infty)$ if there is a metric on $\mathbf{X}$ as in 1.20 such that $p_n \rightarrow p_\infty$. This is called pointed Gromov–Hausdorff convergence. For example, the sequence $(\mathcal{X}_n, p_n) = ([0, \infty), 0)$ converges to $([0, \infty), 0)$, while $(\mathcal{X}_n, p_n) = ([0, \infty), n)$ converges to $(\mathbb{R}, 0)$ as $n \rightarrow \infty$.

M Gromov–Hausdorff metric

In this section we define the metric on the set of *isometry classes* of compact metric spaces. Further, the term *metric space* might also stand for its *isometry class*.

Loosely speaking, the distance between (isometry classes of) compact metric spaces will be defined as the minimal value r such that both spaces can be isometrically embedded into a third space in such

a way that the Hausdorff distance between the images is not greater than r .

Observe that we *know* a real number x if we can give a yes/no answer to the question $x < r$ for any r . Keep this in mind when reading the following definition.

1.21. Definition. *The Gromov-Hausdorff distance $|\mathcal{X} - \mathcal{Y}|_{\text{GH}}$ between compact metric spaces $\mathcal{X}$ and $\mathcal{Y}$ is defined by the following relation.*

Given $r > 0$, we say that $|\mathcal{X} - \mathcal{Y}|_{\text{GH}} < r$ if and only if there exists a metric space $\mathcal{W}$ and subspaces $\mathcal{X}'$ and $\mathcal{Y}'$ in $\mathcal{W}$ that are isometric to $\mathcal{X}$ and $\mathcal{Y}$, respectively, such that $|\mathcal{X}' - \mathcal{Y}'|_{\text{Haus } \mathcal{W}} < r$. (Here $|\mathcal{X}' - \mathcal{Y}'|_{\text{Haus } \mathcal{W}}$ denotes the Hausdorff distance between sets $\mathcal{X}'$ and $\mathcal{Y}'$ in $\mathcal{W}$.)

These distances actually define a metric on (isometry classes of) compact metric spaces; for a proof we refer to [19, 102]. The resulting metric is called the Gromov-Hausdorff metric; the corresponding metric space will be denoted by GH. This means in particular that if $|\mathcal{X} - \mathcal{Y}|_{\text{GH}} = 0$ for compact metric spaces $\mathcal{X}$ and $\mathcal{Y}$, then they are isometric. Moreover, each of the references above provides the following statement.

1.22. Proposition. *GH is a complete metric space.*

The proofs of the mentioned statements are not quite straightforward. The following two exercises should help the reader who wants to reconstruct these proofs without reading the mentioned textbooks.

1.23. Exercise. *Let f be a distance noncontracting map from a compact metric space $\mathcal{K}$ to itself. Show that f is an isometry; that is, it is a distance-preserving bijection.*

1.24. Exercise. *Show that a surjective short map (that is, distance-nonexpanding) from a compact metric space to itself is an isometry.*

Two convergences. Notice that now we have two notions of convergence of metric spaces. One is defined in the previous section; it works for proper metric spaces and its limit might not be uniquely defined. The other is defined as convergence in GH; the limit in GH is unique if it exists. In particular, these convergences are different, but if we restrict our attention to nonempty uniformly bounded compact spaces, then these two convergences are essentially the same. Let us sketch why.

Suppose $|\mathcal{X}_n - \mathcal{X}_\infty|_{\text{GH}} \rightarrow 0$ as $n \rightarrow \infty$; that is, $\mathcal{X}_n$ converges to $\mathcal{X}_\infty$ in GH. Then there is a metric on $\mathcal{V}_n = \mathcal{X}_n \sqcup \mathcal{X}_\infty$ such that

the restriction of the metric on each $\mathcal{X}_n$ and $\mathcal{X}_\infty$ coincides with its original metric, and $|\mathcal{X}_n - \mathcal{X}_\infty|_{\text{Haus } \mathcal{V}_n} < \varepsilon_n$ for some sequence $\varepsilon_n \rightarrow 0$. Gluing all $\mathcal{V}_n$ along $\mathcal{X}_\infty$, we get the required space $\mathbf{X}$, which defines convergence.

Once the convergence is fixed, we can talk about limits of sequences $x_n \in \mathcal{X}_n$ in $\mathcal{X}_\infty$ as $n \rightarrow \infty$, as well as weak limits of a sequence of Borel measures μ_n on $\mathcal{X}_n$ and so on — all these limits are defined as the corresponding limits in the ambient space $\mathbf{X}$.

Now suppose we have a Gromov–Hausdorff convergence $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ with the common space $\mathbf{X}$. Assume that all spaces $\mathcal{X}_1, \mathcal{X}_2, \dots$, as well as $\mathcal{X}_\infty$, are compact, and $\mathcal{X}_\infty$ is nonempty. (The latter condition means that the convergence is nontrivial.) Observe that in this case $|\mathcal{X}_n - \mathcal{X}_\infty|_{\text{Haus } \mathbf{X}} \rightarrow 0$ as $n \rightarrow \infty$, and therefore $|\mathcal{X}_n - \mathcal{X}_\infty|_{\text{GH}} \rightarrow 0$. In other words, $\mathcal{X}_n$ converges to $\mathcal{X}_\infty$ in GH.

N Almost isometries

1.25. Definition. *Let $\mathcal{X}$ and $\mathcal{Y}$ be metric spaces. A map $f: \mathcal{X} \rightarrow \mathcal{Y}$ is called an ε -isometry if the following two conditions hold:*

- (a) *$f(\mathcal{X})$ is an ε -net in $\mathcal{Y}$; that is, for any $y \in \mathcal{Y}$ there is $x \in \mathcal{X}$ such that*

$$|f(x) - y|_{\mathcal{Y}} \leq \varepsilon.$$

- (b) *$f(\mathcal{X})$ is ε -distance-preserving; that is,*

$$|f(x) - f(x')|_{\mathcal{Y}} \leq |x - x'|_{\mathcal{X}} \pm \varepsilon$$

for any $x, x' \in \mathcal{X}$.

When dealing with Gromov–Hausdorff convergence, the following lemma allows us to bypass constructing explicit metrics on the disjoint unions of $\mathcal{X}_1, \mathcal{X}_2, \dots$, and $\mathcal{X}_\infty$

1.26. Lemma. *Let $\mathcal{X}_1, \mathcal{X}_2, \dots$, and $\mathcal{X}_\infty$ be compact metric spaces, and let $\varepsilon_n \rightarrow 0+$ as $n \rightarrow \infty$. Suppose that either*

- (a) *for each n there is an ε_n -isometry $f_n: \mathcal{X}_n \rightarrow \mathcal{X}_\infty$, or*
 (b) *for each n there is an ε_n -isometry $h_n: \mathcal{X}_\infty \rightarrow \mathcal{X}_n$.*

Then there is a Gromov–Hausdorff convergence $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$.

Furthermore, a partial converse also holds.

- (c) *Suppose we have a Gromov–Hausdorff convergence $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$, the spaces $\mathcal{X}_n$ are uniformly bounded, and $\mathcal{X}_\infty$ is nonempty and compact. Then for some sequence $\varepsilon_n \rightarrow 0+$ as $n \rightarrow \infty$ there are ε_n -isometries $f_n: \mathcal{X}_n \rightarrow \mathcal{X}_\infty$ (and $h_n: \mathcal{X}_\infty \rightarrow \mathcal{X}_n$) such that $x_n \in \mathcal{X}_n$ converges to $x_\infty \in \mathcal{X}_\infty$ with respect to the convergence*

$\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ if and only if $f_n(x_n) \rightarrow x_\infty$ (respectively, $|h_n(x_\infty) - x_n|_{\mathcal{X}_n} \rightarrow 0$) as $n \rightarrow \infty$.

Proof. To prove part (a), let us construct a common space $\mathbf{X}$ for the spaces $\mathcal{X}_1, \mathcal{X}_2, \dots$, and $\mathcal{X}_\infty$ by taking the metric ρ on the disjoint union $\mathcal{X}_\infty \sqcup \mathcal{X}_1 \sqcup \mathcal{X}_2 \sqcup \dots$ that is defined the following way:

$$\begin{aligned} |x_n - y_n|_{\mathbf{X}} &= |x_n - y_n|_{\mathcal{X}_n}, \\ |x_\infty - y_\infty|_{\mathbf{X}} &= |x_\infty - y_\infty|_{\mathcal{X}_\infty}, \\ |x_n - x_\infty|_{\mathbf{X}} &= \inf_{y_n \in \mathcal{X}_n} \{ |x_n - y_n|_{\mathcal{X}_n} + \varepsilon_n + |x_\infty - f_n(y_n)|_{\mathcal{X}_\infty} \}, \\ |x_n - x_m|_{\mathbf{X}} &= \inf_{y_\infty \in \mathcal{X}_\infty} \{ |x_n - y_\infty|_{\mathbf{X}} + |x_m - y_\infty|_{\mathbf{X}} \}, \end{aligned}$$

where we assume that $x_\infty, y_\infty \in \mathcal{X}_\infty$, and $x_n, y_n \in \mathcal{X}_n$ for each n . It remains to observe that this indeed defines a metric on $\mathbf{X}$, and $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ in the sense of Hausdorff.

The proof of the second part is analogous; one only needs to change one line in the definition of the metric as follows:

$$|x_n - x_\infty|_{\mathbf{X}} = \inf_{y_\infty \in \mathcal{X}_\infty} \{ |x_n - h_n(y_\infty)|_{\mathcal{X}_n} + \varepsilon_n + |x_\infty - y_\infty|_{\mathcal{X}_\infty} \}.$$

We leave part (c) as an exercise. □

For two metric spaces $\mathcal{X}$ and $\mathcal{Y}$, we write $\mathcal{X} \leq \mathcal{Y} + \varepsilon$ if there is a map $f: \mathcal{X} \rightarrow \mathcal{Y}$ such that

$$|x - x'|_{\mathcal{X}} \leq |f(x) - f(x')|_{\mathcal{Y}} + \varepsilon$$

for any $x, x' \in \mathcal{X}$.

1.27. Exercise. Let $\mathcal{X}_1, \mathcal{X}_2, \dots$, and $\mathcal{X}_\infty$ be nonempty compact metric spaces. Show that there is a Gromov–Hausdorff convergence $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ if and only if for some sequence $\varepsilon_n \rightarrow 0$, we have

$$\mathcal{X}_\infty \leq \mathcal{X}_n + \varepsilon_n \quad \text{and} \quad \mathcal{X}_n \leq \mathcal{X}_\infty + \varepsilon_n.$$

Lemma 1.26 has a natural analogue for pointed convergence. For simplicity, we only state part (a) of the lemma. Parts (b) and (c) can be rephrased similarly.

The proof of the following lemma is analogous to 1.26.

1.28. Lemma. Let $(\mathcal{X}_1, p_1), (\mathcal{X}_2, p_2), \dots$, and $(\mathcal{X}_\infty, p_\infty)$ be pointed metric spaces, and let $\varepsilon(n, R) \rightarrow 0+$ as $n \rightarrow \infty$ for any fixed $R > 0$. Suppose that for each n there is a map $f_n: \mathcal{X}_n \rightarrow \mathcal{X}_\infty$ such that

(a) $f_n(p_n) \rightarrow p_\infty$;

(b) $||f_n(x) - f_n(x')|_{\mathcal{X}_\infty} - |x - x'|_{\mathcal{X}_n}| \leq \varepsilon(n, R)$ for any $x, x' \in B(p_n, R)$;

(c) For any $x \in B(p_\infty, R)$, there is $x_n \in B(p_n, R)$ such that $|x - f_n(x_n)| \leq \varepsilon(n, R)$.

Then there is a pointed Gromov–Hausdorff convergence $(\mathcal{X}_n, p_n) \rightarrow (\mathcal{X}_\infty, p_\infty)$.

O Remarks

In principle, our definition of Gromov–Hausdorff distance can be applied to complete metric spaces that are not necessarily compact. However, according to the following exercise, it only defines a semimetric; that is, zero Gromov–Hausdorff distance does not imply that the spaces are isometric. For that reason it is not in use.

1.29. Exercise.

- (a) Construct two nonisometric proper (noncompact) metric spaces with vanishing Gromov–Hausdorff distance.
- (b) Construct two nonisometric complete geodesic metric spaces of bounded diameter with vanishing Gromov–Hausdorff distance.

Lecture 2

Definitions

Here we will give several definitions of Alexandrov spaces and prove their equivalence.

A Four-point comparison

Recall that $\tilde{\angle}(p_y^x)$ denotes the model angle; see 1G.

Let p, x, y, z be a quadruple of points in a metric space. If the inequality

$$\textcircled{1} \quad \tilde{\angle}(p_y^x)_{\mathbb{E}^2} + \tilde{\angle}(p_z^y)_{\mathbb{E}^2} + \tilde{\angle}(p_x^z)_{\mathbb{E}^2} \leq 2 \cdot \pi$$

holds, then we say that the quadruple meets $\mathbb{E}^2$ -comparison. If the left-hand side is undefined, then we also assume that the comparison holds.

2.1. Exercise. Suppose $\mathbb{E}^2$ -comparison holds for the quadruple p, x_1, x_2, x_3 . Show that it also holds for any quadruple q, y_1, y_2, y_3 such that

$$|q - y_i| \geq |p - x_i| \quad \text{and} \quad |y_i - y_j| \leq |x_i - x_j|$$

for all i and j .

Instead of $\mathbb{E}^2$, we can use $\mathbb{S}^2$ or $\mathbb{H}^2$. This way we get the definition of $\mathbb{S}^2$ - or $\mathbb{H}^2$ -comparisons. Recall that $\tilde{\angle}(p_y^x)_{\mathbb{E}^2}$ and $\tilde{\angle}(p_y^x)_{\mathbb{H}^2}$ are defined if $p \neq x, p \neq y$, but for $\tilde{\angle}(p_y^x)_{\mathbb{S}^2}$ we require in addition that

$$|p - x| + |p - y| + |x - y| < 2 \cdot \pi;$$

if this does not hold for one of the angles, then we assume that $\mathbb{S}^2$ -comparison holds for this quadruple.

More generally, one may apply this definition to the model κ -plane $\mathbb{M}^2(\kappa)$ and define $\mathbb{M}^2(\kappa)$ -comparison for any real κ . However, when considering $\mathbb{M}^2(\kappa)$ -comparison, it is safe to assume that $\kappa = -1, 0$, or 1 ; applying rescaling, the $\mathbb{M}^2(\kappa)$ -comparison can be reduced to these three cases.

2.2. Definition. *A metric space $\mathcal{X}$ has curvature $\geq \kappa$ in the sense of Alexandrov if $\mathbb{M}^2(\kappa)$ -comparison holds for any quadruple of points in $\mathcal{X}$.*

While this definition can be applied to any metric space, we will use it mostly for geodesic spaces that are complete (and often compact or proper). If a complete geodesic space has curvature $\geq \kappa$ in the sense of Alexandrov, then it will be called an $\text{ALEX}(\kappa)$ space; here $\text{ALEX}(\kappa)$ is an adjective. If $\mathcal{X}$ is $\text{ALEX}(\kappa)$ for some κ , then we say that $\mathcal{X}$ is an Alexandrov space.

It is common practice in Alexandrov geometry to write proofs for nonnegative curvature and leave the general curvature bound as an exercise. These generalizations are usually straightforward. We will add notes when they are not. We will also often formulate statements just for $\kappa = 0$ even when they admit straightforward generalizations to arbitrary curvature bounds; see [7] for a more formal treatment.

2.3. Exercise. *Show that $\mathbb{E}^n$ is $\text{ALEX}(0)$.*

2.4. Exercise. *Show that a metric space $\mathcal{X}$ has nonnegative curvature in the sense of Alexandrov if and only if, for any quadruple of points $p, x_1, x_2, x_3 \in \mathcal{X}$ there is a quadruple of points $q, y_1, y_2, y_3 \in \mathbb{E}^2$ such that*

$$|p - x_i|_{\mathcal{X}} \geq |q - y_i|_{\mathbb{E}^2} \quad \text{and} \quad |x_i - x_j|_{\mathcal{X}} \leq |y_i - y_j|_{\mathbb{E}^2}$$

for all i and j .

B Alexandrov's lemma

Recall that $[xy]$ denotes a geodesic from x to y ; set

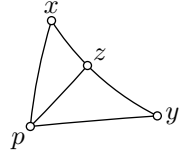
$$]xy[= [xy] \setminus \{x\}, \quad]xy[= [xy] \setminus \{y\}, \quad]xy[= [xy] \setminus \{x, y\}.$$

2.5. Alexandrov's lemma. *Let p, x, y, z be distinct points in a metric space such that $z \in]xy[$. Then the following expressions have the same sign:*

- (a) $\tilde{\angle}(x_y^p) - \tilde{\angle}(x_z^p)$,
- (b) $\tilde{\angle}(z_x^p) + \tilde{\angle}(z_y^p) - \pi$.

The same holds for the hyperbolic and spherical model angles, but in the latter case, one has to assume in addition that

$$|p - x| + |p - y| + |x - y| < 2 \cdot \pi.$$



In the proof we will apply the following statement from elementary geometry, which is an immediate consequence of the cosine law:

2.6. Observation. *Increasing one side in a plane triangle increases its opposite angle, and the other way around. Moreover, the same statement holds for spherical and hyperbolic triangles.*

Proof of 2.5. Consider the model triangle $[\tilde{x}\tilde{p}\tilde{z}] = \tilde{\Delta}(xpz)$. Take a point $\tilde{y}$ on the extension of $[\tilde{x}\tilde{z}]$ beyond $\tilde{z}$ so that $|\tilde{x} - \tilde{y}| = |x - y|$ (and therefore $|\tilde{x} - \tilde{z}| = |x - z|$).

By 2.6, the following expressions have the same sign:

- (i) $\angle[\tilde{x}\tilde{p}\tilde{y}] - \tilde{\angle}(x_p^y)$,
- (ii) $|\tilde{p} - \tilde{y}| - |p - y|$,
- (iii) $\angle[\tilde{z}\tilde{p}\tilde{y}] - \tilde{\angle}(z_p^y)$.

Since

$$\angle[\tilde{x}\tilde{p}\tilde{y}] = \angle[\tilde{x}\tilde{p}\tilde{z}] = \tilde{\angle}(x_p^z)$$

and

$$\angle[\tilde{z}\tilde{p}\tilde{y}] = \pi - \angle[\tilde{z}\tilde{p}\tilde{x}] = \pi - \tilde{\angle}(z_p^x),$$

the statement follows.

The spherical and hyperbolic cases can be proved along the same lines. □

2.7. Exercise! *Assume that p, x, y, z are as in Alexandrov's lemma (2.5). Show that*

$$\tilde{\angle}(p_y^x) \geq \tilde{\angle}(p_z^x) + \tilde{\angle}(p_y^z),$$

with equality if and only if the expressions in (a) and (b) in Alexandrov's lemma vanish.

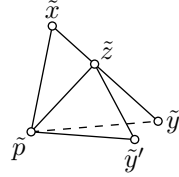
Note that

$$p \in]xy[\quad \Rightarrow \quad \tilde{\angle}(p_y^x) = \pi.$$

Applying this together with Alexandrov's lemma and $\mathbb{E}^2$ -comparison, we get the following.

2.8. Claim. *Let p, x, y, z be points in an $\text{ALEX}(0)$ space. Suppose that $z \in]xy[$. Then*

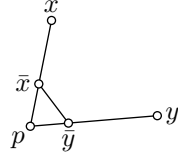
$$\tilde{\angle}(x_y^y) \leq \tilde{\angle}(x_p^z).$$



2.9. Exercise! Let $[p_y^x]$ be a hinge in an $\text{ALEX}(0)$ space. Consider the function

$$f: (|p - \bar{x}|, |p - \bar{y}|) \mapsto \tilde{\angle}(p_{\bar{y}}^{\bar{x}}),$$

where $\bar{x} \in]px[$ and $\bar{y} \in]py[$. Show that f is nonincreasing in each argument.



The statement in the last exercise is called angle-sidelength monotonicity. It implies the following.

2.10. Angle existence and comparison. The angle measure of any hinge in an $\text{ALEX}(0)$ space is defined and is at least as large as the corresponding model angle; that is,

$$\angle[p_y^x] \geq \tilde{\angle}(p_y^x)$$

for any hinge $[p_y^x]$ in an $\text{ALEX}(0)$ space.

2.11. Exercise! Let $[p_y^x]$ be a hinge in an $\text{ALEX}(0)$ space. Suppose $\angle[p_y^x] = 0$; show that $[px] \subset [py]$ or $[py] \subset [px]$.

Conclude that geodesics in an $\text{ALEX}(0)$ space cannot bifurcate; that is, if two geodesics $[px]$ and $[py]$ share a nontrivial arc with an endpoint at p , then $[px] \subset [py]$ or $[py] \subset [px]$.

2.12. Exercise. Let $[xy]$ be a geodesic in an $\text{ALEX}(0)$ space. Suppose $z \in]xy[$. Show that there is a unique geodesic $[xz]$ and $[xz] \subset [xy]$.

Recall that adjacent hinges are defined in 1.12.

2.13. Exercise! Let $[p_z^x]$ and $[p_z^y]$ be adjacent hinges in an $\text{ALEX}(0)$ space. Show that

$$\angle[p_z^x] + \angle[p_z^y] = \pi.$$

2.14. Exercise. Let $\mathcal{A}$ be an $\text{ALEX}(0)$ space. Show that

$$\tilde{\angle}(x_p^y) = \tilde{\angle}(x_p^v) \iff \tilde{\angle}(x_p^y) = \tilde{\angle}(x_p^w)$$

for any points p, x, y, v, w in $\mathcal{A}$ such that $v, w \in]xy[$.

2.15. Exercise (semicontinuity of angles). Let $\mathcal{A}$ be an $\text{ALEX}(0)$ space. Suppose hinges $[x_n y_n]$ in $\mathcal{A}$ converge to a hinge $[x_\infty y_\infty]$; that is, geodesics $[x_n y_n]$ and $[x_n z_n]$ converge to the geodesics $[x_\infty y_\infty]$ and $[x_\infty z_\infty]$ in the sense of Hausdorff. Show that

$$\lim_{n \rightarrow \infty} \angle[x_n y_n] \geq \angle[x_\infty y_\infty].$$

The last inequality might be strict; this happens on the surface of a cube in $\mathbb{E}^3$, which is an $\text{ALEX}(0)$ space by 11.15.

C Hinge comparison

Let $[p_y^x]$ be a hinge in an $\text{ALEX}(0)$ space $\mathcal{A}$. By 2.9, the angle measure $\angle[p_y^x]$ is defined and

$$\angle[p_y^x] \geq \tilde{\angle}(p_y^x).$$

Further, according to 2.13, we have

$$\angle[p_z^x] + \angle[p_z^y] = \pi$$

for adjacent hinges $[p_z^x]$ and $[p_z^y]$ in $\mathcal{A}$.

The following theorem adds a converse.

2.16. Theorem. *A complete geodesic space $\mathcal{A}$ is $\text{ALEX}(0)$ if and only if both of the following conditions hold.*

(a) *For any hinge $[x_y^p]$ in $\mathcal{A}$, the angle $\angle[x_y^p]$ is defined and*

$$\angle[x_y^p] \geq \tilde{\angle}(x_y^p).$$

(b) *For any two adjacent hinges $[p_z^x]$ and $[p_z^y]$ in $\mathcal{A}$, we have*

$$\angle[p_z^x] + \angle[p_z^y] \leq \pi.$$

Recall that, by 2.13, we have equality in (b).

Proof. Recall that the only-if part follows from 2.9 and 2.13; it remains to show the if part.

Consider a point $w \in]pz[$ close to p . From (b), it follows that

$$\angle[w_z^x] + \angle[w_p^x] \leq \pi \quad \text{and} \quad \angle[w_z^y] + \angle[w_p^y] \leq \pi.$$

Since $\angle[w_y^x] \leq \angle[w_p^x] + \angle[w_y^p]$ (see 1.11), we get

$$\angle[w_z^x] + \angle[w_z^y] + \angle[w_y^x] \leq 2 \cdot \pi.$$

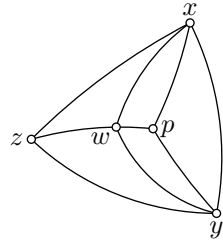
Applying (a),

$$\tilde{\angle}(w_z^x) + \tilde{\angle}(w_z^y) + \tilde{\angle}(w_y^x) \leq 2 \cdot \pi.$$

Passing to the limit as $w \rightarrow p$, we have

$$\tilde{\angle}(p_z^x) + \tilde{\angle}(p_z^y) + \tilde{\angle}(p_y^x) \leq 2 \cdot \pi.$$

□



D Equivalent conditions

The following theorem summarizes 2.8, 2.10, 2.13, and 2.16.

2.17. Theorem. *Let $\mathcal{A}$ be a complete geodesic space. Then the following conditions are equivalent.*

(a) $\mathcal{A}$ is ALEX(0).

(b) (adjacent angle comparison)

$$\tilde{\angle}(z_x^p) + \tilde{\angle}(z_y^p) \leq \pi$$

for any geodesic $[xy]$ and points $z \in]xy[$, $p \neq z$ in $\mathcal{A}$.

(c) (point-on-side comparison)

$$\tilde{\angle}(x_y^p) \leq \tilde{\angle}(x_z^p)$$

for any geodesic $[xy]$ and $z \in]xy[$ in $\mathcal{A}$.

(d) (hinge comparison) the angle $\angle[x_y^p]$ is defined for any hinge $[x_y^p]$ in $\mathcal{A}$. Moreover,

$$\angle[x_y^p] \geq \tilde{\angle}(x_y^p)$$

for any hinge $[x_y^p]$, and

$$\angle[z_y^p] + \angle[z_x^p] \leq \pi$$

for any adjacent hinges $[z_y^p]$ and $[z_x^p]$.

Moreover, the implications $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d)$ hold in any space, not necessarily a geodesic one.

2.18. Advanced exercise. *Construct a complete geodesic space $\mathcal{X}$ that is not ALEX(0), but satisfies the following weaker version of the adjacent angle comparison (2.17b).*

For any three points $p, x, y \in \mathcal{X}$, there is a geodesic $[xy]$ such that for any $z \in]xy[$

$$\tilde{\angle}(z_x^p) + \tilde{\angle}(z_y^p) \leq \pi.$$

2.19. Exercise. *Let $\mathcal{W}$ be $\mathbb{R}^n$ with the metric induced by a norm. Show that if $\mathcal{W}$ is ALEX(0), then $\mathcal{W}$ is isometric to the Euclidean space $\mathbb{E}^n$.*

E Function comparison

Real-to-real functions. Choose $\lambda \in \mathbb{R}$. Let $s: \mathbb{I} \rightarrow \mathbb{R}$ be a locally Lipschitz function defined on an interval $\mathbb{I} \subset \mathbb{R}$. The following statements are equivalent; if one (and therefore any) of them holds for s , then we say that s is λ -concave.

- ◊ We have the inequality $s'' \leq \lambda$, where the second derivative s'' is understood in the sense of distributions.
- ◊ The function $t \mapsto s(t) - \lambda \cdot \frac{t^2}{2}$ is concave.
- ◊ The Jensen inequality

$$s(a \cdot t_0 + (1-a) \cdot t_1) \geq a \cdot s(t_0) + (1-a) \cdot s(t_1) + \frac{\lambda}{2} \cdot a \cdot (1-a) \cdot (t_1 - t_0)^2$$

holds for any $t_0, t_1 \in \mathbb{I}$ and $a \in [0, 1]$.

- ◊ For any $t_0 \in \mathbb{I}$ there is a quadratic polynomial $\ell = \frac{\lambda}{2} \cdot t^2 + a \cdot t + b$ (it is called a barrier) that supports s at t_0 from above; that is, $\ell(t_0) = s(t_0)$ and $\ell(t) \geq s(t)$ for any t .
- ◊ For any $t_0 \in \mathbb{I}$ there is a quadratic polynomial ℓ (it is called a local barrier) that locally supports s at t_0 from above; that is, $\ell(t_0) = s(t_0)$ and $\ell(t) \geq s(t)$ for any t in a neighborhood of t_0 .

To prove the equivalence, approximate s by smooth functions, taking convolutions $s_n = s * k_n$ for a suitable sequence of kernels k_n . Observe that all the conditions are equivalent for s_n ; passing to the limit, we get the same for s .

2.20. Exercise. Show that λ -concave functions are one-sided differentiable. That is, if the function $f: \mathbb{I} \rightarrow \mathbb{R}$ is λ -concave, then

(a) the right derivative

$$f^+(t_0) = \lim_{t \rightarrow t_0^+} \frac{f(t) - f(t_0)}{t - t_0}$$

is defined for any $t_0 \in \mathbb{I}$ except at the left endpoint of $\mathbb{I}$.

(b) the left derivative

$$f^-(t_0) = \lim_{t \rightarrow t_0^-} \frac{f(t) - f(t_0)}{t - t_0}$$

is defined for any $t_0 \in \mathbb{I}$ except at the right endpoint of $\mathbb{I}$.

Furthermore, show that $f^+(t_0) - f^-(t_0) \leq 0$ for any t_0 in the interior of $\mathbb{I}$.

The following exercise implies that if the function is defined on an open interval, then the Lipschitz condition can be dropped from the definition of λ -concavity.

2.21. Exercise. Suppose a real-to-real function f is defined on an open interval and, for some $\lambda \in \mathbb{R}$, it satisfies the Jensen inequality stated above. Show that f is locally Lipschitz.

Functions on metric spaces. A function on a metric space $\mathcal{A}$ will usually mean a *locally Lipschitz real-valued function defined on an open subset of $\mathcal{A}$* . The domain of a function f will be denoted by $\text{Dom } f$. Thus, when we write $f: \mathcal{A} \rightarrow \mathbb{R}$ we do not assume that f is defined on the whole space $\mathcal{A}$.

We say that such f is λ -concave (briefly, $f'' \leq \lambda$) if for any geodesic $\gamma: \mathbb{I} \rightarrow \text{Dom } f$ the real-to-real function $t \mapsto f \circ \gamma(t)$ is λ -concave.

The following proposition is simple but conceptual — it reduces global comparison to an infinitesimal condition on distance functions.

2.22. Proposition. A complete geodesic space $\mathcal{A}$ is $\text{ALEX}(0)$ if and only if $f'' \leq 1$ for any function f of the form

$$f: x \mapsto \frac{1}{2} \cdot |p - x|^2.$$

Proof. Choose a geodesic γ in $\mathcal{A}$ and two points $x = \gamma(t_0)$, $y = \gamma(t_1)$ for some $t_0 < t_1$. Consider the model triangle $[\tilde{p}\tilde{x}\tilde{y}] = \triangle(pxy)$. Let $\tilde{\gamma}: [t_0, t_1] \rightarrow \mathbb{E}^2$ be the unit-speed parametrization of $[\tilde{x}\tilde{y}]$ from $\tilde{x}$ to $\tilde{y}$. Set

$$\tilde{r}(t) := |\tilde{p} - \tilde{\gamma}(t)|, \quad r(t) := |p - \gamma(t)|.$$

Clearly, $\tilde{r}(t_0) = r(t_0)$ and $\tilde{r}(t_1) = r(t_1)$. By the point-on-side comparison (2.17c), the implication

$$\textbf{1} \quad t_0 \leq t \leq t_1 \quad \Rightarrow \quad \tilde{r}(t) \leq r(t)$$

holds for any γ and $t_0 < t_1$.

Observe that $(\tilde{r}^2)'' \equiv 2$. If $(r^2)'' \leq 2$, then $t \mapsto r^2(t) - \tilde{r}^2(t)$ is concave. Since $\tilde{r}(t_0) = r(t_0)$ and $\tilde{r}(t_1) = r(t_1)$, we get **1**, which proves the if part.

On the other hand, **1** implies the Jensen inequality for the function $h: t \mapsto r(t)^2 - t^2$. Since the subinterval $[t_0, t_1]$ can be chosen arbitrarily, we conclude that $h'' \leq 0$, or, equivalently, $(r^2)'' \leq 2$, and the only-if part follows. $\square$

Semiconcave functions. Let f be a locally Lipschitz real-valued function defined on an open subset $\text{Dom } f$ of an Alexandrov space $\mathcal{A}$. Suppose φ is a continuous function defined on $\text{Dom } f$. We will

write $f'' \leq \varphi$ if for any point $x \in \text{Dom } f$ and any $\varepsilon > 0$ there is a neighborhood $U \ni x$ such that the restriction $f|_U$ is $(\varphi(x) + \varepsilon)$ -concave.

If $f'' \leq \varphi$ for some continuous function φ , then f is called semi-concave.

2.23. Exercise. Let f be a distance function on an $\text{ALEX}(0)$ space $\mathcal{A}$; that is, $f(x) \equiv |p - x|$ for some $p \in \mathcal{A}$. Show that $f'' \leq \frac{1}{f}$. In particular, f is semiconcave in $\mathcal{A} \setminus \{p\}$.

Proposition 2.22 admits the following generalization, which can be proved along the same lines, but the formulas get longer.

2.24. Proposition. A complete geodesic space $\mathcal{A}$ is $\text{ALEX}(\mp 1)$ if and only if $f'' \leq \pm f$ for any function of the type $f = \cosh \circ \text{dist}_p$ (respectively, $f = -\cos \circ \text{dist}_p|_{\text{B}(p, \pi)}$).

The geometric meaning of these inequalities remains the same: distance functions on an $\text{ALEX}(\kappa)$ space are more concave than distance functions in $\mathbb{M}^2(\kappa)$.

F Remarks

Alexandrov's lemma has the following useful variation; see [7, 10.2] or [4, 3.3].

2.25. Overlap lemma. Let $\tilde{x}^1, \tilde{x}^2, \tilde{x}^3, \tilde{p}^1, \tilde{p}^2$, and $\tilde{p}^3$ be points in $\mathbb{E}^2$, $\mathbb{S}^2$, or $\mathbb{H}^2$. Assume that, for any permutation $\{i, j, k\}$ of $\{1, 2, 3\}$, we have

- (i) $|\tilde{p}^i - \tilde{x}^k| = |\tilde{p}^j - \tilde{x}^k|$,
- (ii) $\tilde{p}^i$ and $\tilde{x}^i$ lie in the same closed half-space determined by $[\tilde{x}^j \tilde{x}^k]$,
- (iii) no pair of triangles $[\tilde{p}^i \tilde{x}^j \tilde{x}^k]$ overlaps.

Then

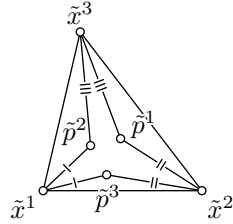
$$\angle \tilde{p}^1 + \angle \tilde{p}^2 + \angle \tilde{p}^3 > 2 \cdot \pi,$$

where $\angle \tilde{p}^i := \angle[\tilde{p}^i \tilde{x}^j \tilde{x}^k]$ for a permutation $\{i, j, k\}$ of $\{1, 2, 3\}$.

The condition (b) in 2.16 might not be necessary. This is a long-standing open problem possibly dating back to Alexandrov [19, footnote in 4.1.5]. Let us state it formally.

2.26. Open question. Let $\mathcal{A}$ be a complete geodesic space (you can also assume that $\mathcal{A}$ is homeomorphic to $\mathbb{S}^2$ or $\mathbb{R}^2$) such that for any hinge $[x_y^p]$ in $\mathcal{A}$, the angle $\angle[x_y^p]$ is defined and

$$\angle[x_y^p] \geq \tilde{\angle}(x_y^p).$$



Is it true that $\mathcal{A}$ is an Alexandrov space?

One could define Alexandrov spaces as complete *length* spaces with curvature $\geq \kappa$ (in the sense of Alexandrov). This condition is more natural and general, but many statements can be reduced to the geodesic case. In particular, any complete length space $\mathcal{A}$ with curvature $\geq \kappa$ *can be isometrically embedded into an $\text{ALEX}(\kappa)$ space* — the ultrapower of $\mathcal{A}$; see [7, 4.11+8.4]. Also, by Plaut's theorem, any point p in $\mathcal{A}$ can be connected by geodesics to *most* points in $\mathcal{A}$; see [7, 8.11] and compare to 6.19c.

Our 4-point comparison in Section 2A is closely related to the so-called CAT comparison, which defines an *upper* curvature bound in the sense of Alexandrov; this is the subject of our previous book [6]. In both comparisons, we check certain conditions on the 6 distances between pairs of points in a 4-point set. Such conditions for n -point subsets were considered by Michael Gromov [40, Section 1.19₊], and they have been actively studied since then [32, 41, 65–68, 70, 72, 100, 120].

Lecture 3

Globalization

The globalization theorem states that every locally Alexandrov space is globally Alexandrov. We prove it for compact spaces and indicate the proof of the general case.

A Globalization

A complete geodesic metric space $\mathcal{A}$ is locally ALEX(0) if any point $p \in \mathcal{A}$ admits a neighborhood $U \ni p$ such that the $\mathbb{E}^2$ -comparison holds for any quadruple of points in U .

It is straightforward to obtain a local version of Theorem 2.17 for locally ALEX(0) spaces; it gives a list of equivalent properties that hold in sufficiently small neighborhoods of any point. In particular, the analog of 2.17d implies that *the angle measure of any hinge in a locally ALEX(0) space is well defined*.

3.1. Globalization theorem. *Any compact locally ALEX(0) space is ALEX(0).*

Proof of 3.1 modulo the key lemma (3.2). Note that condition 2.16b holds in $\mathcal{A}$ (the necessity of 2.16b uses only the local condition). It remains to check 2.16a; that is,

$$\textcircled{1} \quad \angle[x_y^p] \geq \tilde{\angle}(x_y^p)$$

for any hinge $[x_y^p]$ in $\mathcal{A}$.

Inequality $\textcircled{1}$ holds for hinges in a small neighborhood of any point; this can be proved the same way as 2.10 and 2.13, applying the local version of the $\mathbb{E}^2$ -comparison. Since $\mathcal{A}$ is compact, there is $\varepsilon > 0$ such

that ❶ holds whenever $|x - p| + |p - y| < \varepsilon$. Applying the key lemma several times, we get that ❶ holds for any given hinge. $\square$

3.2. Key lemma. *Let $\mathcal{A}$ be a locally ALEX(0) space. If the comparison*

$$\angle[x_q^p] \geq \tilde{\angle}(x_q^p)$$

holds for any hinge $[x_q^p]$ such that $|x - p| + |x - q| < \frac{2}{3} \cdot \ell$, then it holds for any hinge $[x_q^p]$ such that $|x - p| + |x - q| < \ell$.

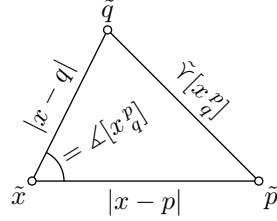
Let $[x_q^p]$ be a hinge in $\mathcal{A}$. Denote by $\tilde{\gamma}[x_q^p]$ its model side; this is the opposite side in the plane triangle with the same angle and adjacent side lengths as in $[x_q^p]$.

More precisely,

$$\tilde{\gamma}[x_q^p]_{\mathcal{A}} := |\tilde{p} - \tilde{q}|_{\mathbb{E}^2}$$

if $[\tilde{x}_{\tilde{q}}^{\tilde{p}}]_{\mathbb{E}^2}$ is the model hinge of $[x_q^p]_{\mathcal{A}}$; that is,

$$\begin{aligned} \angle[\tilde{x}_{\tilde{q}}^{\tilde{p}}]_{\mathbb{E}^2} &= \angle[x_q^p]_{\mathcal{A}}, \\ |\tilde{x} - \tilde{p}|_{\mathbb{E}^2} &= |x - p|_{\mathcal{A}}, \\ |\tilde{x} - \tilde{q}|_{\mathbb{E}^2} &= |x - q|_{\mathcal{A}}. \end{aligned}$$



Note that

$$\tilde{\gamma}[x_q^p] \geq |p - q| \quad \Leftrightarrow \quad \angle[x_q^p] \geq \tilde{\angle}(x_q^p).$$

Proof. Consider a hinge $[x_q^p]$ such that

$$\text{❷} \quad \frac{2}{3} \cdot \ell \leq |p - x| + |x - q| < \ell.$$

It is sufficient to prove that

$$\text{❸} \quad \tilde{\gamma}[x_q^p] \geq |p - q|$$

First, let us describe a construction of a new hinge $[x'^p_q]$ with several properties described below; in particular,

$$\text{❹} \quad |p - x| + |x - q| \geq |p - x'| + |x' - q|,$$

and

$$\text{❺} \quad \tilde{\gamma}[x_q^p] \geq \tilde{\gamma}[x'^p_q].$$

Construction. Assume $|x - q| \geq |x - p|$; otherwise, switch the roles of p and q . Take $x' \in [xq]$ such that

$$\textcircled{6} \quad |p - x| + 3 \cdot |x - x'| = \frac{2}{3} \cdot \ell.$$

Choose a geodesic $[x'p]$ and consider the hinge $[x'p]_q$ formed by $[x'p]$ and $[x'q] \subset [xq]$. The triangle inequality implies $\textcircled{4}$. Furthermore,

$$|p - x| + |x - x'| < \frac{2}{3} \cdot \ell, \quad |p - x'| + |x' - x| < \frac{2}{3} \cdot \ell.$$

In particular, the assumption of the lemma implies that

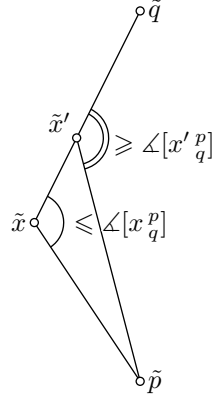
$$\textcircled{7} \quad \angle[x'p]_q \geq \tilde{\angle}(x'p_x) \quad \text{and} \quad \angle[x'p]_x \geq \tilde{\angle}(x'p_x).$$

Let $[\tilde{x}\tilde{x}'\tilde{p}] = \tilde{\Delta}(xx'p)$. Take $\tilde{q}$ on the extension of $[\tilde{x}\tilde{x}']$ beyond $\tilde{x}'$ such that $|\tilde{x} - \tilde{q}| = |x - q|$ (and therefore $|\tilde{x}' - \tilde{q}| = |x' - q|$). By $\textcircled{7}$,

$$\angle[x'p]_q = \angle[x'p]_x \geq \tilde{\angle}(x'p_x) \Rightarrow \tilde{\gamma}[x'p]_q \geq |\tilde{p} - \tilde{q}|.$$

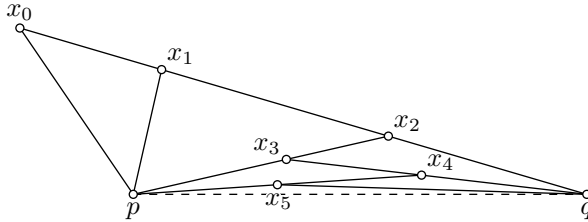
Hence

$$\begin{aligned} \angle[\tilde{x}'\tilde{p}]_q &= \pi - \tilde{\angle}(x'p_x) \geq \\ &\geq \pi - \angle[x'p]_x = \\ &= \angle[x'p]_q, \end{aligned}$$



and $\textcircled{5}$ follows.

Let us continue the proof. Set $x_0 = x$. Applying the above construction inductively, we obtain a sequence of hinges $[x_n p]_q$ with $x_{n+1} = x'_n$. From $\textcircled{5}$, we see that the sequence $s_n = \tilde{\gamma}[x_n p]_q$ is nonincreasing.



The process terminates at x_n if $|p - x_n| + |x_n - q| < \frac{2}{3} \cdot \ell$. In this case, by the assumptions of the lemma, $\tilde{\gamma}[x_n p]_q \geq |p - q|$. Since the sequence s_n is nonincreasing, inequality $\textcircled{3}$ follows.

From now on, we assume that the process does not terminate. Let us prove the following claim.

⑧ The distances $|x_n - x_{n+1}|$, $|x_n - p|$, and $|x_n - q|$ are bounded away from zero for all large n .

Set

$$\begin{aligned} a_n &= \min\{|p - x_n|, |q - x_n|\}, \\ b_n &= \max\{|p - x_n|, |q - x_n|\}, \\ r_n &= |p - x_n| + |x_n - q| = a_n + b_n. \end{aligned}$$

By the triangle inequality, r_n is a nonincreasing sequence. Since the process does not terminate, we have $\frac{2}{3}\cdot\ell \leq r_n < \ell$ for all n .

By ⑥, $|x_n - x_{n+1}| = \frac{1}{3}(\frac{2}{3}\cdot\ell - a_n)$. Since $a_n + b_n = r_n < \ell$ and $a_n \leq b_n$, we have $a_n \leq \frac{1}{2}\cdot\ell$. Hence

$$\textcircled{9} \quad \frac{2}{9}\cdot\ell \geq |x_n - x_{n+1}| \geq \frac{1}{18}\cdot\ell,$$

which proves the claim for $|x_n - x_{n+1}|$.

Note that

$$a_{n+1} = b_n - |x_n - x_{n+1}| \quad \text{or} \quad b_{n+1} = b_n - |x_n - x_{n+1}|.$$

In the latter case,

$$\begin{aligned} a_{n+1} &= r_{n+1} - b_{n+1} = \\ &= (r_{n+1} - r_n) + (r_n - b_n) + |x_n - x_{n+1}| \\ &= -(r_n - r_{n+1}) + a_n + |x_n - x_{n+1}|. \end{aligned}$$

Since r_n does not increase, it must converge; so $r_n - r_{n+1} \rightarrow 0$ as $n \rightarrow \infty$. Therefore, $r_n - r_{n+1} \leq \frac{1}{100}\cdot\ell$ for all large n . Observe that $b_n \geq \frac{1}{2}\cdot r_n$, and therefore $b_n \geq \frac{1}{3}\cdot\ell$. Taking ⑨ into account, we get that in both cases $a_{n+1} \geq \frac{1}{100}\cdot\ell$ for all large n , which finishes the proof of the claim. $\triangle$

Since $r_n - r_{n+1} \rightarrow 0$, ⑧ implies that $\tilde{\angle}(x_n \overset{p_n}{x_{n+1}}) \rightarrow \pi$, where $p_n = p$ if $x_{n+1} \in [x_n q]$, and otherwise $p_n = q$ (this describes switching the roles of p and q in the construction of x'). Since $\angle[x_n \overset{p_n}{x_{n+1}}] \geq \tilde{\angle}(x_n \overset{p_n}{x_{n+1}})$, we have $\angle[x_n \overset{p_n}{x_{n+1}}] \rightarrow \pi$ as $n \rightarrow \infty$.

It follows that

$$r_n - s_n = |p - x_n| + |x_n - q| - \tilde{\gamma}[x_n \overset{p}{q}] \rightarrow 0.$$

By the triangle inequality, $r_n \geq |p - q|$; therefore,

$$\lim_{n \rightarrow \infty} s_n \geq |p - q|.$$

Finally, the monotonicity of the sequence $s_n = \tilde{\gamma}[x_n \overset{p}{q}]$ implies ③. $\square$

B On the general case

3.3. Theorem. *Any locally $\text{ALEX}(\kappa)$ space is $\text{ALEX}(\kappa)$.*

This is a more general version of our globalization theorem 3.1. It holds for an arbitrary curvature bound κ and replaces compactness with completeness. A proof can be found in [7, 2F]; it is based on the idea of 3.1, but requires a few additional twists that we are about to discuss.

Trading compactness for completeness. Assume for a moment that $\kappa = 0$. To replace compactness by completeness, we need the following statement.

3.4. Exercise. *Let $\mathcal{X}$ be a complete metric space. Suppose $r: \mathcal{X} \rightarrow \mathbb{R}$ is a positive continuous function. Show that for any $\varepsilon > 0$, there exists a point $p \in \mathcal{X}$ such that*

$$r(x) > (1 - \varepsilon) \cdot r(p)$$

for every $x \in \overline{B}[p, \frac{1}{\varepsilon} \cdot r(p)]$.

Define $r(x)$ to be the maximal radius of a ball centered at x in which the comparison condition holds (this definition should be made precise, but the details are not important here). Note that if $r(x) = \infty$ for one (and therefore for every) point x , then the theorem follows immediately. To prove the theorem, we need to choose a point p satisfying the conclusion of the exercise and apply the key lemma.

On general curvature bounds. The case $\kappa \leq 0$ in 3.3 can be proved in the same way as the case $\kappa = 0$, but the case $\kappa > 0$ requires extra care. This is because model triangles may be undefined in $\mathbb{M}^2(\kappa)$ when $\kappa > 0$.¹

Meaning of “curvature $\geq \kappa$ ”. Recall that an $\text{ALEX}(\kappa)$ space is defined as a complete geodesic space having “curvature $\geq \kappa$ in the sense of Alexandrov.” This suggests that the following claim should hold.

3.5. Claim. *If $K > \kappa$, then any $\text{ALEX}(K)$ space is $\text{ALEX}(\kappa)$.*

The proof of this statement uses 3.3.

Proof. By 1.8,

$$\tilde{\Delta}(x_z^y)_{\mathbb{M}^2(\kappa)} \leq \tilde{\Delta}(x_z^y)_{\mathbb{M}^2(K)}$$

¹Recall that, according to the definitions in Section 1G, if $\tilde{\Delta}(p_r^q)_{\mathbb{M}^2(\kappa)}$, and hence $\tilde{\Delta}(pqr)_{\mathbb{M}^2(\kappa)}$, are defined for some $\kappa > 0$, then $|p - q| + |q - r| + |r - p| < 2 \cdot \pi / \sqrt{\kappa}$.

whenever both sides are defined. This completes the proof in the case $K \leq 0$. In general, however, the comparison is guaranteed only for sufficiently small triangles; in particular, any $\text{ALEX}(K)$ space is locally $\text{ALEX}(\kappa)$. Therefore, 3.3 completes the proof. $\square$

While the expression “curvature bounded below by κ ” makes sense for geodesic spaces, it does not make much sense for general metric spaces. For example, if $\mathcal{X}$ is the set $\{p, x_1, x_2, x_3\}$ with the metric defined by $|p - x_i| = \pi$ and $|x_i - x_j| = 2 \cdot \pi$ for all $i \neq j$, then $\mathcal{X}$ has no defined spherical model angles. Therefore, formally speaking, $\mathcal{X}$ has curvature ≥ 1 . On the other hand, $\mathcal{X}$ does not have curvature ≥ 0 , since

$$\tilde{\angle}(p_{x_2}^{x_1})_{\mathbb{E}^2} = \tilde{\angle}(p_{x_3}^{x_2})_{\mathbb{E}^2} = \tilde{\angle}(p_{x_3}^{x_1})_{\mathbb{E}^2} = \pi.$$

Use this example as a warning when working with a non-intrinsic metric.

3.6. Exercise! Let p and q be points in an $\text{ALEX}(1)$ space $\mathcal{A}$. Suppose $|p - q| > \pi$, and let m be the midpoint of $[pq]$. Show that for any hinge $[m_p^x]$, we have either $\angle[m_p^x] = 0$ or $\angle[m_p^x] = \pi$.

Conclude that $\mathcal{A}$ is isometric to a line segment or a circle.

3.7. Exercise. Suppose $\mathcal{A}$ is $\text{ALEX}(1)$ and $\text{diam } \mathcal{A} \leq \pi$. Show that

$$|x - y| + |y - z| + |z - x| \leq 2 \cdot \pi$$

for any triple of points $x, y, z \in \mathcal{A}$.

C Remarks

The globalization theorem is also known as the *generalized Toponogov theorem*. Its 2-dimensional case was proved by Paolo Pizzetti [103] and rediscovered by Alexandr Alexandrov [12]. Victor Toponogov [119] proved it for Riemannian manifolds of all dimensions. For Alexandrov spaces of all dimensions, the theorem first appears in the paper by Michael Gromov, Yuriy Burago, and Grigory Perelman [20]. They prove globalization for *complete length spaces*. Another version for *noncomplete but geodesic spaces* is due to the third author [94].

We took the proof from our book [7], but present it in a less general form. This proof is based on simplifications obtained by Conrad Plaut [104] and Dmitry Burago, Yuriy Burago, and Sergei Ivanov [19]. The same proof was discovered independently by Urs Lang and Viktor Schroeder [63]. Another simplified argument was found by Katsuhiko Shiohama [114]. Alternative proofs were found by Shengqi Hu, Xiaole Su, and Yusheng Wang [51, 52].

Lecture 4

Calculus

This lecture discusses the first-order derivatives in Alexandrov spaces, including spaces of directions, tangent spaces, differentials, and gradients.

A Space of directions

Let $\mathcal{A}$ be an Alexandrov space. By 2.9, the angle measure of any hinge in $\mathcal{A}$ is defined. Given $p \in \mathcal{A}$, consider the set $\mathfrak{S}_p$ of all nontrivial geodesics starting at p . By 1.11, the triangle inequality holds for $\angle$ on $\mathfrak{S}_p$, that is, $(\mathfrak{S}_p, \angle)$ forms a semimetric space; that is, $\angle$ behaves like a metric, but might vanish for distinct geodesics.

The metric space corresponding to $(\mathfrak{S}_p, \angle)$ is called the space of geodesic directions at p ; it will be denoted by Σ'_p or $\Sigma'_p \mathcal{A}$. The elements of Σ'_p are called geodesic directions at p . Each geodesic direction is formed by an equivalence class of geodesics starting from p for the equivalence relation

$$[px] \sim [py] \quad \Leftrightarrow \quad \angle[p_y^x] = 0;$$

the direction of $[px]$ is denoted by $\uparrow_{[px]}$. By 2.11,

$$[px] \sim [py] \quad \Leftrightarrow \quad ([px] \subset [py] \quad \text{or} \quad [px] \supset [py]).$$

The completion of Σ'_p is called the space of directions at p and is denoted by Σ_p or $\Sigma_p \mathcal{A}$. The elements of Σ_p are called directions at p .

4.1. Exercise. *Let $\mathcal{A}$ be an Alexandrov space. Assume that a sequence of geodesics $[px_n]$ converges to a geodesic $[px_\infty]$ in the sense of Hausdorff, and $x_\infty \neq p$. Suppose Σ_p is compact. Show that $\uparrow_{[px_n]} \rightarrow \uparrow_{[px_\infty]}$ as $n \rightarrow \infty$.*

B Tangent space

The Euclidean cone $\mathcal{V} = \text{Cone } \mathcal{X}$ over a metric space $\mathcal{X}$ is defined as the metric space with the underlying set of equivalence classes in $[0, \infty) \times \mathcal{X}$ with the equivalence relation “ $\sim$ ” given by $(0, p) \sim (0, q)$ for any points $p, q \in \mathcal{X}$, and its metric is defined by the cosine rule:

$$|(s, p) - (t, q)|_{\mathcal{V}} = \sqrt{s^2 + t^2 - 2 \cdot s \cdot t \cdot \cos \theta},$$

where $\theta = \min\{\pi, |p - q|_{\mathcal{X}}\}$.

Note that

$$\text{Cone } \mathbb{S}^n \stackrel{\text{iso}}{=} \mathbb{E}^{n+1},$$

here “ $\stackrel{\text{iso}}{=}$ ” stands for “is isometric to”.

Next we define multiplication by scalar, scalar product, norm, and angles on Euclidean cones, extending the usual definitions from Euclidean space.

The point v in $\mathcal{V}$ that corresponds to $(t, x) \in [0, \infty) \times \mathcal{X}$ will be denoted by $t \cdot x$; the space $\mathcal{X}$ might be identified with the unit sphere in $\mathcal{V}$; so $x = 1 \cdot x$. More generally, for $v, w \in \mathcal{V}$, we will write $w = s \cdot v$ if v corresponds to $(t, x) \in [0, \infty) \times \mathcal{X}$ and w corresponds to $(s \cdot t, x) \in [0, \infty) \times \mathcal{X}$. The point in $\mathcal{V}$ formed by the equivalence class of $\{0\} \times \mathcal{X}$ is called the origin of the cone and is denoted by 0 or $0_{\mathcal{V}}$. For $v \in \mathcal{V}$ the distance $|0 - v|_{\mathcal{V}}$ is called the norm of v and is denoted by $|v|$ or $|v|_{\mathcal{V}}$. The angle measure $\angle(v, w)$ and the scalar product $\langle v, w \rangle$ of $v = s \cdot p$ and $w = t \cdot q$ are defined by

$$\angle(v, w) := \min\{\pi, |p - q|_{\mathcal{X}}\} \quad \text{and} \quad \langle v, w \rangle := |v| \cdot |w| \cdot \cos(\angle(v, w)).$$

If $v = 0$ or $w = 0$, then $\angle(v, w)$ is undefined and $\langle v, w \rangle := 0$.

4.2. Exercise. *Show that $\text{Cone } \mathcal{X}$ is geodesic if and only if $\mathcal{X}$ is π -geodesic; that is, any two points $x, y \in \mathcal{X}$ such that $|x - y|_{\mathcal{X}} < \pi$ can be joined by a geodesic in $\mathcal{X}$.*

Tangent space. The Euclidean cone $\text{Cone } \Sigma_p$ over the space of directions Σ_p is called the tangent space at p and is denoted by T_p or $T_p \mathcal{A}$. The elements of $T_p \mathcal{A}$ will be called tangent vectors at p (despite the fact that T_p is not a vector space). The space of directions Σ_p can be (and will be) identified with the unit sphere in T_p ; that is, with the set $\{v \in T_p : |v| = 1\}$.

4.3. Proposition. *Any tangent space to an Alexandrov space has nonnegative curvature in the sense of Alexandrov.*

Halbeisen’s example [7, 13.6] shows that the tangent space T_p at some point of an Alexandrov space might fail to be geodesic; in this case T_p is *not* ALEX(0).

Proof. Consider the tangent space $T_p = \text{Cone } \Sigma_p$ of an Alexandrov space $\mathcal{A}$ at a point p . We need to show that the $\mathbb{E}^2$ -comparison holds for a given quadruple $v_0, v_1, v_2, v_3 \in T_p$.

Recall that the space of geodesic directions Σ'_p is dense in Σ_p . It follows that the subcone $T'_p = \text{Cone } \Sigma'_p$ is dense in T_p . Therefore, we can assume that $v_0, v_1, v_2, v_3 \in T'_p$.

For each i , choose a geodesic γ_i from p in the direction of v_i ; reparametrize each γ_i so that it has speed $|v_i|$. Since the angles are defined, we have

$$\bullet \quad |\gamma_i(\varepsilon) - \gamma_j(\varepsilon)|_{\mathcal{A}} = \varepsilon \cdot |v_i - v_j|_{T_p} + o(\varepsilon)$$

for $\varepsilon > 0$. The quadruple $\gamma_0(\varepsilon), \gamma_1(\varepsilon), \gamma_2(\varepsilon), \gamma_3(\varepsilon)$ meets the $\mathbb{M}^2(\kappa)$ -comparison. After rescaling all the distances by $\frac{1}{\varepsilon}$, it becomes the $\mathbb{M}^2(\varepsilon^2 \cdot \kappa)$ -comparison. Passing to the limit as $\varepsilon \rightarrow 0$ and applying $\bullet$, we get the $\mathbb{E}^2$ -comparison for v_0, v_1, v_2, v_3 . $\square$

4.4. Exercise! Let p be a point in an Alexandrov space $\mathcal{A}$, and let $\lambda_n \rightarrow \infty$. Suppose Σ_p is compact. Show that there is a pointed Gromov–Hausdorff convergence $(\lambda_n \cdot \mathcal{A}, p) \rightarrow (T_p, 0)$. Moreover, the convergence can be chosen so that for any geodesic γ that starts at p , we have

$$\iota_n \circ \gamma(t/\lambda_n) \rightarrow t \cdot \xi,$$

where ι_n sends a point in $\mathcal{A}$ to the corresponding point in $\lambda_n \cdot \mathcal{A}$ and $\xi \in \Sigma_p$ is the direction of γ at 0.

C Differential

Let f be a semiconcave function defined in a neighborhood of a point p in an Alexandrov space $\mathcal{A}$ (recall that all our functions are locally Lipschitz). Choose a geodesic γ that starts at p ; let $\xi \in \Sigma_p$ be its direction. Define

$$(\mathbf{d}_p f)(\xi) := (f \circ \gamma)^+(0),$$

where $(f \circ \gamma)^+$ denotes the right derivative of $(f \circ \gamma)$; it is defined since f is semiconcave.

By the following exercise, $\mathbf{d}_p f$ is a Lipschitz function on Σ'_p . It follows that the function $\mathbf{d}_p f: \Sigma'_p \rightarrow \mathbb{R}$ can be uniquely extended to a Lipschitz function $\mathbf{d}_p f: \Sigma_p \rightarrow \mathbb{R}$. Furthermore, we can extend it to the tangent space by setting

$$(\mathbf{d}_p f)(r \cdot \xi) := r \cdot (\mathbf{d}_p f)(\xi)$$

for any $r \geq 0$ and $\xi \in \Sigma_p$. The resulting function $\mathbf{d}_p f: T_p \rightarrow \mathbb{R}$ is Lipschitz; it is called the differential of f at p .

4.5. Exercise. Let f be a semiconcave function on an Alexandrov space. Suppose γ_1 and γ_2 are geodesics that start at $p \in \text{Dom } f$; denote by θ the angle between γ_1 and γ_2 at p . Show that

$$|(f \circ \gamma_1)^+(0) - (f \circ \gamma_2)^+(0)| \leq L \cdot \theta,$$

where L is the Lipschitz constant of f in a neighborhood of p .

4.6. Exercise (First variation formula)! Let p and q be distinct points in an Alexandrov space $\mathcal{A}$.

- (a) Show that $(\mathbf{d}_p \text{dist}_q)(v) \leq -\langle \uparrow_{[pq]}, v \rangle$ for any $v \in T_p$.
- (b) Suppose $\mathcal{A}$ is proper. Let $\uparrow_p^q$ be the set of all directions of geodesics from p to q . Show that

$$(\mathbf{d}_p \text{dist}_q)(v) = -\max_{\xi \in \uparrow_p^q} \langle \xi, v \rangle$$

for any $v \in T_p$.

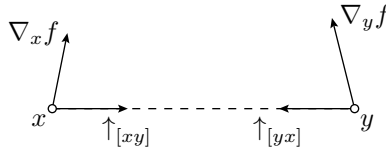
D Gradient

The following definition generalizes the notion of gradient of a smooth function to semiconcave functions on Alexandrov spaces. This generalization is not trivial even for concave functions on Euclidean space. We suggest keeping this example in mind while reading further.

4.7. Definition. Let f be a semiconcave function on an Alexandrov space and $p \in \text{Dom } f$. A tangent vector $g \in T_p$ is called a gradient of f at p (briefly, $g = \nabla_p f$) if

- (a) $(\mathbf{d}_p f)(w) \leq \langle g, w \rangle$ for any $w \in T_p$, and
- (b) $(\mathbf{d}_p f)(g) = \langle g, g \rangle$.

The following exercise provides a property of gradients that will play a key role in the proof of the first distance estimate (5.6).



4.8. Exercise! Let f be a λ -concave function on an Alexandrov space. Suppose that gradients $\nabla_x f$ and $\nabla_y f$ are defined. Show that

$$\langle \uparrow_{[xy]}, \nabla_x f \rangle + \langle \uparrow_{[yx]}, \nabla_y f \rangle + \lambda \cdot |x - y| \geq 0.$$

4.9. Proposition. *Suppose that a semiconcave function f is defined in a neighborhood of a point p in an Alexandrov space. Then the gradient $\nabla_p f$ is uniquely defined.*

Moreover, if $\mathbf{d}_p f \leq 0$, then we have $\nabla_p f = 0$; otherwise, $\nabla_p f = s \cdot \bar{\xi}$, where $s = \mathbf{d}_p f(\bar{\xi})$ and $\bar{\xi} \in \Sigma_p$ is the direction that maximizes the value $\mathbf{d}_p f(\xi)$ for $\xi \in \Sigma_p$.

4.10. Key lemma. *Let f be a semiconcave function that is defined in a neighborhood of a point p in an Alexandrov space $\mathcal{A}$. Assume*

$$s = \sup \{ (\mathbf{d}_p f)(\xi) : \xi \in \Sigma_p \} \geq 0.$$

Then for any $u, v \in T_p$, we have

$$s \cdot \sqrt{|u|^2 + 2 \cdot \langle u, v \rangle + |v|^2} \geq (\mathbf{d}_p f)(u) + (\mathbf{d}_p f)(v).$$

Suppose that $T_p \stackrel{\text{iso}}{=} \mathbb{E}^m$ and the differential $\mathbf{d}_p f$ is a concave function. Let $w = \frac{u+v}{2}$. Then

$$\textcircled{1} \quad 2 \cdot |w| = \sqrt{|u|^2 + 2 \cdot \langle u, v \rangle + |v|^2}$$

and

$$\textcircled{2} \quad 2 \cdot (\mathbf{d}_p f)(w) \geq (\mathbf{d}_p f)(u) + (\mathbf{d}_p f)(v),$$

which implies the lemma.

In general, T_p is not geodesic (and not even a length space), so concavity of $\mathbf{d}_p f$ does not make sense. The key lemma, however, says that in a certain sense $\mathbf{d}_p f$ behaves as a concave function. The proof mimics the calculations above inside $\mathcal{A}$. Instead of vectors (u, v, w) we choose a triple of points (u_t, v_t, w_t) in $\mathcal{A}$ and prove the approximate versions of $\textcircled{1}$ and $\textcircled{2}$; see $\textcircled{4}$ and $\textcircled{5}$, respectively. The main difficulty can be seen in the following exercise; technically it lies in the proof of $\textcircled{4}$ below.

4.11. Exercise! *Let p and q be distinct points in an Alexandrov space $\mathcal{A}$. Suppose the geodesic $[pq]$ can be extended beyond q .*

Show that

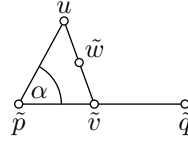
$$(\mathbf{d}_p \text{dist}_q)(v) = -\langle \uparrow_{[pq]}, v \rangle$$

for any $v \in T_p$.

Proof of 4.10. We will assume that $\mathcal{A}$ is $\text{ALEX}(0)$ and f is concave; the general case requires only minor modifications. Furthermore, we can assume that $u \neq 0$, $v \neq 0$, and $\alpha = \angle(u, v) > 0$; otherwise, the statement is trivial.

Consider a model configuration of five points: $\tilde{p}, \tilde{u}, \tilde{v}, \tilde{q}, \tilde{w} \in \mathbb{E}^2$ such that

- ◊ $\angle[\tilde{p}\tilde{u}\tilde{v}] = \alpha$,
- ◊ $|\tilde{p} - \tilde{u}| = |u|$,
- ◊ $|\tilde{p} - \tilde{v}| = |v|$,
- ◊ $\tilde{q}$ lies on an extension of $[\tilde{p}\tilde{v}]$ so that $\tilde{v}$ is the midpoint of $[\tilde{p}\tilde{q}]$,
- ◊ $\tilde{w}$ is the midpoint between $\tilde{u}$ and $\tilde{v}$.



Note that

$$\textcircled{3} \quad |\tilde{p} - \tilde{w}| = \frac{1}{2} \cdot \sqrt{|u|^2 + 2 \cdot \langle u, v \rangle + |v|^2}.$$

Since the space of geodesic directions Σ'_p is dense in Σ_p , we can assume that there are geodesics in the directions of u and v . Choose such geodesics γ_u and γ_v and assume that they are parametrized with speeds $|u|$ and $|v|$, respectively. For all small $t > 0$, consider points $u_t, v_t, q_t, w_t \in \mathcal{A}$ such that

- ◊ $v_t = \gamma_v(t)$ and $q_t = \gamma_v(2 \cdot t)$,
- ◊ $u_t = \gamma_u(t)$,
- ◊ w_t is the midpoint of $[u_t v_t]$.

Clearly

$$|p - u_t| = t \cdot |u|, \quad |p - v_t| = t \cdot |v|, \quad |p - q_t| = 2 \cdot t \cdot |v|.$$

Since $\angle(u, v)$ is defined, we have

$$|u_t - v_t| = t \cdot |\tilde{u} - \tilde{v}| + o(t), \quad |u_t - q_t| = t \cdot |\tilde{u} - \tilde{q}| + o(t).$$

From the point-on-side comparison (2.17c), we have

$$\tilde{\angle}(v_t \overset{p}{w_t}) \geq \tilde{\angle}(v_t \overset{p}{u_t}) = \angle[\tilde{v} \overset{\tilde{p}}{\tilde{u}}] + \frac{o(t)}{t}$$

and

$$\tilde{\angle}(v_t \overset{q_t}{w_t}) \geq \tilde{\angle}(v_t \overset{q_t}{u_t}) = \angle[\tilde{v} \overset{\tilde{q}}{\tilde{u}}] + \frac{o(t)}{t}.$$

Clearly, $\angle[\tilde{v} \overset{\tilde{p}}{\tilde{u}}] + \angle[\tilde{v} \overset{\tilde{q}}{\tilde{u}}] = \pi$. From the adjacent angle comparison (2.17b), $\tilde{\angle}(v_t \overset{p}{u_t}) + \tilde{\angle}(v_t \overset{q_t}{u_t}) \leq \pi$. Hence $\tilde{\angle}(v_t \overset{p}{w_t}) \rightarrow \angle[\tilde{v} \overset{\tilde{p}}{\tilde{w}}]$ as $t \rightarrow 0+$ and thus

$$\textcircled{4} \quad |p - w_t| = t \cdot |\tilde{p} - \tilde{w}| + o(t).$$

Without loss of generality, we can assume that $f(p) = 0$. Since f is concave, we have

$$\begin{aligned} \textcircled{5} \quad 2 \cdot f(w_t) &\geq f(u_t) + f(v_t) = \\ &= t \cdot [(\mathbf{d}_p f)(u) + (\mathbf{d}_p f)(v)] + o(t). \end{aligned}$$

Applying concavity of f , we have

$$\begin{aligned} (\mathbf{d}_p f)(\uparrow_{[pw_t]}) &\geq \frac{f(w_t)}{|p - w_t|} \geq \\ &\geq \frac{t \cdot [(\mathbf{d}_p f)(u) + (\mathbf{d}_p f)(v)] + o(t)}{2 \cdot t \cdot |\tilde{p} - \tilde{w}| + o(t)}. \end{aligned}$$

By ❸, the key lemma follows. $\square$

Proof of 4.9; uniqueness. If $g, g' \in T_p$ are two gradients of f , then

$$\langle g, g \rangle = (\mathbf{d}_p f)(g) \leq \langle g, g' \rangle, \quad \langle g', g' \rangle = (\mathbf{d}_p f)(g') \leq \langle g, g' \rangle.$$

Therefore,

$$|g - g'|^2 = \langle g, g \rangle - 2 \cdot \langle g, g' \rangle + \langle g', g' \rangle \leq 0.$$

It follows that $g = g'$.

Existence. If $\mathbf{d}_p f \leq 0$, then one can take $\nabla_p f = 0$.

Suppose $s = \sup \{ (\mathbf{d}_p f)(\xi) : \xi \in \Sigma_p \} > 0$. It is sufficient to show that there is $\bar{\xi} \in \Sigma_p$ such that

$$\text{❹} \quad (\mathbf{d}_p f)(\bar{\xi}) = s.$$

Indeed, suppose $\bar{\xi}$ exists. Applying 4.10 for $u = \bar{\xi}$, $v = \varepsilon \cdot w$ with $\varepsilon \rightarrow 0+$, we get

$$(\mathbf{d}_p f)(w) \leq \langle w, s \cdot \bar{\xi} \rangle$$

for any $w \in T_p$; that is, $s \cdot \bar{\xi}$ is the gradient at p .

Existence of $\bar{\xi}$ is obvious if Σ_p is compact. In the general case, take a sequence of directions $\xi_n \in \Sigma_p$, such that $(\mathbf{d}_p f)(\xi_n) \rightarrow s$. Applying 4.10 for $u = \xi_n$ and $v = \xi_m$, we get

$$s \geq \frac{(\mathbf{d}_p f)(\xi_n) + (\mathbf{d}_p f)(\xi_m)}{\sqrt{2 + 2 \cdot \cos \angle(\xi_n, \xi_m)}}.$$

Therefore $\angle(\xi_n, \xi_m) \rightarrow 0$ as $n, m \rightarrow \infty$; that is, $\xi_1, \xi_2, \dots$ is a Cauchy sequence. Clearly, $\bar{\xi} = \lim_n \xi_n$ meets ❹. $\square$

4.12. Exercise! Let f and g be locally Lipschitz semiconcave functions defined in a neighborhood of a point p in an Alexandrov space. Show that

$$|\nabla_p f - \nabla_p g|_{T_p}^2 \leq s \cdot (|\nabla_p f| + |\nabla_p g|),$$

where

$$s = \sup \{ |(\mathbf{d}_p f)(\xi) - (\mathbf{d}_p g)(\xi)| : \xi \in \Sigma_p \}.$$

Conclude that if the sequence of restrictions $\mathbf{d}_p f_n|_{\Sigma_p}$ converges uniformly, then $\nabla_p f_n$ converges as $n \rightarrow \infty$. Here we assume that all functions $f_1, f_2, \dots$ are semiconcave and Lipschitz near p .

4.13. Exercise! Let f be a locally Lipschitz λ -concave function on an Alexandrov space $\mathcal{A}$.

(a) Suppose $s \geq 0$. Show that $|\nabla_x f| > s$ if and only if for some point y we have

$$f(y) - f(x) > s \cdot \ell + \lambda \cdot \frac{\ell^2}{2},$$

where $\ell = |x - y|$.

(b) Show that $x \mapsto |\nabla_x f|$ is lower semicontinuous; that is,

$$|\nabla_{x_\infty} f| \leq \varliminf_{x_n \rightarrow x_\infty} |\nabla_{x_n} f|.$$

Lecture 5

Gradient flow

Here we define the gradient flows of semiconcave functions on Alexandrov spaces and prove related distance estimates.

A Velocity of curve

Let α be a curve in an Alexandrov space $\mathcal{A}$, and let $p = \alpha(t_0)$. If for any choice of geodesics $[p\alpha(t_0 + \varepsilon)]$ the vectors

$$\frac{1}{\varepsilon} \cdot |p - \alpha(t_0 + \varepsilon)| \cdot \uparrow_{[p\alpha(t_0 + \varepsilon)]}$$

converge as $\varepsilon \rightarrow 0+$, then their limit in T_p is called the right derivative of α at t_0 ; it will be denoted by $\alpha^+(t_0)$. In addition, we assume that $\alpha^+(t_0) := 0$ if $\frac{1}{\varepsilon} \cdot |p - \alpha(t_0 + \varepsilon)| \rightarrow 0$ as $\varepsilon \rightarrow 0+$.

The tangent vector $v = |p - x| \cdot \uparrow_{[px]}$ will be called a logarithm of x at p (briefly, $v = \log_p x$). In general the logarithm is a multivalued function from $\mathcal{A}$ to T_p ; so $v = \log_p x$ and $w = \log_p x$ do *not* imply $v = w$. Note that $\gamma^+(0) = \log_p x$ for any geodesic path γ from p to x .

5.1. Claim. *Let α be a curve in an Alexandrov space $\mathcal{A}$. Suppose f is a semiconcave Lipschitz function defined in a neighborhood of $p = \alpha(0)$, and $\alpha^+(0)$ is defined. Then $(f \circ \alpha)^+(0)$ exists and*

$$(f \circ \alpha)^+(0) = (\mathbf{d}_p f)(\alpha^+(0)).$$

Proof. Without loss of generality, we can assume that $f(p) = 0$. Suppose f and therefore $\mathbf{d}_p f$ are L -Lipschitz.

Let γ be a geodesic with a constant-speed reparametrization that starts from p , and such that the distance $s = |\alpha^+(0) - \gamma^+(0)|_{T_p}$ is

small. By the definition of differential,

$$(f \circ \gamma)^+(0) = \mathbf{d}_p f(\gamma^+(0)).$$

By comparison and the definition of α^+ ,

$$|\alpha(\varepsilon) - \gamma(\varepsilon)|_{\mathcal{A}} \leq s \cdot \varepsilon + o(\varepsilon)$$

for $\varepsilon > 0$. Therefore,

$$|f \circ \alpha(\varepsilon) - f \circ \gamma(\varepsilon)| \leq L \cdot s \cdot \varepsilon + o(\varepsilon).$$

Suppose $(f \circ \alpha)^+(0)$ is defined. Then

$$|(f \circ \alpha)^+(0) - (f \circ \gamma)^+(0)| \leq L \cdot s.$$

Since $\mathbf{d}_p f$ is L -Lipschitz, we also get

$$|\mathbf{d}_p f(\alpha^+(0)) - \mathbf{d}_p f(\gamma^+(0))| \leq L \cdot s.$$

It follows that the needed identity holds up to error $2 \cdot L \cdot s$. The statement follows since $s > 0$ can be chosen arbitrarily.

The same argument is applicable if in place of $(f \circ \alpha)^+(0)$ we use any limit of $\frac{1}{\varepsilon_n} \cdot [f \circ \alpha(\varepsilon_n) - f(p)]$ for a sequence $\varepsilon_n \rightarrow 0+$. It proves that all such limits coincide; in particular, $(f \circ \alpha)^+(0)$ is defined and equals $(\mathbf{d}_p f)(\alpha^+(0))$. $\square$

B Gradient curves

5.2. Definition. *Let f be a semiconcave function on an Alexandrov space $\mathcal{A}$.*

A locally Lipschitz curve $\alpha: [t_{\min}, t_{\max}) \rightarrow \text{Dom } f$ will be called an f -gradient curve if

$$\alpha^+(t) = \nabla_{\alpha(t)} f$$

for any $t \in [t_{\min}, t_{\max})$.

A complete proof of the following theorem is given in [7, 16.15]; it mimics the proof of the standard Picard theorem on the existence and uniqueness of solutions of ordinary differential equations. The uniqueness will follow from the first distance estimate (5.6) proved in the next section. We omit the proof of existence.

5.3. Picard theorem. *Let $f: \mathcal{A} \rightarrow \mathbb{R}$ be a locally Lipschitz and λ -concave function on an Alexandrov space $\mathcal{A}$. Then for any $p \in \text{Dom } f$, there are a unique $t_{\max} \in (0, \infty]$ and an f -gradient curve*

$\alpha: [0, t_{\max}) \rightarrow \mathcal{A}$ with $\alpha(0) = p$ such that for any sequence $t_n \rightarrow t_{\max}-$, the sequence $\alpha(t_n)$ does not have a limit point in $\text{Dom } f$.

According to the theorem, the future of a gradient curve is determined by its present, but its past might *not* be determined by the present. Indeed, consider the function $f: x \mapsto -|x|$ on $\mathbb{R}$. The tangent space $T_x \mathbb{R}$ can be identified with $\mathbb{R}$. Note that $\nabla_x f = -\text{sgn } x$; that is,

$$\nabla_x f = \begin{cases} 1 & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x > 0. \end{cases}$$

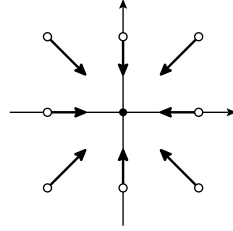
So, the f -gradient curves go to the origin with unit speed and then stand there forever. In particular, if α is an f -gradient curve that starts at x , then $\alpha(t) = 0$ for any $t \geq |x|$.

Here is a slightly more interesting example; it shows that gradient curves can merge even in the region where $|\nabla f| \neq 0$.

5.4. Example. Consider the function $f: (x, y) \mapsto -|x| - |y|$ on the (x, y) -plane. It is concave, and its gradient field is sketched on the figure.

Let α be an f -gradient curve that starts at (x, y) for $x > y > 0$. Then

$$\alpha(t) = \begin{cases} (x - t, y - t) & \text{for } 0 \leq t \leq x - y, \\ (x - t, 0) & \text{for } x - y \leq t \leq x, \\ (0, 0) & \text{for } x \leq t. \end{cases}$$



C Distance estimates

5.5. Lemma. Let α be a gradient curve of a λ -concave function f defined on the whole Alexandrov space. Choose a point p ; let $\ell(t) := \text{dist}_p \circ \alpha(t)$ and $q = \alpha(t_0)$. Assume $p \neq q$. Then

$$\ell^+(t_0) \leq - (f(p) - f(q) - \frac{\lambda}{2} \cdot \ell^2(t_0)) / \ell(t_0)$$

Proof. Let γ be the unit-speed parametrization of $[qp]$ from q to p , so

$q = \gamma(0)$. Then

$$\begin{aligned}
 \ell^+(t_0) &= (\mathbf{d}_q \text{dist}_p)(\nabla_q f) \leq && \text{(by 5.1)} \\
 &\leq -\langle \uparrow_{[qp]}, \nabla_q f \rangle \leq && \text{(by 4.6a)} \\
 &\leq -\mathbf{d}_q f(\uparrow_{[qp]}) = && \text{(by 4.7)} \\
 &= -(f \circ \gamma)^+(0) \leq \\
 &\leq -(f(p) - f(q) - \frac{\lambda}{2} \cdot \ell^2(t_0)) / \ell(t_0)
 \end{aligned}$$

The last two lines follow by the definition of differential, and the concavity of $t \mapsto f \circ \gamma(t) - \frac{\lambda}{2} \cdot t^2$. $\square$

The following estimate implies uniqueness in the Picard theorem (5.3).

5.6. First distance estimate. *Let f be a λ -concave locally Lipschitz function defined on the whole Alexandrov space $\mathcal{A}$. Then*

$$|\alpha(t) - \beta(t)| \leq e^{\lambda \cdot t} \cdot |\alpha(0) - \beta(0)|$$

for any $t \geq 0$ and any two f -gradient curves α and β .

Moreover, the statement holds for a locally Lipschitz λ -concave function defined in an open domain if there is a geodesic $[\alpha(t) \beta(t)]$ in $\text{Dom } f$ for any t .

Proof. Fix a choice of a geodesic $[\alpha(t) \beta(t)]$ for each t . Let $\ell(t) = |\alpha(t) - \beta(t)|$. Note that

$$\ell^+(t) \leq -\langle \uparrow_{[\alpha(t)\beta(t)]}, \nabla_{\alpha(t)} f \rangle - \langle \uparrow_{[\beta(t)\alpha(t)]}, \nabla_{\beta(t)} f \rangle \leq \lambda \cdot \ell(t).$$

Here one has to apply 5.5 for distance from the midpoint m of $[\alpha(t) \beta(t)]$, then apply the triangle inequality and 4.8.

Since ℓ is Lipschitz, it is differentiable almost everywhere. Integrating this inequality, we get the result. $\square$

The following exercise describes a global geometric property of a gradient curve without direct reference to its function. It is based on the notion of self-contracting curves introduced by Aris Daniilidis, Olivier Ley, and Stéphane Sabourau [26].

5.7. Exercise. *Let α be a gradient curve of a concave function on an Alexandrov space. Show that*

$$|\alpha(t_1) - \alpha(t_3)|_{\mathcal{A}} \geq |\alpha(t_2) - \alpha(t_3)|_{\mathcal{A}}$$

if $t_1 \leq t_2 \leq t_3$.

5.8. Exercise. Let f be a concave function defined on an Alexandrov space $\mathcal{A}$. Suppose $\hat{\alpha}: [0, \ell] \rightarrow \mathcal{A}$ is an arc-length reparametrization of an f -gradient curve. Show that $f \circ \hat{\alpha}$ is concave.

The following exercise implies that gradient curves for a uniformly converging sequence of λ -concave functions converge to the gradient curves of the limit function.

5.9. Exercise! Let f and g be λ -concave functions on an Alexandrov space $\mathcal{A}$. Suppose $\alpha, \beta: [0, t_{\max}) \rightarrow \mathcal{A}$ are respectively f - and g -gradient curves. Assume $|f - g| < \varepsilon$; let $\ell: t \mapsto |\alpha(t) - \beta(t)|$. Show that

$$\ell^+ \leq \lambda \cdot \ell + \frac{2 \cdot \varepsilon}{\ell}.$$

Conclude that if $\alpha(0) = \beta(0)$ and $t_{\max} < \infty$, then

$$|\alpha(t) - \beta(t)| \leq c \cdot \sqrt{\varepsilon \cdot t}$$

for some constant $c = c(t_{\max}, \lambda)$.

D Gradient flow

Let f be a semiconcave function defined on an open subset of an Alexandrov space $\mathcal{A}$. If there is an f -gradient curve α such that $\alpha(0) = x$ and $\alpha(t) = y$, then we will write

$$\text{Flow}_f^t(x) = y.$$

The partially defined map Flow_f^t from $\mathcal{A}$ to itself is called the f -gradient flow for time t . Note that

$$\text{Flow}_f^{t_1+t_2} = \text{Flow}_f^{t_1} \circ \text{Flow}_f^{t_2}.$$

In other words, the gradient flow is a partial action of the *semigroup* $([0, \infty), +)$ on $\mathcal{A}$.

From the first distance estimate 5.6, it follows that for any $t \geq 0$, the domain of definition of Flow_f^t is an open subset of $\mathcal{A}$. For sufficiently nice functions, the gradient flow is globally defined. For example, if f is a λ -concave function and it is defined on the whole space $\mathcal{A}$, then $\text{Flow}_f^t(x)$ is defined for all $x \in \mathcal{A}$ and $t \geq 0$; see [7, 16.19].

Using this new terminology, we can reformulate several statements about gradient curves. From the first distance estimate, we have the following.

5.10. Proposition. Let f be a semiconcave function defined on an Alexandrov space $\mathcal{A}$. Then the map $x \mapsto \text{Flow}_f^t(x)$ is locally Lipschitz.

Moreover, if f is λ -concave, then Flow_f^t is $e^{\lambda \cdot t}$ -Lipschitz.

The next proposition follows from 5.9.

5.11. Proposition. *Let $\mathcal{A}$ be an Alexandrov space. Suppose $f_n: \mathcal{A} \rightarrow \mathbb{R}$ is a sequence of λ -concave functions that uniformly converges to $f_\infty: \mathcal{A} \rightarrow \mathbb{R}$. Then for any $x \in \mathcal{A}$ and $t \geq 0$, we have*

$$\text{Flow}_{f_n}^t(x) \rightarrow \text{Flow}_{f_\infty}^t(x)$$

as $n \rightarrow \infty$.

This proposition can be generalized to a converging sequence $\mathcal{A}_n \rightarrow \mathcal{A}_\infty$ of spaces and a converging sequence of functions $f_n: \mathcal{A}_n \rightarrow \mathbb{R}$; see [7, 16.21].

E Gradient exponent

One of the technical difficulties in Alexandrov geometry comes from nonextendability of geodesics. In particular, the exponential map, $\exp_p: T_p \rightarrow \mathcal{A}$, if defined in the usual way, can be undefined in an arbitrarily small neighborhood of the origin.

The gradient exponential map

$$\text{gexp}_p: T_p \rightarrow \mathcal{A},$$

which we are about to introduce, essentially solves this problem. It shares many properties with the ordinary exponential map and is better in certain respects, even in the Riemannian universe.

Let p be a point in an $\text{ALEX}(0)$ space $\mathcal{A}$. Consider the function $f = \frac{1}{2} \cdot \text{dist}_p^2$. Recall that Flow_f^t denotes the gradient flow. Let us define the *gradient exponential map* as the limit

$$\text{gexp}_p(v) = \lim_{n \rightarrow \infty} \text{Flow}_f^{t_n}(x_n),$$

where the sequences $x_n \in \mathcal{A}$ and $t_n \geq 0$ are chosen so that $t_n \rightarrow \infty$ and $e^{t_n} \cdot \log_p x_n \rightarrow v$ as $n \rightarrow \infty$.

More intuitively, suppose $i_\lambda: \lambda \cdot \mathcal{A} \rightarrow \mathcal{A}$ sends a point in the rescaled copy $\lambda \cdot \mathcal{A}$ to the corresponding point in $\mathcal{A}$. By the first distance estimate (5.6), the map

❶
$$\text{Flow}_f^t \circ i_{e^t}: e^t \cdot \mathcal{A} \rightarrow \mathcal{A}$$

is short for any $t \geq 0$. If we have a pointed Gromov–Hausdorff convergence $(e^{t_n} \cdot \mathcal{A}, p) \rightarrow (T_p, o_p)$, then $\text{gexp}_p: T_p \rightarrow \mathcal{A}$ is the limit of

$\text{Flow}_f^{t_n} \circ i_{e^{t_n}}$. This way we get that gexp_p is short as a limit of short maps. This observation is generalized in the following proposition.

5.12. Proposition. *Let $\mathcal{A}$ be a proper $\text{ALEX}(0)$ space. Then for any $p \in \mathcal{A}$ the gradient exponent $\text{gexp}_p: T_p \rightarrow \mathcal{A}$ is uniquely defined. Moreover, gexp_p is a short map and*

$$\text{gexp}_p(\gamma^+(0)) = \gamma(1)$$

for any geodesic path γ that starts at p .

The last statement implies that

$$\text{gexp}_p \circ \log_p = \text{id},$$

so it is appropriate to use the term *exponent* for gexp .

Proof. Note that $f'' \leq 1$. Since the space is proper we can choose a limit in **1**.

Let γ be a geodesic that starts at p . Observe that $t \mapsto \gamma \circ \exp(t)$ is an f -gradient curve. By the first distance estimate, we have that Flow_f^t is e^t -Lipschitz. This implies that any limit in **1** has the same value; that is, gexp_p is uniquely defined.

Again, since Flow_f^t is e^t -Lipschitz, we get that gexp_p is short. $\square$

F Remarks

The idea of using gradient flows in Alexandrov geometry was inspired by the success of Sharafutdinov's retraction in comparison geometry [113]. The gradient flow was introduced by the third author to construct quasigeodesics with given initial data [86, 95, 96]. It turned out that gradient flow and gradient exponent are better tools than quasigeodesics. These tools quickly found applications in other types of singular spaces [13, 54, 76, 77, 83, 112].

For a general lower curvature bound κ , the construction of gradient exponent has to be modified; it is denoted by gexp_p^κ [7, 16.36]. It is done by taking limits of appropriately reparameterized gradient curves of the modified distance function.

For $\kappa = -1$ we have that $\text{gexp}_p^{-1}(\gamma^+(0)) = \gamma(1)$ for any geodesic path γ that starts at p and

$$|\text{gexp}_p^{-1} v - \text{gexp}_p^{-1} w|_{\mathcal{A}} \leq \tilde{\gamma}[0_w^v]_{\mathbb{H}^2}.$$

In other words gexp_p^{-1} is short if we equip T_p with the hyperbolic cone metric.

Similarly, for $\kappa = 1$ we have $\text{gexp}_p^1(\gamma^+(0)) = \gamma(1)$ for any geodesic path γ that starts at p and

$$|\text{gexp}_p^1 v - \text{gexp}_p^1 w|_{\mathcal{A}} \leq \tilde{\gamma}[0^v_w]_{\mathbb{S}^2},$$

but this time all this holds only if $|v|, |w| \leq \frac{\pi}{2}$ and $\text{length } \gamma \leq \frac{\pi}{2}$.

The gradient exponential map in a Riemannian manifold (M, g) coincides with the Riemannian exponential map before the cut locus, but *is different* from the Riemannian exponential after that. Moreover, beyond the cut locus gexp_p^κ depends on κ while the Riemannian exponential map does not. The following exercise is meant to show that this technique can prove something nontrivial even for Riemannian manifolds.

5.13. Exercise. *Let (M, g) be a complete m -dimensional Riemannian manifold with sectional curvature at least 1. Assume M is not homeomorphic to $\mathbb{S}^m$. Show that there is a short onto map $\mathbb{S}^m \rightarrow (M, g)$.*

Lecture 6

Line splitting

In this lecture, we prove the line splitting theorem and apply it in tangent spaces of Alexandrov spaces.

A Busemann function

A half-line is a geodesic defined on the real half-line $[0, \infty)$; that is, a distance-preserving map from $[0, \infty)$ to a metric space.

If $\gamma: [0, \infty) \rightarrow \mathcal{X}$ is a half-line, then the limit

$$\bullet \quad \text{bus}_\gamma(x) = \lim_{t \rightarrow \infty} |\gamma(t) - x| - t$$

is called the Busemann function of γ . It behaves like a shifted distance function from the ideal point of γ .

6.1. Proposition. *For any half-line γ in a metric space $\mathcal{X}$, its Busemann function $\text{bus}_\gamma: \mathcal{X} \rightarrow \mathbb{R}$ is defined. Moreover, bus_γ is 1-Lipschitz and $\text{bus}_\gamma \circ \gamma(t) = -t$ for any t .*

Proof. Since $t = |\gamma(0) - \gamma(t)|$, the triangle inequality implies that

$$t \mapsto |\gamma(t) - x| - t$$

is a nonincreasing function, and

$$|\gamma(t) - x| - t \geq -|\gamma(0) - x|$$

for any $x \in \mathcal{X}$. Therefore, the limit in $\bullet$ is defined, and it has to be 1-Lipschitz as a limit of 1-Lipschitz functions.

The last statement follows since $|\gamma(t) - \gamma(t_0)| = t - t_0$ for $t > t_0$. $\square$

6.2. Exercise! *Show that any Busemann function on an ALEX(0) space is concave.*

B Splitting theorem

A line is a distance-preserving map from $\mathbb{R}$ to a metric space. In other words, a line is a geodesic defined on the whole real line $\mathbb{R}$.

6.3. Exercise! *Let γ be a line in a metric space $\mathcal{X}$. Show that for any point x we have*

$$\text{bus}_+(x) + \text{bus}_-(x) \geq 0,$$

where bus_+ and bus_- are the Busemann functions associated with half-lines $\gamma : [0, \infty) \rightarrow \mathcal{X}$ and $\gamma : (-\infty, 0] \rightarrow \mathcal{X}$ respectively.

Let A and B be two subsets of a metric space $\mathcal{X}$. We say that $\mathcal{X}$ is a direct sum of A and B , or briefly,

$$\mathcal{X} = A \oplus B$$

if there are retractions $\text{proj}_A : \mathcal{X} \rightarrow A$ and $\text{proj}_B : \mathcal{X} \rightarrow B$ such that

$$|x - y|^2 = |\text{proj}_A(x) - \text{proj}_A(y)|^2 + |\text{proj}_B(x) - \text{proj}_B(y)|^2$$

for any two points $x, y \in \mathcal{X}$.

Note that if $\mathcal{X} = A \oplus B$, then $\mathcal{X}$ is isometric to $A \times B$.

6.4. Exercise. *Suppose $\mathcal{X} = A \oplus B$. Show that*

- (a) *A intersects B at a single point, and*
- (b) *both sets A and B are convex in $\mathcal{X}$, meaning that any geodesic with the endpoints in A (or B) lies in A (respectively, B).*

6.5. Line splitting theorem. *If γ is a line in an $\text{ALEX}(0)$ space $\mathcal{A}$, then*

$$\mathcal{A} = \mathcal{A}' \oplus \gamma(\mathbb{R})$$

for some subset $\mathcal{A}' \subset \mathcal{A}$.

6.6. Corollary. *Any $\text{ALEX}(0)$ space $\mathcal{A}$ splits isometrically as*

$$\mathcal{A} = \mathcal{A}' \oplus H$$

where $H \subset \mathcal{A}$ is a subset isometric to a Hilbert space, and $\mathcal{A}' \subset \mathcal{A}$ is a convex subset that contains no lines.

The following lemma is closely related to the first distance estimate (5.6); it is also a limit case of 5.12. The proof is similar.

6.7. Lemma. *Suppose $f: \mathcal{A} \rightarrow \mathbb{R}$ is a concave 1-Lipschitz function on an ALEX(0) space $\mathcal{A}$. Consider two f -gradient curves α and β . Then for any $t, s \geq 0$ we have*

$$|\alpha(s) - \beta(t)|^2 \leq |p - q|^2 + 2 \cdot (f(p) - f(q)) \cdot (s - t) + (s - t)^2,$$

where $p = \alpha(0)$ and $q = \beta(0)$.

Proof. Since f is 1-Lipschitz it follows that $|\nabla f| \leq 1$. Therefore

$$f \circ \beta(t) \leq f(q) + t$$

for any $t \geq 0$.

Set $\ell(t) = |p - \beta(t)|$. Applying 5.5, we get

$$\begin{aligned} (\ell^2)^+(t) &\leq 2 \cdot (f \circ \beta(t) - f(p)) \leq \\ &\leq 2 \cdot (f(q) + t - f(p)). \end{aligned}$$

Therefore

$$\ell^2(t) - \ell^2(0) \leq 2 \cdot (f(q) - f(p)) \cdot t + t^2.$$

This proves the needed inequality in the case $s = 0$. Combining it with the first distance estimate (5.6), we get the result in the case $s \leq t$. The case $s \geq t$ follows by switching the roles of s and t . $\square$

Proof of 6.5. Consider two Busemann functions, bus_+ and bus_- , associated with the half-lines $\gamma: [0, \infty) \rightarrow \mathcal{A}$ and $\gamma: (-\infty, 0] \rightarrow \mathcal{A}$ respectively; that is,

$$\text{bus}_\pm(x) := \lim_{t \rightarrow \infty} |\gamma(\pm t) - x| - t.$$

According to 6.2, both bus_+ and bus_- are concave.

By 2.22, $\varphi: t \mapsto \text{dist}_x^2 \circ \gamma(t)$ is 2-concave. In particular, $\varphi(t) \leq t^2 + at + b$ for some constants $a, b \in \mathbb{R}$; see 2E. Therefore,

$$|\gamma(t) - x| + |\gamma(-t) - x| - 2 \cdot t \leq \sqrt{t^2 + at + b} + \sqrt{t^2 - at + b} - 2 \cdot t$$

for all large t . Passing to the limit as $t \rightarrow \infty$, we get that $\text{bus}_+(x) + \text{bus}_-(x) \leq 0$. By 6.3, $\text{bus}_+(x) + \text{bus}_-(x) \geq 0$; hence

$$\text{bus}_+(x) + \text{bus}_-(x) = 0$$

for any $x \in \mathcal{A}$. In particular, the functions bus_+ and bus_- are affine; that is, they are convex and concave at the same time.

For any x ,

$$\begin{aligned} |\nabla_x \text{bus}_\pm| &= \sup \{ \mathbf{d}_x \text{bus}_\pm(\xi) : \xi \in \Sigma_x \} = \\ &= \sup \{ -\mathbf{d}_x \text{bus}_\mp(\xi) : \xi \in \Sigma_x \} \equiv \\ &\equiv 1. \end{aligned}$$

A curve α is a $\text{bus}_\pm$ -gradient curve if and only if α is a geodesic such that $(\text{bus}_\pm \circ \alpha)^+ = 1$. Indeed, if α is a geodesic, then $(\text{bus}_\pm \circ \alpha)^+ \leq 1$ and the equality holds only if $\nabla_\alpha \text{bus}_\pm = \alpha^+$. Now suppose $\nabla_\alpha \text{bus}_\pm = \alpha^+$. Then $|\alpha^+| \leq 1$ and $(\text{bus}_\pm \circ \alpha)^+ = 1$; therefore

$$\begin{aligned} |t_0 - t_1| &\geq |\alpha(t_0) - \alpha(t_1)| \geq \\ &\geq |\text{bus}_\pm \circ \alpha(t_0) - \text{bus}_\pm \circ \alpha(t_1)| = \\ &= |t_0 - t_1|. \end{aligned}$$

It follows that for any $t > 0$, the $\text{bus}_\pm$ -gradient flows commute; that is,

$$\text{Flow}_{\text{bus}_+}^t \circ \text{Flow}_{\text{bus}_-}^t = \text{id}_{\mathcal{A}}.$$

Setting

$$\text{Flow}^t := \begin{cases} \text{Flow}_{\text{bus}_-}^t & \text{if } t \geq 0 \\ \text{Flow}_{\text{bus}_+}^{-t} & \text{if } t \leq 0 \end{cases}$$

defines an $\mathbb{R}$ -action on $\mathcal{A}$.

Consider the level set $\mathcal{A}' = \text{bus}_+^{-1}\{0\} = \text{bus}_-^{-1}\{0\}$; it is a closed convex subset of $\mathcal{A}$, and therefore forms an Alexandrov space. Let $h: \mathcal{A}' \times \mathbb{R} \rightarrow \mathcal{A}$ be the map defined by $h: (x, t) \mapsto \text{Flow}^t(x)$. Note that $\text{Flow}^{-t}(y) \in \mathcal{A}'$ for $t = \text{bus}_-(y) = -\text{bus}_+(y)$; hence h is onto.

Applying 6.7 for $\text{Flow}_{\text{bus}_+}^t$ and $\text{Flow}_{\text{bus}_-}^t$ shows that h is distance nonexpanding and noncontracting at the same time; that is, h is an isometry. $\square$

6.8. Exercise. Suppose $\mathcal{X}$ is a complete geodesic space. Show that $\text{Cone } \mathcal{X}$ is $\text{ALEX}(0)$ if and only if $\mathcal{X}$ is $\text{ALEX}(1)$ and $\text{diam } \mathcal{X} \leq \pi$.

Recall that according to our definition any circle or closed real interval is $\text{ALEX}(1)$. Therefore, the condition $\text{diam } \mathcal{X} \leq \pi$ is necessary. Nevertheless, according to 3.6, most $\text{ALEX}(1)$ spaces have diameter at most π .

C Anti-sum

The following lemma is a corollary of 4.12. It will be used to prove basic properties of tangent spaces.

6.9. Anti-sum lemma. Let p be a point in an Alexandrov space $\mathcal{A}$. Given two vectors $u, v \in T_p$, there is a unique vector $w \in T_p$ such that

$$\langle u, x \rangle + \langle v, x \rangle + \langle w, x \rangle \geq 0$$

for any $x \in T_p$, and

$$\langle u, w \rangle + \langle v, w \rangle + \langle w, w \rangle = 0.$$

6.10. Exercise! Suppose $u, v, w \in T_p$ are as in 6.9. Show that

$$|w|^2 \leq |u|^2 + |v|^2 + 2 \cdot \langle u, v \rangle.$$

If the tangent space T_p was $\text{ALEX}(0)$, then the lemma would follow from the existence of the gradient, applied to the function $T_p \rightarrow \mathbb{R}$ defined by $x \mapsto -(\langle u, x \rangle + \langle v, x \rangle)$, which is concave by 6.2. This idea cannot work as is since T_p might fail to be geodesic (see Halbeisen's example [7, 13.6]), but it works after a proper revision.

Applying the above lemma for $u = v$, we have the following statement.

6.11. Existence of polar vector. Let $\mathcal{A}$ be an Alexandrov space and $p \in \mathcal{A}$. Given a vector $u \in T_p$, there is a unique vector $u^* \in T_p$ such that $\langle u^*, u^* \rangle + \langle u, u^* \rangle = 0$ and u^* is polar to u ; that is,

$$\langle u^*, x \rangle + \langle u, x \rangle \geq 0$$

for any $x \in T_p$.

Proof of 6.9. By 4.11, we can choose two sequences of points a_n, b_n such that

$$(\mathbf{d}_p \text{dist}_{a_n})(x) = -\langle \uparrow_{[pa_n]}, x \rangle \quad \text{and} \quad (\mathbf{d}_p \text{dist}_{b_n})(x) = -\langle \uparrow_{[pb_n]}, x \rangle$$

for any $x \in T_p$. Furthermore, we can assume that $\uparrow_{[pa_n]} \rightarrow u/|u|$ and $\uparrow_{[pb_n]} \rightarrow v/|v|$ as $n \rightarrow \infty$.

Consider a sequence of functions

$$f_n = |u| \cdot \text{dist}_{a_n} + |v| \cdot \text{dist}_{b_n}.$$

Note that

$$(\mathbf{d}_p f_n)(x) = -|u| \cdot \langle \uparrow_{[pa_n]}, x \rangle - |v| \cdot \langle \uparrow_{[pb_n]}, x \rangle$$

for any $x \in T_p$. Thus, $\mathbf{d}_p f_n$ uniformly converges to $x \mapsto -\langle u, x \rangle - \langle v, x \rangle$ on Σ_p . According to 4.12, the sequence $\nabla_p f_n$ converges as $n \rightarrow \infty$. Let

$$w = \lim_{n \rightarrow \infty} \nabla_p f_n.$$

By the definition of gradient,

$$\begin{aligned} \langle w, w \rangle &= \lim_{n \rightarrow \infty} \langle \nabla_p f_n, \nabla_p f_n \rangle = & \langle w, x \rangle &= \lim_{n \rightarrow \infty} \langle \nabla_p f_n, x \rangle \geq \\ &= \lim_{n \rightarrow \infty} (\mathbf{d}_p f_n)(\nabla_p f_n) = & &\geq \lim_{n \rightarrow \infty} (\mathbf{d}_p f_n)(x) = \\ &= -\langle u, w \rangle - \langle v, w \rangle, & &= -\langle u, x \rangle - \langle v, x \rangle. \end{aligned}$$

The proof of uniqueness is left to the reader; it is very similar to the proof of uniqueness of gradients (page 41). $\square$

D Linear subspace

6.12. Definition. Let $\mathcal{A}$ be an Alexandrov space, $p \in \mathcal{A}$ and $u, v \in T_p$. We say that vectors u and v are opposite to each other (briefly, $u + v = 0$) if

- ◇ $|u| = |v| = 0$ or
- ◇ $\angle(u, v) = \pi$ and $|u| = |v|$.

The subcone

$$\text{Lin}_p = \{ v \in T_p : \exists w \in T_p \text{ such that } w + v = 0 \}$$

will be called the linear subspace of T_p .

Soon we will introduce a natural linear structure on Lin_p , which will justify its name.

6.13. Proposition. Let p be a point in an Alexandrov space. Given two vectors $u, v \in T_p$, the following statements are equivalent:

- (a) $u + v = 0$;
- (b) $\langle u, x \rangle + \langle v, x \rangle = 0$ for any $x \in T_p$;
- (c) $\langle u, \xi \rangle + \langle v, \xi \rangle = 0$ for any $\xi \in \Sigma_p$.

Proof. The equivalence $(b) \Leftrightarrow (c)$ is trivial.

The condition $u + v = 0$ is equivalent to $\langle u, u \rangle = -\langle u, v \rangle = \langle v, v \rangle$; thus, $(b) \Rightarrow (a)$. It remains to prove $(a) \Rightarrow (b)$.

Suppose (a) holds. We can assume that $|u| = |v| \neq 0$ since otherwise (b) is trivial. Since $\angle(u, v) = \pi$ and T_p is a metric cone, the half-lines in directions of u and v form a line in T_p . By 4.3, T_p has nonnegative curvature, and by 2.13 we have $\angle[0_x^u] + \angle[0_x^v] = \pi$, which implies (b). $\square$

6.14. Exercise! Let u, v , and w be tangent vectors at a point of an Alexandrov space. Assume $u + v = 0$ and $u + w = 0$. Show that $v = w$.

Let $u \in \text{Lin}_p$; that is, $u + v = 0$ for some $v \in T_p$. Given $s < 0$, let

$$s \cdot u := (-s) \cdot v.$$

So we can multiply any vector in Lin_p by any real number (positive and negative). By 6.14, this multiplication is uniquely defined. By 6.13, we have the identity

$$\langle -v, x \rangle = -\langle v, x \rangle.$$

6.15. Exercise! Suppose u, v , and w are as in 6.9. Show that

$$\langle u, x \rangle + \langle v, x \rangle + \langle w, x \rangle = 0$$

for any $x \in \text{Lin}_p$.

6.16. Exercise! Let $\mathcal{A}$ be an Alexandrov space, $p \in \mathcal{A}$ and $u \in T_p$. Suppose $u^* \in T_p$ is as in 6.11; that is,

$$\langle u^*, u^* \rangle + \langle u, u^* \rangle = 0 \quad \text{and} \quad \langle u^*, x \rangle + \langle u, x \rangle \geq 0$$

for any $x \in T_p$. Show that $u = -u^*$ if and only if $|u| = |u^*|$.

6.17. Theorem. Let p be a point in an Alexandrov space. Then Lin_p is isometric to a Hilbert space.

Proof. Lin_p is a closed subset of T_p ; in particular, it is complete.

If we know that any two vectors in Lin_p can be connected by a geodesic in Lin_p , then the statement follows from the splitting theorem (6.5). By Menger's lemma (1.6), it is sufficient to show that any pair $x, y \in \text{Lin}_p$ has a midpoint $w \in \text{Lin}_p$.

Choose $w \in T_p$ to be the anti-sum of $u = -\frac{1}{2} \cdot x$ and $v = -\frac{1}{2} \cdot y$; see 6.9. By 6.10 and 6.15,

$$\begin{aligned} |w|^2 &\leq \frac{1}{4} \cdot |x|^2 + \frac{1}{4} \cdot |y|^2 + \frac{1}{2} \cdot \langle x, y \rangle, \\ \langle w, x \rangle &= \frac{1}{2} \cdot |x|^2 + \frac{1}{2} \cdot \langle x, y \rangle, \\ \langle w, y \rangle &= \frac{1}{2} \cdot |y|^2 + \frac{1}{2} \cdot \langle x, y \rangle. \end{aligned}$$

It follows that

$$\begin{aligned} |x - w|^2 &= |x|^2 + |w|^2 - 2 \cdot \langle w, x \rangle \leq \\ &\leq \frac{1}{4} \cdot |x|^2 + \frac{1}{4} \cdot |y|^2 - \frac{1}{2} \cdot \langle x, y \rangle = \\ &= \frac{1}{4} \cdot |x - y|^2. \end{aligned}$$

That is, $|x - w| \leq \frac{1}{2} \cdot |x - y|$. Similarly, we get $|y - w| \leq \frac{1}{2} \cdot |x - y|$. Therefore w is a midpoint of x and y . In addition, we get the equality

$$|w|^2 = \frac{1}{4} \cdot |x|^2 + \frac{1}{4} \cdot |y|^2 + \frac{1}{2} \cdot \langle x, y \rangle.$$

It remains to show that $w \in \text{Lin}_p$. Let w^* be the polar vector provided by 6.11. Note that

$$|w^*| \leq |w|, \quad \langle w^*, x \rangle + \langle w, x \rangle = 0, \quad \text{and} \quad \langle w^*, y \rangle + \langle w, y \rangle = 0.$$

The same calculation as above shows that w^* is a midpoint of $-x$ and $-y$ and

$$|w^*|^2 = \frac{1}{4} \cdot |x|^2 + \frac{1}{4} \cdot |y|^2 + \frac{1}{2} \cdot \langle x, y \rangle = |w|^2.$$

By 6.16, $w = -w^*$; hence $w \in \text{Lin}_p$. □

6.18. Lemma. *Given a point p in an Alexandrov space $\mathcal{A}$, let $f = \text{dist}_p$, and let S be the subset of points $x \in \mathcal{A}$ such that $|\nabla_x f| = 1$. Then S is a dense G -delta set; that is, a countable intersection of open sets.*

Proof. Given a positive integer n , consider the set

$$S_n = \left\{ x \in \mathcal{A} : |\nabla_x f| > 1 - \frac{1}{n} \right\}.$$

By 4.13a, S_n is open.

Choose a point $q \neq p$. Observe that $|\nabla_x f| = 1$ for any point $x \in]pq[$. It follows that S_n is dense in $\mathcal{A}$. Since $S = \bigcap_n S_n$, the lemma follows from the Baire category theorem. $\square$

6.19. Exercise! *Let p , f , and S be as in 6.18.*

(a) *Show that*

$$\nabla_x f + \uparrow_{[xp]} = 0$$

for any $x \in S$; in particular, $\uparrow_{[xp]} \in \text{Lin}_x$.

(b) *Show that if $|\nabla_x f| = 1$, then $(\mathbf{d}_x f)(w) = \langle \nabla_x f, w \rangle$ for every $w \in T_x$.*

(c) *Show that for any $x \in S$ there is a unique geodesic $[px]$.*

This exercise implies the following.

6.20. Corollary. *Given a countable set of points X in an Alexandrov space $\mathcal{A}$, there is a G -delta dense set $S \subset \mathcal{A}$ such that $\uparrow_{[sx]} \in \text{Lin}_s$ for any $s \in S$ and $x \in X$.*

E Remarks

The history of the splitting theorem starts with Stefan Cohn-Vossen [25], who proved it in the 2-dimensional case. For Riemannian manifolds of higher dimensions, it was proved by Victor Toponogov [119]. Then it was generalized by Anatoliy Milka [79] to Alexandrov spaces; historically, it was the first result about Alexandrov spaces of dimension higher than 2. Nearly the same proof is used in [19, 1.5].

Generalizations to Riemannian manifolds with nonnegative Ricci curvature were obtained by Jeff Cheeger and Detlef Gromoll [24]. This was further generalized by Jeff Cheeger and Toby Colding for the limits of Riemannian manifolds with almost nonnegative Ricci curvature [22]. Nicola Gigli generalized it further to the so-called RCD spaces (spaces with synthetically defined Ricci-curvature bound) [34, 35]. Jost-Hinrich Eschenburg obtained an analogous result for Lorentzian manifolds [31].

The presented proof is close in spirit to the proof given by Cheeger and Gromoll [24]; it is taken from our book [7].

6.21. Open question. *Let p be a point in an Alexandrov space $\mathcal{A}$. Suppose that $0 \neq v \in \text{Lin}_p$. Is it true that the tangent space T_p splits in the direction of v ?*

6.22. Open question. *Let $\mathcal{A}$ be a proper Alexandrov space. Is it true that for any $p \in \mathcal{A}$, the tangent space T_p is a length space?*

Lecture 7

Dimension and volume

A Linear dimension

Let $\mathcal{A}$ be an Alexandrov space. We define its linear dimension $\text{LinDim } \mathcal{A}$ as the least upper bound of integers m such that the Euclidean space $\mathbb{E}^m$ is isometric to a subspace of the tangent space $T_p \mathcal{A}$ at *some* point $p \in \mathcal{A}$.

If not stated otherwise, dimension means linear dimension. In 7F and 7G, we will show that linear dimension of Alexandrov spaces coincides with all reasonable notions of dimension; thereafter, we will use the notation $\dim \mathcal{A}$ for $\text{LinDim } \mathcal{A}$.

7.1. ($n+1$)-comparison. *Let $\mathcal{A}$ be an $\text{ALEX}(0)$ space. Then for any finite set of points $p, x_1, \dots, x_n \in \mathcal{A}$, there exists a model configuration $\tilde{p}, \tilde{x}_1, \dots, \tilde{x}_n \in \mathbb{E}^m$ for some m such that*

$$|\tilde{p} - \tilde{x}_i|_{\mathbb{E}^m} = |p - x_i|_{\mathcal{A}} \quad \text{and} \quad |\tilde{x}_i - \tilde{x}_j|_{\mathbb{E}^m} \geq |x_i - x_j|_{\mathcal{A}}$$

for any i and j . Moreover, we can assume that $m \leq \text{LinDim } \mathcal{A}$.

If $\mathcal{A}$ is an $\text{ALEX}(\pm 1)$ space, then the same holds if we replace $\mathbb{E}^m$ by $\mathbb{S}^m$ or $\mathbb{H}^m$, respectively; in the $\text{ALEX}(1)$ case one has to assume in addition that $\text{diam } \mathcal{A} \leq \pi$.

Proof. By 6.20, we can choose a point p' arbitrarily close to p so that $\text{Lin}_{p'} \ni \uparrow_{[p'x_i]}$ for any i . Let us identify $\mathbb{E}^m$ with a subspace of $\text{Lin}_{p'}$ spanned by $\uparrow_{[p'x_1]}, \dots, \uparrow_{[p'x_n]}$. Note that $m \leq \text{LinDim } \mathcal{A}$.

Set $\tilde{p}' = 0 \in \mathbb{E}^m$ and $\tilde{x}_i = |p' - x_i| \cdot \uparrow_{[p'x_i]} \in \mathbb{E}^m$ for every i . Note that

$$|\tilde{p}' - \tilde{x}_i|_{\mathbb{E}^m} = |p' - x_i|_{\mathcal{A}}$$

for every i . Applying the comparison $\angle[p'_{x_j}] \geq \tilde{\angle}(p'_{x_j})$, we get

$$|\tilde{x}_i - \tilde{x}_j|_{\mathbb{E}^m} \geq |x_i - x_j|_{\mathcal{A}}$$

for all i and j . Passing to a limit configuration as $p' \rightarrow p$ we get the result.

If $\mathcal{A}$ is an ALEX(-1) space, then the same argument gives $\tilde{p}, \tilde{x}_1, \dots, \tilde{x}_n \in \mathbb{E}^m$ such that

$$|\tilde{p} - \tilde{x}_i|_{\mathbb{E}^m} = |p - x_i|_{\mathcal{A}} \quad \text{and} \quad \tilde{\Upsilon}[\tilde{p} \tilde{x}_i]_{\mathbb{H}^2} \geq |x_i - x_j|_{\mathcal{A}}.$$

It remains to observe that $\mathbb{E}^m$ with the metric defined by $|\tilde{v} - \tilde{w}| := \tilde{\Upsilon}[\tilde{p} \tilde{v}]_{\mathbb{H}^2}$ is isometric to $\mathbb{H}^m$.

The ALEX(1) case is analogous, but one has to pull x_i toward p to ensure that $|p - x_i| < \pi$ for each i , and then pass to an additional limit. $\square$

7.2. Exercise! Let $\mathcal{A}$ be an ALEX(0) space. Assume that $\mathbb{E}^m$ is isometric to a subspace of the tangent space $T_p \mathcal{A}$ at some point $p \in \mathcal{A}$ and $\text{LinDim } \mathcal{A} = m < \infty$. Show that $T_p \mathcal{A} \stackrel{\text{iso}}{=} \mathbb{E}^m$.

7.3. Exercise! Show that a 1-dimensional Alexandrov space is homeomorphic to a 1-dimensional manifold, possibly with nonempty boundary.

7.4. Exercise. Let $\mathcal{A}$ be an ALEX(0) space. Show that $\text{LinDim } \mathcal{A} \geq m$ if and only if there are $m+2$ points $p, a_0, \dots, a_m \in \mathcal{A}$ such that $\tilde{\angle}(p_{a_j}^{a_i}) > \frac{\pi}{2}$ for any pair $i \neq j$.

B Space of directions

Recall that a metric space $\mathcal{X}$ is ℓ -geodesic if any two points $x, y \in \mathcal{X}$ such that $|x - y| < \ell$ can be connected by a geodesic.

7.5. Theorem. For any point p in a finite-dimensional Alexandrov space, the space of directions Σ_p is compact and π -geodesic.

By 4.4, this theorem implies the following.

7.6. Corollary. Let p be a point in a finite-dimensional Alexandrov space $\mathcal{A}$, and let $\lambda_n \rightarrow \infty$. Then there is a pointed Gromov-Hausdorff convergence $(\lambda_n \cdot \mathcal{A}, p) \rightarrow (T_p, 0)$.

Proof of 7.5. Choose $\varepsilon > 0$; suppose $\mathcal{A}$ is m -dimensional. Assume $\text{pack}_\varepsilon \Sigma_p \geq n$; that is, we can choose n directions $\xi_1, \dots, \xi_n \in \Sigma_p$ such

that $\angle(\xi_i, \xi_j) > \varepsilon$ for any $i \neq j$. Without loss of generality, we may assume that each direction is geodesic; that is, there is a point $x_i \in \mathcal{A}$ such that $\xi_i = \uparrow_{[px_i]}$.

Choose small $r > 0$ and $y_i \in [px_i]$ such that $|p - y_i| = r$ for each i . Since r is small, we can assume that $\angle(p_{y_j}^{y_i}) > \varepsilon$ for $i \neq j$. By 6.20, we can choose p' arbitrarily close to p such that $\uparrow_{[p'y_i]} \in \text{Lin}_{p'}$ for any i ; in particular, all directions $\uparrow_{[p'y_i]}$ lie in a unit sphere of dimension at most $m-1$. Since $|p' - p|$ is small, $\tilde{\angle}(p_{y_j}^{y_i}) > \varepsilon$ if $i \neq j$. By comparison,

$$\angle[p_{y_j}^{y_i}] > \varepsilon.$$

Therefore, $\text{pack}_\varepsilon \Sigma_p \leq \text{pack}_\varepsilon \mathbb{S}^{m-1}$.

Since $\mathbb{S}^{m-1}$ is compact, $\text{pack}_\varepsilon \mathbb{S}^{m-1} < \infty$ for any $\varepsilon > 0$. By the definition, the space of directions Σ_p is complete. Applying 1.2, we get that Σ_p is compact.

It remains to prove the following claim.

❶ *If Σ_p is compact, then it is π -geodesic.*

Choose two geodesic directions $\xi = \uparrow_{[px]}$ and $\zeta = \uparrow_{[py]}$; let

$$\alpha = \frac{1}{2} \cdot \angle[p_y^x] = \frac{1}{2} \cdot |\xi - \zeta|_{\Sigma_p}.$$

Assume $\alpha < \frac{\pi}{2}$.

It is sufficient to construct an almost midpoint $\mu = \uparrow_{[pz]}$ of ξ and ζ in Σ_p ; that is, we need to show that for any $\varepsilon > 0$ there is a geodesic $[pz]$ such that

$$\angle[p_z^x] \leq \alpha + \varepsilon \quad \text{and} \quad \angle[p_z^y] \leq \alpha + \varepsilon.$$

Indeed, once this is done, the compactness of Σ_p can be used to get an actual midpoint for any two directions in Σ_p , and Menger's lemma (1.6) will finish the proof.

Choose a sequence of small positive numbers $r_n \rightarrow 0$. Consider sequences $x_n \in [px]$ and $y_n \in [py]$ such that $|p - x_n| = |p - y_n| = r_n$. Let m_n be a midpoint of $[x_n y_n]$. Since $\alpha < \frac{\pi}{2}$, $m_n \neq p$.

Since Σ_p is compact, we can pass to a subsequence of r_n such that $\uparrow_{[pm_n]}$ converges; denote the limit by μ . Choose a geodesic $[pz]$ that runs at a small angle to μ . It remains to show that $\uparrow_{[pz]}$ is the needed almost midpoint.

Evidently, $\tilde{\angle}(p_{m_n}^{x_n}) = \tilde{\angle}(p_{m_n}^{y_n})$. By 2.7, we have

$$\tilde{\angle}(p_{m_n}^{x_n}) + \tilde{\angle}(p_{m_n}^{y_n}) \leq \tilde{\angle}(p_{y_n}^{x_n}).$$

For each large n , choose a point $z_n \in [pz]$ such that $|p - z_n| = |p - m_n|$. By construction, for all large n , we have $\angle[p_{m_n}^z] \approx 0$ with

arbitrarily small given error. By comparison, the value $\frac{|z_n - m_n|}{|p - z_n|}$ can be assumed to be arbitrarily small for all large n . Applying this observation and the definition of angle measure, we also get that the following approximations

$$\begin{aligned}\tilde{\angle}(p_{y_n}^{x_n}) &\approx \angle[p_{y_n}^{x_n}], \\ \tilde{\angle}(p_{m_n}^{x_n}) &\approx \tilde{\angle}(p_{z_n}^{x_n}) \approx \angle[p_{z_n}^{x_n}], \\ \tilde{\angle}(p_{y_n}^{m_n}) &\approx \tilde{\angle}(p_{y_n}^{z_n}) \approx \angle[p_{y_n}^{z_n}],\end{aligned}$$

hold with arbitrary given error and all large n . It follows that $\uparrow_{[pz]}$ is an almost midpoint of $\uparrow_{[px]}$ and $\uparrow_{[py]}$, as required. $\square$

Remark. In the above proof, the angles $\angle[p_z^x]$ and $\angle[p_z^y]$ have lower bounds by the comparison, but we needed upper bounds. These were extracted from the definition of angle measure and the compactness of the space of directions. Halbeisen's example [7, 13.6] shows that it cannot be done without the compactness condition.

7.7. Exercise! Let f be a semiconcave function defined in an open set $\text{Dom } f$ in a finite-dimensional Alexandrov space. Show that the differential $d_p f$ is a concave function $T_p \rightarrow \mathbb{R}$ for any $p \in \text{Dom } f$.

C Right-inverse theorem

7.8. Right-inverse theorem. Suppose $p, a_0, \dots, a_m$ are points in an Alexandrov space $\mathcal{A}$ such that

$$\tilde{\angle}(p_{a_j}^{a_i}) > \frac{\pi}{2}$$

for any $i \neq j$. Then the map $f: \mathcal{A} \rightarrow \mathbb{R}^m$ defined by

$$f: x \mapsto (|a_1 - x|, \dots, |a_m - x|)$$

has a continuous right inverse Φ defined in a neighborhood of $f(p)$ such that $p = \Phi \circ f(p)$.

In the proof we construct a local right inverse Φ of f around $f(p)$. The construction uses gradient flow for a suitably chosen family of functions. The proof can be extracted from the following exercise; more details are given in the hints.

7.9. Exercise! Suppose $p, a_0, \dots, a_m \in \mathcal{A}$ and $f: \mathcal{A} \rightarrow \mathbb{R}^m$ are as in 7.8. Assume $\varepsilon > 0$ is sufficiently small. Given $\mathbf{y} = (y_1, \dots, y_m) \in \mathbb{R}^m$, consider the function on $\mathcal{A}$ defined by

$$h_{\mathbf{y}}(x) := \min\{0, |a_1 - x| - y_1, \dots, |a_m - x| - y_m\} + \varepsilon \cdot |a_0 - x|.$$

(a) Show that for some fixed $r > 0$ and λ , the function $h_{\mathbf{y}}$ is λ -concave in $B(p, r)$, and the following hold at any $x \in B(p, r)$:

- (i) $(\mathbf{d}_x \text{dist}_{a_i})(\nabla_x h_{\mathbf{y}}) < -\varepsilon^2$ if $|a_i - x| > y_i$ and
- (ii) $(\mathbf{d}_x \text{dist}_{a_i})(\nabla_x h_{\mathbf{y}}) > \varepsilon^2$ if

$$|a_i - x| - y_i = \min_{j>0} \{|a_j - x| - y_j\} < 0.$$

(b) Let $\alpha_{\mathbf{y}}$ be an $h_{\mathbf{y}}$ -gradient curve that starts at p . Use (a) to show that

$$f[\alpha_{\mathbf{y}}(t_0)] = \mathbf{y}$$

if $\frac{1}{\varepsilon^2} \cdot |f(p) - \mathbf{y}| \leq t_0 \leq \frac{r}{2}$.

(c) Let $t_0: \mathbf{y} \mapsto \frac{1}{\varepsilon^2} \cdot |f(p) - \mathbf{y}|$. Use 5.9 to show that the map

$$\Phi: \mathbf{y} \mapsto \alpha_{\mathbf{y}} \circ t_0(\mathbf{y})$$

is continuous (in fact Hölder) in $\Omega = B(f(p), \frac{\varepsilon^2}{2} \cdot r) \subset \mathbb{R}^m$ and $f \circ \Phi(\mathbf{y}) = \mathbf{y}$ for any $\mathbf{y} \in \Omega$. (This finishes the proof of 7.8.)

7.10. Corollary. Let $\mathcal{A}$ be an Alexandrov space, and let m be an integer. If $\text{LinDim } \mathcal{A} \geq m$, then

$$\dim \text{Lin}_s \geq m$$

for any point s in a dense G -delta subset of points in $\mathcal{A}$.

Moreover, if $m = \text{LinDim } \mathcal{A}$, then the set S of points $s \in \mathcal{A}$ such that $T_s \mathcal{A} \xrightarrow{\text{iso}} \mathbb{E}^m$ is a dense G -delta set in $\mathcal{A}$.

It is worth mentioning that the set $S \subset \mathcal{A}$ is also convex; see [98, 1.10].

Proof. Assume $\mathcal{A}$ is ALEX(0). By the right-inverse theorem and 7.4 there is a ball $B = \overline{B}[p, r]_{\mathcal{A}}$ and $c > 0$ such that

$$\text{pack}_{\varepsilon} B \geq \frac{c}{\varepsilon^m}$$

for all small $\varepsilon > 0$.

In particular, given a radius R we can choose $\varepsilon > 0$ and an integer N such that

$$\text{pack}_{\varepsilon} \overline{B}(0, R + r)_{\mathbb{E}^{m-1}} < N \leq \text{pack}_{\varepsilon} B.$$

Applying 6.20 to an ε -packing $x_1, \dots, x_N$ in B , we get that

$$\dim \text{Lin}_s \geq m$$

for any point s in a dense G-delta subset of $B(p, R)$. Since $R > 0$ is arbitrary, the first statement follows.

To prove the last statement, apply 7.2.

The general case can be reduced to $\text{ALEX}(-1)$ using rescaling, and it can be done essentially in the same way; one only has to change $\mathbb{E}^{m-1}$ to $\mathbb{H}^{m-1}$. $\square$

D Distance chart

7.11. Theorem. *Let $p, a_0, \dots, a_m$ be points in an m -dimensional Alexandrov space $\mathcal{A}$ such that*

$$\tilde{\angle}(p_{a_j}^{a_i}) > \frac{\pi}{2}$$

for any $i \neq j$. Then the map $f: \mathcal{A} \rightarrow \mathbb{R}^m$ defined by

$$f: x \mapsto (|a_1 - x|, \dots, |a_m - x|)$$

gives a bi-Lipschitz embedding of a neighborhood Ω of p ; the restriction $f|_{\Omega}$ is called a distance chart at p .

The following exercise provides a guide to the proof of the theorem.

7.12. Exercise! *Suppose $p, a_0, \dots, a_m \in \mathcal{A}$ and $f: \mathcal{A} \rightarrow \mathbb{R}^m$ are as in 7.8. Show that there is $\varepsilon > 0$ such that for any two points x, y in a sufficiently small neighborhood of p , one of the inequalities holds in each of the following two strings of m inequalities*

$$\begin{aligned} \angle[x_{a_1}^y] &< \frac{\pi}{2} - \varepsilon, \dots, \angle[x_{a_m}^y] < \frac{\pi}{2} - \varepsilon; \\ \angle[y_{a_1}^x] &< \frac{\pi}{2} - \varepsilon, \dots, \angle[y_{a_m}^x] < \frac{\pi}{2} - \varepsilon. \end{aligned}$$

Use this together with the right-inverse theorem (7.8) to prove 7.11.

E Volume

Fix a positive integer m . The m -dimensional Hausdorff measure of a Borel set B in a metric space will be called its m -volume; it will be denoted by $\text{vol}_m B$. We assume that the Hausdorff measure is calibrated so that the unit cube in $\mathbb{E}^m$ has unit volume.

This definition will be applied mostly to subsets of m -dimensional Alexandrov spaces. In this case, we may write $\text{vol } B$ instead of $\text{vol}_m B$.

7.13. Bishop–Gromov inequality. *Let $\mathcal{A}$ be an $\text{ALEX}(0)$ space.*

- (a) For any integer $k \geq 0$, any point $p \in \mathcal{A}$ and any radius $r > 0$, we have

$$\text{vol}_k B(p, r)_{\mathcal{A}} \leq \text{vol}_k B(0, r)_{T_p \mathcal{A}}.$$

Further, suppose $\text{LinDim } \mathcal{A} = m < \infty$.

- (b) For any point $p \in \mathcal{A}$ and any radius $r > 0$ we have

$$\text{vol}_m B(p, r) \leq \omega_m \cdot r^m,$$

where ω_m denotes the volume of the unit ball in $\mathbb{E}^m$.

- (c) Moreover, the function

$$r \mapsto \frac{\text{vol}_m B(p, r)}{r^m}$$

is nonincreasing.

Proof. Given $x \in \mathcal{A}$, choose a geodesic path γ_x from p to x . Recall that $\gamma_x^+(0) = \log_p x$ (here $\log_p: \mathcal{A} \rightarrow T_p$ is a multivalued function; so, $\gamma_x^+(0)$ might not be the only value of $\log_p x$). By comparison, $\log_p$ is distance-noncontracting. Note that $\log_p(B(p, r)_{\mathcal{A}}) \subset B(0, r)_{T_p}$; this proves part (a).

If $T_p \stackrel{\text{iso}}{=} \mathbb{E}^m$, then (a) implies (b). In the general case, by 7.10, we can find a point p' arbitrarily close to p such that $T_{p'} \stackrel{\text{iso}}{=} \mathbb{E}^m$. If $\varepsilon > |p - p'|$, then $B(p, r) \subset B(p', r + \varepsilon)$. Therefore,

$$\text{vol } B(p, r) \leq \omega_m \cdot (r + \varepsilon)^m$$

for any $\varepsilon > 0$. Hence (b) follows.

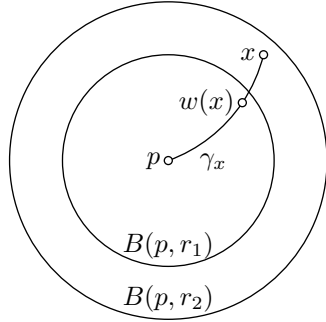
Now, suppose $0 < r_1 < r_2$. Consider the map $w: \mathcal{A} \rightarrow \mathcal{A}$ defined by $w: x \mapsto \gamma_x(\frac{r_1}{r_2})$. (This map mimics the dilation with center at p and coefficient $\frac{r_1}{r_2}$.) By comparison,

$$|w(x) - w(y)| \geq \frac{r_1}{r_2} \cdot |x - y|.$$

Observe that $B(p, r_1) \supset w[B(p, r_2)]$. Therefore,

$$\text{vol } B(p, r_1) \geq \left(\frac{r_1}{r_2}\right)^m \cdot \text{vol } B(p, r_2).$$

□



The following exercise generalizes the Bishop–Gromov inequality to the $\text{ALEX}(-1)$ case; it is sufficient for most applications. A more exact statement for $\text{ALEX}(\kappa)$ spaces is given in 7.21.

7.14. Exercise! Let $\mathcal{A}$ be a finite-dimensional $\text{ALEX}(-1)$ space; so $\text{LinDim } \mathcal{A} = m < \infty$. Show that

$$\text{vol } B(p, r) \leq \omega_m \cdot (\sinh r)^m,$$

where ω_m denotes the volume of the unit ball in $\mathbb{E}^m$. Moreover, the function

$$r \mapsto \frac{\text{vol } B(p, r)}{(\sinh r)^m}$$

is nonincreasing.

7.15. Exercise! Show that any finite-dimensional Alexandrov space is proper.

7.16. Exercise. Show that any nonempty open set in an Alexandrov space of finite dimension m has positive m -volume.

F Packings

7.17. Exercise. Let $\mathcal{A}$ be an m -dimensional $\text{ALEX}(\kappa)$ space and $m < \infty$.

- (a) Show that for any two balls B_1 and B_2 in $\mathcal{A}$ with positive radii there is $\delta > 0$ such that the inequality $\text{pack}_{\delta, \varepsilon} B_1 \geq \text{pack}_\varepsilon B_2$ holds for all sufficiently small $\varepsilon > 0$.
- (b) Show that there are constants $c_1 = c_1(m)$ and $c_2 = c_2(m, r, \kappa)$ such that

$$c_1 \cdot \text{vol } B \leq \varepsilon^m \cdot \text{pack}_\varepsilon B \leq c_2 \cdot \text{vol } B$$

for any ball $B = \overline{B}[p, r]_{\mathcal{A}}$ with $r > 0$ and all small $\varepsilon > 0$.

The exercise implies that for Alexandrov spaces, Minkowski dimension (also known as box-counting dimension) coincides with linear dimension. Namely, let $\mathcal{A}$ be an Alexandrov space. Given a point $p \in \mathcal{A}$ and $r > 0$, let

$$m(p, r) := \inf \left\{ \alpha \in \mathbb{R} : \varepsilon^\alpha \cdot \text{pack}_\varepsilon \overline{B}[p, r] \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0 \right\}.$$

Then $m(p, r) = \text{LinDim } \mathcal{A}$; in particular, $m(p, r)$ is an integer, and it does not depend on the choice of p and r .

G Other dimensions

Now we want to show that *all reasonable definitions of dimension give the same result for Alexandrov spaces*. More precisely, we have the following theorem; compare to [7, 15.16]. We refer to [53] for definitions of Lebesgue covering dimension and Hausdorff dimension, which will be denoted by TopDim and HausDim , respectively.

7.18. Theorem. *For any Alexandrov space $\mathcal{A}$, we have*

$$\text{LinDim } \mathcal{A} = \text{TopDim } \mathcal{A} = \text{HausDim } \mathcal{A}.$$

Proof. Suppose $\text{LinDim } \mathcal{A} = \infty$. By the right-inverse theorem (7.8), $\mathcal{A}$ contains a compact subset K with arbitrarily large $\text{TopDim } K$. In particular,

$$\text{TopDim } \mathcal{A} = \infty.$$

By Szpilrajn's theorem, $\text{HausDim } K \geq \text{TopDim } K$. Thus we also have

$$\text{HausDim } \mathcal{A} = \infty.$$

Now suppose $\text{LinDim } \mathcal{A} = m < \infty$. By the Bishop–Gromov inequality (7.13 and 7.14),

$$\text{HausDim } \mathcal{A} \leq m.$$

Since $\mathcal{A}$ is proper (7.15), Szpilrajn's theorem implies that

$$\text{TopDim } \mathcal{A} \leq \text{HausDim } \mathcal{A} \leq m.$$

Finally, by the right-inverse theorem (7.8), $m \leq \text{TopDim } \mathcal{A}$. $\square$

7.19. Exercise. *Let Ω be an open subset of an Alexandrov space $\mathcal{A}$. Show that*

$$\text{LinDim } \mathcal{A} = \text{LinDim } \Omega = \text{TopDim } \Omega = \text{HausDim } \Omega.$$

7.20. Exercise! *Let p be a point in a finite-dimensional Alexandrov space $\mathcal{A}$. Prove the following:*

- (a) *The tangent space T_p is a proper ALEX(0) space.*
- (b) $\text{LinDim } \Sigma_p + 1 = \text{LinDim } T_p = \text{LinDim } \mathcal{A}$.
- (c) *If $\text{LinDim } \mathcal{A} > 1$, then Σ_p is geodesic.*

Part (b) of the exercise opens a way to use induction on dimension, which can be used for finite-dimensional Alexandrov spaces; in Lecture 9, we will use it to define boundary. The condition $\text{LinDim } \mathcal{A} > 1$ in (c) is necessary; indeed $\Sigma_0 \mathbb{R}$ is a two-point space which is not geodesic.

H Remarks

Halbeisen's example [7, 46] shows that compactness of the space of directions is essential in the proof that the space of directions is π -geodesic (7.5).

The proof of the following version of the Bishop–Gromov inequality requires additionally the so-called *coarea formula* for Alexandrov spaces. The weaker inequality from 7.14 is sufficient for the sequel.

7.21. Optimal Bishop–Gromov inequality. *Given a point p in an m -dimensional $\text{ALEX}(\kappa)$ space, consider the function*

$$v: r \mapsto \text{vol}_m B(p, r);$$

denote by $\tilde{v}(r)$ the volume of an r -ball in $\mathbb{M}^m(\kappa)$. Then

$$v(r) \leq \tilde{v}(r)$$

for any $r > 0$. Moreover, the function

$$r \mapsto \frac{v(r)}{\tilde{v}(r)}$$

is nonincreasing. If $\kappa > 0$, then one has to assume that $r < \frac{\pi}{\sqrt{\kappa}}$.

This inequality was originally proved for Riemannian manifolds with lower Ricci curvature. The first part is also known as Bishop's inequality; it is due to Richard Bishop [17], [16, Corollary 4, p. 256]. The second part is due to Michael Gromov [39].

Theorem 7.18 was essentially proved by Conrad Plaut [104]. At that time, it was not known whether

$$\text{LinDim } \mathcal{A} = \infty \quad \Rightarrow \quad \text{TopDim } \mathcal{A} = \infty$$

for any Alexandrov space $\mathcal{A}$. The latter implication was proved by Grigory Perelman and the third author [86]; our proof is slightly different.

Lecture 8

Limit spaces

This lecture discusses the main source of applications of Alexandrov geometry. We start with an observation that the curvature bound of Alexandrov spaces survives under a Gromov–Hausdorff limit. Further, we present Perelman’s construction of strictly concave functions and prove Gromov’s selection theorem. We then use all this to prove the homotopy stability theorem (8.15) and the homotopy finiteness theorem (8.16).

A Survival of curvature bounds

8.1. Theorem. *Let $\mathcal{X}_n \rightarrow \mathcal{X}_\infty$ be a convergence in the sense of Gromov–Hausdorff. Suppose that for each n , the space $\mathcal{X}_n$ has curvature $\geq \kappa$ in the sense of Alexandrov. Then the same holds for $\mathcal{X}_\infty$.*

Proof. Let us verify the 4-point comparison for a quadruple of points $p_\infty, x_\infty, y_\infty, z_\infty \in \mathcal{X}_\infty$. We can assume that all the comparison angles $\angle(p_\infty x_\infty y_\infty)_{\mathbb{M}^2(\kappa)}, \angle(p_\infty y_\infty z_\infty)_{\mathbb{M}^2(\kappa)}, \angle(p_\infty x_\infty z_\infty)_{\mathbb{M}^2(\kappa)}$ are defined since otherwise there is nothing to check.

By the definition of Gromov–Hausdorff convergence, we can choose points $p_n, x_n, y_n, z_n \in \mathcal{X}_n$ that converge to $p_\infty, x_\infty, y_\infty, z_\infty \in \mathcal{X}_\infty$, respectively. In particular, the 6 distances between p_n, x_n, y_n, z_n converge to the distances between the corresponding pairs of $p_\infty, x_\infty, y_\infty, z_\infty$.

Since $\mathbb{M}^2(\kappa)$ -comparison holds for $p_n, x_n, y_n, z_n \in \mathcal{X}_n$, passing to the limit, we get the $\mathbb{M}^2(\kappa)$ -comparison for $p_\infty, x_\infty, y_\infty, z_\infty$. $\square$

8.2. Exercise! *A sequence $\mathcal{A}_1, \mathcal{A}_2, \dots$ of $\text{ALEX}(\kappa)$ spaces converges to $\mathcal{A}_\infty$ in the sense of Gromov–Hausdorff. Show that $\mathcal{A}_\infty$ is $\text{ALEX}(\kappa)$.*

and

$$\dim \mathcal{A}_\infty \leq \varliminf_{n \rightarrow \infty} \dim \mathcal{A}_n.$$

B Gromov's selection theorem

8.3. Theorem. *Let m be a positive integer, and let $D, \kappa \in \mathbb{R}$. Then any sequence of m -dimensional $\text{ALEX}(\kappa)$ spaces with diameters at most D has a converging subsequence in the sense of Gromov–Hausdorff.*

Recall that GH denotes the space of compact metric spaces with the Gromov–Hausdorff metric; see Section 1M.

Proof of 8.3. Suppose $\kappa = 0$. Denote by $\mathbf{K}$ the set of all isometry classes of $\text{ALEX}(0)$ spaces with dimension $\leq m$ and diameter $\leq D$. By 8.2, $\mathbf{K}$ is a closed subset of GH.

Choose a space $\mathcal{A} \in \mathbf{K}$; suppose $x_1, \dots, x_n \in \mathcal{A}$ is a collection of points such that $|x_i - x_j| > \varepsilon$ for all $i \neq j$. Note that the balls $B_i = B(x_i, \frac{\varepsilon}{2})$ do not overlap.

By 7.8, $\text{vol } \mathcal{A} > 0$. By the Bishop–Gromov inequality, $\text{vol } \mathcal{A} < \infty$, and if $\varepsilon < D$, then

$$\text{vol } B_i \geq \left(\frac{\varepsilon}{2 \cdot D}\right)^m \cdot \text{vol } \mathcal{A}$$

for any i . It follows that $n \leq \left(\frac{2 \cdot D}{\varepsilon}\right)^m$; that is,

$$\text{pack}_\varepsilon \mathcal{A} \leq N(\varepsilon) := \left(\frac{2 \cdot D}{\varepsilon}\right)^m$$

for all small $\varepsilon > 0$.

Choose a maximal ε -packing $x_1, \dots, x_n \in \mathcal{A}$. By 1.2, $\mathcal{F}_\varepsilon := \{x_1, \dots, x_n\}$ is an ε -net of $\mathcal{A}$. Observe that $|\mathcal{F}_\varepsilon - \mathcal{A}|_{\text{GH}} \leq \varepsilon$.

Let $\mathbf{F}_\varepsilon$ be the set of all metric spaces with at most $N(\varepsilon)$ points and diameter $\leq D$. Note that the set $\mathbf{F}_\varepsilon$ is a compact subset of GH.

Summarizing, for any $\varepsilon > 0$ we can find a compact ε -net $\mathbf{F}_\varepsilon \subset \text{GH}$ of $\mathbf{K}$. Since GH is complete (1.22), it remains to apply 1.1b.

We finished the proof of the case $\kappa = 0$. In the general case, applying rescaling, we can assume that $\kappa = -1$ and then argue as before, using 7.14 instead of 7.13. $\square$

8.4. Exercise. *Show that any sequence of m -dimensional $\text{ALEX}(\kappa)$ spaces with marked points contains a convergent subsequence in the sense of pointed Gromov–Hausdorff convergence (see Section 1L).*

8.5. Exercise! *Show that if $\mathcal{A}_n$ is a sequence of m -dimensional $\text{ALEX}(0)$ spaces with diameter $\leq D$, and volume $\geq v_0 > 0$, then its Gromov–Hausdorff limit $\mathcal{A}_\infty$ (if it exists) has dimension m .*

C Controlled concavity

Alexandrov spaces have plenty of semiconcave functions, including squared distance functions and their convex combinations. The following theorem provides a source of strictly concave functions defined on small open sets of finite-dimensional Alexandrov spaces.

8.6. Theorem. *Let $\mathcal{A}$ be a complete finite-dimensional Alexandrov space. Then for any point $p \in \mathcal{A}$, there is a strictly concave function f defined in an open neighborhood of p .*

Moreover, given a nonzero vector $v \in T_p$, the differential, $d_p f$, can be chosen arbitrarily close to $x \mapsto -\langle v, x \rangle$.

Idea of the proof. Suppose f is a semiconcave function. One can improve concavity of f by passing to a composition $\varphi \circ f$ with a very concave real-to-real function φ such that $\varphi' \approx 1$. Namely, if γ is a geodesic such that $|(f \circ \gamma)^\pm|$ is bounded away from zero, then $\varphi \circ f \circ \gamma$ can be made as concave as you wish by choosing an appropriate φ . For smooth real-to-real functions f and φ this follows from the chain rule $(\varphi \circ f)''(x) = \varphi''(f(x)) \cdot (f'(x))^2 + \varphi'(f(x)) \cdot f''(x)$. This trick does not help if $|(f \circ \gamma)^\pm| \approx 0$, but it does not make concavity worse; so $\varphi \circ f$ is very concave in most directions and at least as concave as f in all directions. In the proof we consider a sum $\varphi \circ f_1 + \dots + \varphi \circ f_N$ for a suitably chosen sequence of functions $f_1, \dots, f_N$, so that for any geodesic γ in a small neighborhood of p the value $|(f_i \circ \gamma)^\pm|$ is bounded away from zero for some i .

Before getting into the proof, we need to refresh material from 2E and 7F.

8.7. Lemma. *Let f be a locally Lipschitz function defined on a real interval $\mathbb{I}$. Then f is λ -concave if and only if for any $t_0 \in \mathbb{I}$ there is a smooth (local) upper barrier $\bar{f}$ of f at t_0 such that $f''(t_0) \leq \lambda$.*

Moreover, in the only-if part, we can assume, in addition, that $\bar{f}'(t_0)$ takes a given value in the interval $[f^+(t_0), f^-(t_0)]$ (by 2.20, this interval is not empty).

Proof. Passing to the function $t \mapsto f(t) - \lambda \cdot \frac{t^2}{2}$, we can assume that $\lambda = 0$.

The only-if part follows since a concave function is supported by the linear function $t \mapsto f(t_0) + c \cdot (t - t_0)$ for any $c \in [f^+(t_0), f^-(t_0)]$.

On the other hand, suppose f is not concave. Then adding a linear function to f we can assume that there is a subinterval $[a, b] \subset \mathbb{I}$ such that

$$f(a) > 0, \quad f(b) > 0, \quad \text{but} \quad f(t) < 0 \quad \text{for some} \quad t \in [a, b].$$

Let s be the minimal value such that $f(t) \geq \frac{s}{2} \cdot (t-a) \cdot (t-b)$ for any $t \in [a, b]$. Note that $s > 0$ and $f(t_0) = \frac{s}{2} \cdot (t_0-a) \cdot (t_0-b)$ for some $t_0 \in (a, b)$; that is, $t \mapsto \frac{s}{2} \cdot (t-a) \cdot (t-b)$ is a local lower barrier of f at t_0 . Therefore, $f''(t_0) \geq s > 0$ for any upper barrier $\bar{f}$ of f at t_0 , and the if part follows. $\square$

8.8. Exercise! Let f be a Lipschitz real-to-real function, and let φ be a smooth function defined in a neighborhood of $x_0 = f(t_0)$. Suppose that f is λ_1 -concave for some $\lambda_1 \geq 0$ and there are some constants $c_1, c_2, c_3 \in \mathbb{R}$ such that (1) $|f^+(t_0)| > c_1 > 0$ (or $|f^-(t_0)| > c_1 > 0$), (2) $c_2 > \varphi'(x_0) \geq 0$, and (3) $0 > c_3 \geq \varphi''(x_0)$. Let $\lambda_2 = c_2 \cdot \lambda_1 + c_1^2 \cdot c_3$. Show that $\varphi \circ f$ is λ_2 -concave in a neighborhood of t_0 .

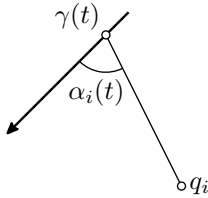
8.9. Exercise! Let Ω be a nonempty open set in an m -dimensional Alexandrov space $\mathcal{A}$; assume $m < \infty$. Given a point p , there are $r > 0$ and $c > 0$ such that for any sufficiently small $\varepsilon > 0$ there are at least c/ε^{m-1} points $q_1, \dots, q_N \in \Omega$ such that $|p - q_i| = r$ for any i and $\angle[p_{q_j}^{q_i}] > \varepsilon$ if $i \neq j$.

Proof of 8.6. Suppose $r > 0$ is small and c is large (these values will be specified a bit later). Consider the real-to-real function

$$\varphi_{r,c}(x) = (x-r) - c \cdot (x-r)^2/r;$$

so,

$$\varphi_{r,c}(r) = 0, \quad \varphi'_{r,c}(r) = 1, \quad \varphi''_{r,c}(r) = -\frac{2 \cdot c}{r}.$$



Let γ be a geodesic. Given points $q_1, \dots, q_N$, set

$$\alpha_i(t) = \angle(\gamma^+(t), \uparrow_{[\gamma(t)q_i]}),$$

$$\lambda_i(t) = \frac{3 - c \cdot \cos^2[\alpha_i(t)]}{r},$$

$$h_i = \varphi_{r,c} \circ \text{dist}_{q_i} \circ \gamma.$$

Suppose that $|q_i - \gamma(t_0)|$ is sufficiently close to r . Then by 8.8 we have that h_i is $\lambda_i(t_0)$ -concave in a neighborhood of t_0 .

Assume that $|p - q_i| = r$ for each i , and $\varepsilon_1 > 0$ is small. If the geodesic γ lies in $B(p, \varepsilon_1)$, then from above we have that $\sum h_i$ is $\sum \lambda_i(t_0)$ -concave in a neighborhood of any t_0 .

Suppose that there is $\varepsilon_2 > 0$ such that

❶

$$\max_i \{|\alpha_i(t) - \frac{\pi}{2}|\} \geq \varepsilon_2 > 0$$

for any geodesic γ in $B(p, \varepsilon_1)$. If $c > 3 \cdot N / \sin^2 \varepsilon_2$, then $\sum \lambda_i(t_0) < 0$ and $\sum h_i$ is strictly concave in a neighborhood of t_0 for any choice of geodesic γ in $B(p, \varepsilon_1)$. It follows that

$$f = \sum_i \varphi_{r,c} \circ \text{dist}_{q_i}$$

is strictly concave in $B(p, \varepsilon_1)$.

It remains to choose $r, \varepsilon_1, \varepsilon_2 > 0$ and points $q_1, \dots, q_N$ such that $|p - q_i| = r$ for each i , and **1** holds.

Let us assume that $\mathcal{A}$ is ALEX(0). Applying 8.9 to a small open ball around p and small $\varepsilon_3 > 0$, we can choose points $q_1, \dots, q_N$ such that $N \geq c_1 / \varepsilon_3^{m-1}$ and $|p - q_i| = r$ for some small $r > 0$ and $\angle(p_{q_i}^{q_j}) > \varepsilon_3$ (here $c_1 > 0$). On the other hand, suppose γ runs from x to y . Let us apply the $(n+1)$ -point comparison (7.1) to $\gamma(t)$, x , y and $q_1, \dots, q_N$; denote the obtained points by $\tilde{x}, \tilde{y}, \tilde{q}_1, \dots, \tilde{q}_N$. If $\varepsilon_1 \ll \varepsilon_3$, $\varepsilon_2 \ll \varepsilon_3$, and $|\alpha_i(t) - \frac{\pi}{2}| < \varepsilon_2$ for each i , then the points $\tilde{q}_1, \dots, \tilde{q}_N$ lie very close to an r -sphere in a hyperplane in $\mathbb{E}^m$ that is perpendicular to $[\tilde{x}\tilde{y}]$; in particular, $N \leq c_2(m) / \varepsilon_3^{m-2}$. Therefore, for small $\varepsilon_3 > 0$ and yet smaller ε_1 and ε_2 , points $q_1, \dots, q_N$ meet **1**.

To prove the last statement, choose a geodesic $[pz]$ in the direction sufficiently close to v and apply 8.9 to a small ball centered on a point $\tilde{z} \in [pz]$ close to p .

The ALEX(κ)-case can be reduced to the ALEX(-1)-case by rescaling, and it is done the same way. The only difference is that the points $\tilde{q}_1, \dots, \tilde{q}_N$ lie very close to an r -sphere in a hyperplane in $\mathbb{H}^m$. $\square$

Given a point p a strictly concave function f can be chosen so that $f(p) = 0$ and $(d_p f)(x) \approx -|x|$. In particular, p is the maximal point of f . It can be constructed by taking the minimum of the functions in the theorem. Then the set $K = \{x \in \mathcal{A} : f(x) \geq -\varepsilon\}$ forms a closed convex neighborhood of p for any small $\varepsilon > 0$, so we get the following.

8.10. Corollary. *Any point p of a finite-dimensional Alexandrov space admits an arbitrarily small convex closed neighborhood K and a strictly concave function f defined in a neighborhood of K such that p is the maximum point of f and $f|_{\partial K} = 0$.*

8.11. Exercise. *Construct an Alexandrov space $\mathcal{A}$ such that there is no open set $\Omega \subset \mathcal{A}$ with a strictly concave function $f: \Omega \rightarrow \mathbb{R}$.*

8.12. Exercise. *Let $p, a_0, \dots, a_m$ be points in an m -dimensional Alexandrov space $\mathcal{A}$ such that $|p - a_i| = r$ for any i and $\angle(p_{a_j}^{a_i}) > \frac{\pi}{2}$ for any $i \neq j$, and let $\varphi_{c,r}$ be as in the proof of 8.6, with c large. Show that the function $f = \varphi_{c,r} \circ \text{dist}_{a_1} + \dots + \varphi_{c,r} \circ \text{dist}_{a_m}$ is strictly concave in a neighborhood of p .*

D Liftings

Suppose that the Gromov–Hausdorff distance $|\mathcal{A} - \mathcal{A}'|_{\text{GH}}$ is sufficiently small, so we may think that both spaces $\mathcal{A}$ and $\mathcal{A}'$ lie at a small Hausdorff distance in an ambient metric space $\mathcal{W}$. In particular, we may choose a small $\varepsilon > 0$, so that for any point $p \in \mathcal{A}$, there is a point $p' \in \mathcal{A}'$ such that $|p - p'|_{\mathcal{W}} < \varepsilon$; the point p' will be called a *lifting* (or ε -*lifting*) of p in $\mathcal{A}'$. We may choose a lifting $p' \in \mathcal{A}'$ for every point $p \in \mathcal{A}$; in this case, the map $p \mapsto p'$ is called a (ε) -*lifting map*.

Let us emphasize that liftings are not uniquely defined, and the lifting map is not assumed to be continuous. Also, to talk about liftings, we have to choose $\varepsilon > 0$, as well as the inclusions $\mathcal{A}, \mathcal{A}' \hookrightarrow \mathcal{W}$.

Let $\mathcal{A}$ be a compact m -dimensional $\text{ALEX}(\kappa)$ space. Suppose $\mathcal{A}'$ is another compact m -dimensional $\text{ALEX}(\kappa)$ space such that $|\mathcal{A} - \mathcal{A}'|_{\text{GH}}$ is sufficiently small — smaller than some $\varepsilon = \varepsilon(\mathcal{A}) > 0$. If the construction of the previous section is performed in $\mathcal{A}$, then it can be repeated in $\mathcal{A}'$ for the liftings of all points and the same function φ . It produces a strictly concave function defined in a controlled neighborhood of the lifting p' of p .

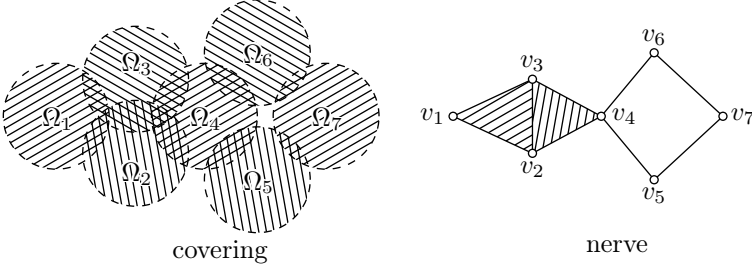
The results of this and related constructions will also be called liftings; say we can talk about a lifting from $\mathcal{A}$ to $\mathcal{A}'$ of a function provided by 8.6 (if the Gromov–Hausdorff distance $|\mathcal{A} - \mathcal{A}'|_{\text{GH}}$ is small, then these liftings are strictly concave) and a lifting of a convex neighborhood from 8.10. Here one cannot use 8.6 and 8.10 as black boxes — one has to understand the construction, but it is straightforward.

8.13. Exercise. *Give an example of Gromov–Hausdorff convergence of spaces $\mathcal{A}_n \rightarrow \mathcal{A}_\infty$ such that all $\mathcal{A}_n$ and $\mathcal{A}_\infty$ are compact finite-dimensional $\text{ALEX}(0)$ and for any small $\varepsilon > 0$ there is no continuous ε -lifting map $\mathcal{A}_\infty \rightarrow \mathcal{A}_n$ for any large n .*

E Nerves

Let $\{\Omega_1, \dots, \Omega_k\}$ be a finite open cover of a compact metric space $\mathcal{X}$. Consider an abstract simplicial complex $\mathcal{N}$, with one vertex v_i for each set Ω_i such that a simplex with vertices $v_{i_1}, \dots, v_{i_m}$ is included in $\mathcal{N}$ if the intersection $\Omega_{i_1} \cap \dots \cap \Omega_{i_m}$ is nonempty. The obtained simplicial complex $\mathcal{N}$ is called the *nerve* of the covering $\{\Omega_i\}$. Evidently, $\mathcal{N}$ is a finite simplicial complex — it is a subcomplex of a simplex with the vertices $\{v_1, \dots, v_k\}$. Recall that Star_{v_i} denotes the union of all simplices in $\mathcal{N}$ that share the vertex v_i .

The next statement follows from [48, 4G.3].



8.14. Nerve theorem. Let $\{\Omega_1, \dots, \Omega_k\}$ be an open cover of a compact metric space $\mathcal{X}$ and let $\mathcal{N}$ be the corresponding nerve with vertices $\{v_1, \dots, v_k\}$. Suppose that every nonempty finite intersection $\Omega_{\alpha_1} \cap \dots \cap \Omega_{\alpha_m}$ is contractible. Then $\mathcal{X}$ is homotopy equivalent to the nerve $\mathcal{N}$ of the cover.

Moreover, homotopy equivalences $a: \mathcal{X} \rightarrow \mathcal{N}$ and $b: \mathcal{N} \rightarrow \mathcal{X}$ can be chosen so that if $x \in \Omega_i$, then $a(x) \in \text{Star}_{v_i}$, and if $y \in \mathcal{N}$ lies in the simplex with vertices $v_{i_1}, \dots, v_{i_m}$, then $b(y) \in \Omega_{i_1} \cup \dots \cup \Omega_{i_m}$.

F Homotopy stability

8.15. Theorem. Let $\mathcal{A}_1, \mathcal{A}_2, \dots$, and $\mathcal{A}_\infty$ be compact m -dimensional $\text{ALEX}(\kappa)$ spaces, and $m < \infty$. Suppose $\mathcal{A}_n \rightarrow \mathcal{A}_\infty$ as $n \rightarrow \infty$ in the sense of Gromov–Hausdorff. Then $\mathcal{A}_\infty$ is homotopy equivalent to $\mathcal{A}_n$ for all large n .

Moreover, given $\varepsilon > 0$ there are maps $h_n: \mathcal{A}_\infty \rightarrow \mathcal{A}_n$ that are homotopy equivalences and ε -liftings for all large n .

Applying this theorem with Gromov’s selection theorem (8.3) and Exercise 8.5, we get the following.

8.16. Theorem. Given $\kappa \in \mathbb{R}, D > 0, v_0 > 0$ and a positive integer m , there are only finitely many homotopy types of m -dimensional $\text{ALEX}(\kappa)$ spaces with diameter $\leq D$, and volume $\geq v_0$.

Proof of 8.16 modulo 8.15. Assume the contrary. Then we can choose a sequence of m -dimensional $\text{ALEX}(\kappa)$ spaces $\mathcal{A}_1, \mathcal{A}_2, \dots$ that have different homotopy types and satisfy the assumptions of the theorem. By Gromov’s selection theorem, we can assume that $\mathcal{A}_n$ converges to some space $\mathcal{A}_\infty$ in the sense of Gromov–Hausdorff.

By 8.4, $\dim \mathcal{A}_\infty = m$. It remains to apply 8.15. □

Proof of 8.15. Since $\mathcal{A}_\infty$ is compact, applying 8.10, we can find a finite open cover of $\mathcal{A}_\infty$ by convex open sets $\Omega_1, \dots, \Omega_k$ such that for each Ω_i

there is a strictly concave function f_i that is defined in a neighborhood of the closure $\bar{\Omega}_i$ and such that $f_i|_{\partial\Omega_i} = 0$.

Subtracting from functions f_i some small value $\varepsilon > 0$, we can ensure that $\bigcap_{i \in S} \Omega_i \neq \emptyset$ if and only if $\bigcap_{i \in S} \bar{\Omega}_i \neq \emptyset$.

Suppose that $W = \bigcap_{i \in S} \Omega_i \neq \emptyset$. Then W is contractible. In other words, the cover $\{\Omega_1, \dots, \Omega_k\}$ meets the assumptions of the nerve theorem (8.14).

Indeed, the function

$$f_S := \min_{i \in S} f_i$$

is strictly concave and it vanishes on the boundary of W . The f_S -gradient flow $(t, x) \mapsto \text{Flow}_{f_S}^t(x)$ defines a homotopy $[0, \infty) \times W \rightarrow W$. By the first distance estimate (5.6), $\text{Flow}_{f_S}^t(x)$ converges to the (necessarily unique) maximum point of f_S . Therefore, in the obtained homotopy we can reparametrize $[0, \infty)$ by $[0, 1)$ and extend it continuously to $[0, 1]$; thus we get that W is contractible.

The functions f_i and sets Ω_i can be lifted to $\mathcal{A}_n$ while keeping their properties for all large n . More precisely, there are liftings $f_{i,n}$ of all f_i to $\mathcal{A}_n$ which are strictly concave for all large n and such that $\bar{\Omega}_{i,n} = \{x \in \mathcal{A}_n : f_{i,n}(x) \geq 0\}$ is a compact convex set and $\Omega_{i,n} = \{x \in \mathcal{A}_n : f_{i,n}(x) > 0\}$ is an open convex set for each i .

Notice that $\{\Omega_{1,n}, \dots, \Omega_{k,n}\}$ is an open cover of $\mathcal{A}_n$ for all large n . Indeed, suppose we have $p_n \in \mathcal{A}_n \setminus (\Omega_{1,n} \cup \dots \cup \Omega_{k,n})$ for arbitrarily large n . Since $\mathcal{A}_\infty$ is compact, there is a limit point $p_\infty \in \mathcal{A}_\infty$ for a subsequence of p_n . But $p_\infty \in \Omega_i$ for some i and therefore $p_n \in \Omega_{i,n}$ for arbitrarily large n — a contradiction.

In a similar fashion we can show that if n is large then any collection $\{\Omega_{i,n}\}_{i \in S}$ has a common point in $\mathcal{A}_n$ if and only if $\{\Omega_i\}_{i \in S}$ has a common point in $\mathcal{A}_\infty$. Here we have to use that $\bigcap_{i \in S} \Omega_i \neq \emptyset$ if and only if $\bigcap_{i \in S} \bar{\Omega}_i \neq \emptyset$.

It follows that for any large n the covers

- ◇ $\{\Omega_1, \dots, \Omega_k\}$ of $\mathcal{A}_\infty$ and
- ◇ $\{\Omega_{1,n}, \dots, \Omega_{k,n}\}$ of $\mathcal{A}_n$

have the same nerve. By the nerve theorem (8.14), $\mathcal{A}_n$ and $\mathcal{A}_\infty$ are homotopy equivalent for all large n . $\square$

Note that the proof above implies the following.

8.17. Theorem. *Any compact finite-dimensional Alexandrov space is homotopy equivalent to a finite simplicial complex.*

G Remarks

Originally, Gromov's selection theorem was proved for Riemannian manifolds with a lower bound on Ricci curvature [39]. It motivates the study of limits of manifolds with lower Ricci curvature bounds and their synthetic generalizations, the so-called $CD(K, m)$ spaces; CD stands for curvature-dimension condition. This theory has significant applications in Alexandrov geometry; in particular, it provides a version of the Liouville theorem about phase-space volume of geodesic flow in Alexandrov spaces [18].

The construction of a strictly concave function (8.6) is due to Grigory Perelman [85, 88]. A more exact version of this construction is given in the survey of the third author [96, 7.2]. Let us mention the following closely related result of Artem Nepechiy [82]: *given a point p in an Alexandrov space there is a 2-convex function that approximates dist_p^2 up to second order at p .*

The fact that the controlled concavity survives after lifting (see Section 8D) was observed and used by the first author [57, 4.2]. Functions with controlled concavity can be lifted to a collapsing sequence; in this case the resulting function is not concave but it has some interesting properties. In particular, if the limit space has dimension m , then the Morse index of the lifting is at least m ; this property implies a partial answer to an old question of *which finite-dimensional Alexandrov spaces can be approximated by Riemannian manifolds with a fixed lower curvature bound*; see [58].

Furthermore, such a lifted function has the following property (let us call it gap-convexity) which, as far as we know, did not find an application: *If f is a controllably concave function and f_n are liftings of f to a collapsing sequence of Alexandrov spaces with uniform lower curvature bound, then for any $\varepsilon > 0$ and all large n , the restriction of f_n to any geodesic of length at least ε is concave.* (Recall that our geodesics are always length-minimizing.)

Let us list some results that can be proved by applying Gromov's selection theorem in the same fashion as in the proof of the homotopy-type finiteness theorem (8.16).

8.18. Betti-number theorem. *For any $\kappa \in \mathbb{R}$, $D > 0$ and a positive integer m , there is a constant $c = c(m, D, \kappa)$ such that*

$$\beta_0(M) + \cdots + \beta_m(M) \leq c$$

for any closed m -dimensional Riemannian manifold M with sectional curvature $\geq \kappa$ and diameter $\leq D$. Here $\beta_i(M)$ denotes the i -th Betti number of M .

Gromov's original proof [37] of the Betti-number theorem did not use Alexandrov geometry directly. However, it is easy to produce a proof following the steps above (no one bothered to write it so far). The following result was proved by the third author [99], and it uses the same technique.

8.19. Scalar curvature bound. *Given $\kappa \in \mathbb{R}$, $D > 0$ and a positive integer m , there is a constant $c = c(m, D, \kappa)$ such that*

$$\int_M \text{Sc} \leq c$$

for any closed m -dimensional Riemannian manifold M with sectional curvature $\geq \kappa$ and diameter $\leq D$. Here Sc denotes the scalar curvature.

The following theorem is a more exact version of 8.15. Its close relative (9.2) will play an important role in the following lecture.

8.20. Stability theorem. *Let $\mathcal{A}_1, \mathcal{A}_2, \dots$, and $\mathcal{A}_\infty$ be compact m -dimensional $\text{ALEX}(\kappa)$ spaces, and $m < \infty$. Suppose $\mathcal{A}_n \rightarrow \mathcal{A}_\infty$ as $n \rightarrow \infty$ in the sense of Gromov–Hausdorff. Then $\mathcal{A}_\infty$ is homeomorphic to $\mathcal{A}_n$ for all large n .*

Moreover, given $\varepsilon > 0$ there are maps $h_n: \mathcal{A}_\infty \rightarrow \mathcal{A}_n$ that are homeomorphisms and ε -liftings for all large n .

This theorem was proved by Grigory Perelman [87]. The proof was rewritten with more details by the first author [55]; for more exact statements, see [57, 58]. In private conversations, Perelman claimed that the homeomorphisms in the theorem can be assumed to be bi-Lipschitz with constants that depend on $\mathcal{A}_\infty$. Since no proof has been written, this statement should be considered as a conjecture; partial results in this direction were obtained by Mohammad Alattar [2].

Theorem 8.16 was originally proved by Karsten Grove and Peter Petersen [42]. Perelman's stability theorem (8.20) implies the following stronger statement.

8.21. Homeomorphism-type finiteness. *For any real numbers κ , $D > 0$, $v > 0$ and a positive integer m , there are only finitely many homeomorphism types of closed m -dimensional manifolds that admit a Riemannian metric with sectional curvature $\geq \kappa$, volume $\geq v$, and diameter $\leq D$.*

This statement can be improved to diffeomorphism-type finiteness in all dimensions $m \neq 4$. Indeed, for $m \neq 4$, a closed topological m -manifold admits only finitely many smooth structures; see [59] and [81, 118] for cases $m \geq 5$ and $m \leq 3$, respectively.

In all these theorems we prove a statement about m -dimensional manifolds with lower sectional curvature bound following the following steps:

- ◇ Arguing by contradiction, we assume the existence of a sequence of manifolds that violates the statement.
- ◇ Use Gromov's selection theorem to choose a converging subsequence.
- ◇ Use curvature survival to conclude that the limit space is Alexandrov.
- ◇ Try to arrive at a contradiction using Alexandrov geometry.

In principle, the same strategy might work for a sequence of Riemannian manifolds with dimension approaching infinity, but no applications of this kind have been found yet. The following question gives an example where such a strategy might be successful in principle.

8.22. Question. *Is it true that no simply-connected closed manifold of arbitrarily large dimension admits a Riemannian metric with two-sided sectional-curvature bound and diameter bounded by a fixed sufficiently small positive constant?*

If the dimension is bounded, then Gromov's theorem [38] implies that such a manifold can be covered by a compact nil-manifold; in particular, the manifold cannot be simply connected. However, if the dimension is unbounded, then Riemannian manifolds with these conditions may not be covered by compact nil-manifolds; such examples were found by Galina Guzhvina [45].

Lecture 9

Boundary

Here we define the boundary of a finite-dimensional Alexandrov space, discuss its properties, and prove the doubling theorem (9.10d).

A Definition

The definition of boundary will use induction on dimension, and it will work *only* for finite-dimensional Alexandrov spaces.

Suppose $\mathcal{A}$ is a 1-dimensional Alexandrov space. By 7.3, $\mathcal{A}$ is homeomorphic to a 1-dimensional manifold (possibly with nonempty boundary). This allows us to define the boundary $\partial\mathcal{A} \subset \mathcal{A}$ as the boundary of the manifold.

Now assume that the notion of boundary is defined in dimensions $1, \dots, m-1$ where $m \geq 2$. Let $\mathcal{A}$ be an m -dimensional Alexandrov space. We say that $p \in \mathcal{A}$ belongs to the boundary (briefly $p \in \partial\mathcal{A}$) if $\partial\Sigma_p \neq \emptyset$. By 7.5 and 7.20b, Σ_p is an $(m-1)$ -dimensional Alexandrov space; therefore its boundary is already defined and hence this inductive definition makes sense.

9.1. Exercise. *Prove the following statements:*

- (a) *For a closed convex set $K \subset \mathbb{E}^m$ with nonempty interior, the topological boundary of K as a subset of $\mathbb{E}^m$ coincides with the boundary of K described above.*
- (b) *Suppose a finite-dimensional Alexandrov space $\mathcal{A}$ is a direct sum of $\mathcal{A}_1$ and $\mathcal{A}_2$. Then*

$$\partial\mathcal{A} = (\partial\mathcal{A}_1 \oplus \mathcal{A}_2) \cup (\mathcal{A}_1 \oplus \partial\mathcal{A}_2)$$

- (c) If $\text{Cone } \Sigma$ is an $\text{ALEX}(0)$ space of dimension ≥ 2 (by 7.20b this implies that Σ is $\text{ALEX}(1)$), then

$$\partial \text{Cone } \Sigma = \text{Cone } \partial \Sigma,$$

where $\text{Cone } \partial \Sigma := \{ s \cdot \xi \in \text{Cone } \Sigma : \xi \in \partial \Sigma \}$.

B Conic neighborhoods

We are going to use the following result without proof; it is a close relative of Perelman's stability theorem 8.20, and it is proved in [88].

Recall that the logarithm $\log_p: \mathcal{A} \rightarrow T_p$ is a multivalued function defined in 5A.

9.2. Theorem. *For any point p in a finite-dimensional Alexandrov space $\mathcal{A}$ and all sufficiently small $\varepsilon > 0$ there is a homeomorphism $h_\varepsilon: B(p, \varepsilon)_{\mathcal{A}} \rightarrow B(0, \varepsilon)_{T_p}$ such that $0 = h_\varepsilon(p)$.*

Moreover, we may assume that

$$\sup_{x \in B(p, \varepsilon)} \left\{ \frac{1}{\varepsilon} \cdot |\log_p x - h_\varepsilon(x)|_{T_p} \right\} \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0.$$

Note that the last condition implies that h_ε is an $o(\varepsilon)$ -isometry.

The above theorem is often used together with the *uniqueness of conic neighborhoods* stated below.

Suppose that an open neighborhood U of a point x in a metric space $\mathcal{X}$ admits a homeomorphism to $\text{Cone } \Sigma$ such that x is mapped to the origin of the cone. In this case, we say that U is a conic neighborhood of x .

9.3. Uniqueness of conic neighborhoods. *Any two conic neighborhoods of a given point in a metric space are pointed homeomorphic; that is, there is a homeomorphism between neighborhoods that maps the origin of one cone to the origin of the other.*

9.4. Advanced exercise. *Prove 9.3 or read the proof in [62].*

9.5. Exercise! *Suppose $x \mapsto x'$ is a homeomorphism between finite-dimensional Alexandrov spaces $\mathcal{A}$ and $\mathcal{A}'$. Show that*

- (a) $T_x \cong T_{x'}$ (here and below $\cong$ means homeomorphic),
- (b) $\text{Susp } \Sigma_x \cong \text{Susp } \Sigma_{x'}$,
- (c) but in general $\Sigma_x \not\cong \Sigma_{x'}$.

C Topology

The following theorem asserts that the boundary is a topological invariant, despite the heavy use of geometry in its definition.

9.6. Theorem. *Let $\mathcal{A}$ and $\mathcal{A}'$ be homeomorphic finite-dimensional Alexandrov spaces. Then $\dim \mathcal{A} = \dim \mathcal{A}'$ and*

$$\partial \mathcal{A} \neq \emptyset \quad \Leftrightarrow \quad \partial \mathcal{A}' \neq \emptyset.$$

While working on the proof, keep in mind that there are pairs of spaces $\mathcal{K}_1$ and $\mathcal{K}_2$ such that $\mathcal{K}_1 \not\cong \mathcal{K}_2$, but $\mathbb{R} \times \mathcal{K}_1 \cong \mathbb{R} \times \mathcal{K}_2$. For example, one can let $\mathcal{K}_1 = \mathbb{S}^4$ and let $\mathcal{K}_2$ be the suspension over the Poincaré homology sphere; compare to 9.5c.

Let $\mathcal{A}$ be an m -dimensional Alexandrov space and $m < \infty$. Define corank of $\mathcal{A}$ (briefly, $\text{corank } \mathcal{A}$) as the minimal value k such that $\mathcal{A}$ splits isometrically as $\mathbb{R}^{m-k} \times \mathcal{K}$; here $\mathcal{K}$ is a k -dimensional Alexandrov space.

In the following proof we will apply induction on the corank of $\mathcal{A}$.

Proof of 9.6. The first statement follows from 7.18.

Suppose we have a counterexample, say $\partial \mathcal{A} \neq \emptyset$, but $\partial \mathcal{A}' = \emptyset$. Let $k := \text{corank } \mathcal{A}$ and $k' := \text{corank } \mathcal{A}'$. We can assume that the pair (k, k') is minimal in lexicographic order among all counterexamples; in particular, k is minimal. Let $x \mapsto x'$ be a homeomorphism from $\mathcal{A}$ to $\mathcal{A}'$.

Choose $x \in \partial \mathcal{A}$. Since $\partial \mathcal{A}' = \emptyset$, we have $x' \notin \partial \mathcal{A}'$. Note that

$$\text{corank } T_x \leq k \quad \text{and} \quad \text{corank } T_{x'} \leq k'.$$

By 9.5a, $T_x \cong T_{x'}$. Note that $\partial T_x \neq \emptyset$ and $\partial T_{x'} = \emptyset$. Therefore, we may assume that $\mathcal{A}$ and $\mathcal{A}'$ are Euclidean cones and the homeomorphism sends the origin to the origin. The remaining part of the proof is divided into three cases.

Case 1. Suppose $k > 1$. Let $\mathcal{A} \stackrel{\text{iso}}{=} \mathbb{R}^{m-k} \times \mathcal{C}$, where $\mathcal{C}$ is a k -dimensional ALEX(0) cone. Observe that $\text{corank } T_y \leq \text{corank } \mathcal{A}$ for any $y \in \mathcal{A}$ and the equality holds only if y projects to the origin of $\mathcal{C}$.

Since $k > 1$, we can choose $z \neq 0$ that lies in $\partial \mathcal{C}$ (compare to 9.1c). Choose $y \in \mathcal{A}$ that projects to z ; in particular, $\text{corank } T_y < \text{corank } \mathcal{A}$. By 9.5a, $T_y \cong T_{y'}$, $\partial T_y \neq \emptyset$ and $\partial T_{y'} = \emptyset$. The latter contradicts the minimality of k .

Case 2. Suppose $k \leq 1$ and $k' > 1$. Since $\partial \mathcal{A} \neq \emptyset$, we get that $k = 1$; therefore, $\mathcal{A} = \mathbb{R}^{m-1} \times [0, \infty)$.

Let $\mathcal{A}' \stackrel{\text{iso}}{=} \mathbb{R}^{m-k'} \times \mathcal{C}'$, where $\mathcal{C}'$ is a k' -dimensional $\text{ALEX}(0)$ cone. Since $\partial\mathcal{A} \cong \mathbb{R}^{m-1}$, the image of $\partial\mathcal{A}$ in $\mathcal{A}'$ does not lie in $\mathbb{R}^{m-k'} \times \{0\}$. In other words, we can choose $y \in \partial\mathcal{A}$ such that its image $y' \in \mathcal{A}'$ has a nonzero projection in $\mathcal{C}'$. Observe that $T_y \cong T_{y'}$,

$$\text{corank } T_y \leq k = 1, \quad \text{corank } T_{y'} < k', \quad \partial T_y \neq \emptyset, \quad \text{and} \quad \partial T_{y'} = \emptyset,$$

which contradicts minimality of k' .

Case 3. Suppose $k \leq 1$ and $k' \leq 1$. Since $\partial\mathcal{A} \neq \emptyset$, we have $k = 1$. By 7.3, $\mathcal{A} \cong \mathbb{R}^{m-1} \times [0, \infty)$. Note that $\mathcal{A}' \cong \mathbb{R}^m$, and $\mathcal{A} \not\cong \mathcal{A}' - a$ contradiction. $\square$

9.7. Exercise! Let $x \mapsto x'$ be a homeomorphism $\Omega \rightarrow \Omega'$ between open subsets in finite-dimensional Alexandrov spaces $\mathcal{A}$ and $\mathcal{A}'$. Show that $x \in \partial\mathcal{A}$ if and only if $x' \in \partial\mathcal{A}'$.

9.8. Exercise! Show that the boundary of a finite-dimensional Alexandrov space is a closed subset.

D Tangent space

Spaces of directions and tangent spaces of an Alexandrov space have already been defined in 4A and 4B, respectively. Let us extend these definitions to subsets of an Alexandrov space.

Let X be a subset in a finite-dimensional Alexandrov space $\mathcal{A}$. Choose $p \in \mathcal{A}$ and $\xi \in \Sigma_p$. Suppose ξ is a limit of directions $\uparrow_{[px_n]}$ for a sequence $x_1, x_2, \dots \in X$ that converges to p . Then we say that ξ is in the space of directions from p to X ; briefly $\xi \in \Sigma_p X$.

Further, $\text{Cone}(\Sigma_p X)$ will be called the tangent space to X at p ; it will be denoted by $T_p X$.

Note that $\Sigma_p X \subset \Sigma_p \mathcal{A}$ and $T_p X \subset T_p \mathcal{A}$.

9.9. Theorem. For any finite-dimensional Alexandrov space $\mathcal{A}$, we have

$$\partial(\Sigma_p \mathcal{A}) = \Sigma_p(\partial\mathcal{A}) \quad \text{and} \quad \partial(T_p \mathcal{A}) = T_p(\partial\mathcal{A}).$$

Proof. Choose a sequence $x_n \in \partial\mathcal{A}$ such that $x_n \rightarrow p$ and $\uparrow_{[px_n]} \rightarrow \xi$.

Let $\varepsilon_n = 2 \cdot |p - x_n|$, and let $h_{\varepsilon_n}: B(p, \varepsilon_n)_{\mathcal{A}} \rightarrow B(0, \varepsilon_n)_{T_p}$ be the homeomorphisms provided by 9.2; in particular, $\frac{2}{\varepsilon_n} \cdot h_{\varepsilon_n}(x_n) \rightarrow \xi$ as $n \rightarrow \infty$. By 9.7, $h_{\varepsilon_n}(x_n) \in \partial T_p$. By 9.8, $\xi \in \partial T_p$. Therefore,

$$\partial(\Sigma_p \mathcal{A}) \supset \Sigma_p(\partial\mathcal{A}) \quad \text{and} \quad \partial(T_p \mathcal{A}) \supset T_p(\partial\mathcal{A}).$$

Similarly, choose $\xi \in \partial\Sigma_p$. Let $h_{\varepsilon_n} : B(p, \varepsilon_n)_{\mathcal{A}} \rightarrow B(0, \varepsilon_n)_{T_p}$ be the homeomorphisms provided by 9.2 for a sequence $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$. By 9.7, $x_n = h_{\varepsilon_n}^{-1}(\frac{\varepsilon_n}{2} \cdot \xi) \in \partial\mathcal{A}$. By 9.2, $\uparrow_{[px_n]} \rightarrow \xi$. Hence

$$\partial(\Sigma_p\mathcal{A}) \subset \Sigma_p(\partial\mathcal{A}) \quad \text{and} \quad \partial(T_p\mathcal{A}) \subset T_p(\partial\mathcal{A}).$$

□

E Doubling

Let A be a closed subset in a metric space $\mathcal{X}$. The doubling $\mathcal{W}$ of $\mathcal{X}$ across A is two copies of $\mathcal{X}$ glued along A . More precisely, the underlying set of $\mathcal{W}$ is the quotient $\mathcal{X} \times \{0, 1\} / \sim$, where $(a, 0) \sim (a, 1)$ for any $a \in A$ and $\mathcal{W}$ is equipped with the minimal metric such that both maps $\mathcal{X} \rightarrow \mathcal{W}$ defined by $x \mapsto (x, 0)$ and $x \mapsto (x, 1)$ are distance-preserving.

Alternatively, one may say that $\mathcal{W}$ is equipped with the maximal metric such that the projection $\text{proj} : \mathcal{W} \rightarrow \mathcal{X}$ defined by $(x, i) \mapsto x$ is a short map. The metric on $\mathcal{W}$ can also be defined directly:

$$|(x, i) - (y, j)|_{\mathcal{W}} = \begin{cases} |x - y|_{\mathcal{X}} & \text{if } i = j. \\ \inf \{ |x - a|_{\mathcal{X}} + |y - a|_{\mathcal{X}} : a \in A \} & \text{if } i \neq j. \end{cases}$$

9.10. Theorem. *Let $\mathcal{A}$ be a finite-dimensional Alexandrov space with nonempty boundary, and let $f = \frac{1}{2} \cdot \text{dist}_p^2$ for some $p \in \mathcal{A}$. Then*

- (a) *If $\dim \mathcal{A} \geq 2$, then $\text{dist}_{\partial\Sigma_x} \xi \leq \frac{\pi}{2}$ for any $x \in \partial\mathcal{A}$ and $\xi \in \Sigma_x$.
Moreover, if $\text{dist}_{\partial\Sigma_x} \xi = \frac{\pi}{2}$, then $\angle(\xi, \zeta) = \frac{\pi}{2}$ for any $\zeta \in \partial\Sigma_x$.*
- (b) *$\nabla_x f \in \partial T_x$ for any $x \in \partial\mathcal{A}$.*
- (c) *If α is an f -gradient curve that starts at $x \in \partial\mathcal{A}$, then $\alpha(t) \in \partial\mathcal{A}$ for any t . Moreover, if $p \in \partial\mathcal{A}$, then $\text{gexp}_p(v) \in \partial\mathcal{A}$ for any $v \in \partial T_p$.*
- (d) *The doubling $\mathcal{W}$ of $\mathcal{A}$ across $\partial\mathcal{A}$ is an Alexandrov space with the same curvature bound as $\mathcal{A}$.*

Part (d) is called the doubling theorem.

Proof. We will denote by $(a)_m, \dots, (d)_m$ the corresponding statement assuming $m = \dim \mathcal{A}$.

The proof goes by induction on m . Statement $(d)_1$ follows from 7.3 — this is the base. The induction step is a combination of the implications below.

$(d)_{m-1} \Rightarrow (a)_m$. Suppose $m = 2$, then $\dim \Sigma_x = 1$; see 7.20b. By 7.3, Σ_x is isometric to a line segment $[0, \ell]$; we need to show that $\ell \leq \pi$.

Recall that the tangent space $T_x = \text{Cone } \Sigma_x$ is an Alexandrov space; see 7.20. If $\ell > \pi$, then T_x has bifurcating geodesics, which is impossible by 2.11.

Now suppose $m > 2$, so $\dim \Sigma_x > 1$. Assume $\text{dist}_{\partial \Sigma_x} \xi > \frac{\pi}{2}$ for some ξ . By $(d)_{m-1}$, the doubling Ξ of Σ_x is ALEX(1). By 6.8, $\text{Cone } \Xi$ is ALEX(0). Denote by ξ_0 and ξ_1 the points in Ξ that correspond to ξ . Observe that $|\xi_0 - \xi_1|_{\Xi} > \pi$. The latter contradicts 3.6.

Finally, if $\text{dist}_{\partial \Sigma_x} \xi = \frac{\pi}{2}$, then $|\xi_0 - \xi_1|_{\Xi} = \pi$. Therefore, $\text{Cone } \Xi$ contains a line in the directions of ξ_0 and ξ_1 . By the splitting theorem (6.5), $\text{Cone } \Xi$ is a direct product of the line with some subcone; in other words, Ξ is a spherical suspension with poles ξ_0 and ξ_1 . In particular, every point of Ξ lies on distance at most $\frac{\pi}{2}$ from ξ_0 or ξ_1 . The natural projection $\Xi \rightarrow \Sigma_x$ does not increase distances and sends both ξ_0 and ξ_1 to ξ . Therefore, the second statement of $(a)_m$ follows.

$(a)_{m-1} + (a)_m \Rightarrow (b)_m$. We can assume that $s = |\nabla_x f| \neq 0$. By 4.9, $\nabla_x f = s \cdot \bar{\xi}$, where $s = \mathbf{d}_x f(\bar{\xi}) > 0$ and $\bar{\xi} \in \Sigma_x$ is the direction that maximizes $\mathbf{d}_x f(\bar{\xi})$.

Let $\zeta \in \partial \Sigma_x$ be a direction that minimizes the angle $\angle(\bar{\xi}, \zeta)$. It is sufficient to show that $\zeta = \bar{\xi}$.

Assume $\zeta \neq \bar{\xi}$; let $\eta = \uparrow_{[\zeta \bar{\xi}]_{\Sigma_x}}$. By $(a)_m$, $\angle(\bar{\xi}, \zeta) \leq \frac{\pi}{2}$; further, $(a)_{m-1}$ implies that

$$\textcircled{1} \quad \angle(\eta, \nu) \leq \frac{\pi}{2}$$

for any $\nu \in \Sigma_{\zeta} \Sigma_x$ (if $m = 2$, then the last statement is evident).

Let $\varphi: \Sigma_x \rightarrow \mathbb{R}$ be the restriction of $\mathbf{d}_x f$ to Σ_x . Applying 4.6b and $\textcircled{1}$, we get that $(\mathbf{d}_{\zeta} \varphi)(\eta) \leq 0$. Recall that $\mathbf{d}_x f$ is concave (7.7), so $(\mathbf{d}_x f)'' \leq 0$; by straightforward computations, we have that $\varphi'' + \varphi \leq 0$. If $\varphi(\zeta) \leq 0$, then it implies that $\varphi(\bar{\xi}) \leq 0$ — a contradiction to the fact that $s > 0$. If $\varphi(\zeta) > 0$, then $\varphi(\bar{\xi}) < \varphi(\zeta)$ — a contradiction again.

$(b)_m \Rightarrow (c)_m$. Let α be an f -gradient curve and $\ell(t) = \text{dist}_{\partial \mathcal{A}} \alpha(t)$.

Choose t ; let $x = \alpha(t)$ and $y \in \partial \mathcal{A}$ be a closest point to x . By $(b)_m$, we have that $\nabla_y f \in \partial T_y$. Since the distance $|x - y|$ is minimal, we get $\langle \uparrow_{[yx]}, v \rangle \leq 0$ for any $v \in \partial T_y$. In particular,

$$\langle \uparrow_{[yx]}, \nabla_y f \rangle \leq 0.$$

Applying 4.8 to x and y , we get

$$\ell'(t) \leq \ell(t)$$

if the left-hand side is defined. Since ℓ is Lipschitz, ℓ' is defined almost everywhere. Integrating the inequality, we get

$$\ell(t) \leq e^t \cdot \ell(0)$$

for any $t \geq 0$. In particular, if $\ell(0) = 0$, then $\ell(t) = 0$ for any $t \geq 0$. Since $\partial\mathcal{A}$ is closed (9.8), the first statement follows.

It remains to show that $\text{gexp}_p(v) \in \partial\mathcal{A}$ for any $v \in \partial T_p$. Suppose $v = t \cdot u$ and $|u| = 1$. Note that $u \in \partial T_p$. Therefore, we can choose a sequence of points $x_n \in \partial\mathcal{A}$ such that $\uparrow_{[px_n]} \rightarrow u$; let $r_n = |p - x_n|$. From above, the f -gradient curve α_n that starts at x_n remains in $\partial\mathcal{A}$. By the definition of gradient exponent, $\text{gexp}_p(v) = \text{gexp}_p(t \cdot u)$ is a limit of $\alpha_n(\ln \frac{t}{r_n})$. Hence the last statement follows.

$(c)_m + (d)_{m-1} \Rightarrow (d)_m$. We will consider the case $\kappa = 0$; other cases can be done in the same way, but formulas get more complicated.

Denote by $\mathcal{A}_0$ and $\mathcal{A}_1$ the two copies of $\mathcal{A}$ in $\mathcal{W}$; let us keep the notation $\partial\mathcal{A}$ for the common boundary of $\mathcal{A}_0$ and $\mathcal{A}_1$.

❷ *A geodesic in $\mathcal{W}$ has at most one interior point in $\partial\mathcal{A}$ or it lies in $\partial\mathcal{A}$ completely.*

Indeed, assume γ shares at least two points with $\partial\mathcal{A}$, say $x = \gamma(t_1)$ and $y = \gamma(t_2)$ and these are not endpoints of γ . Remove from γ the set $\gamma \cap \mathcal{A}_1$ and replace it by its reflection across $\partial\mathcal{A}$; denote the obtained curve by $\hat{\gamma}$.



Any arc of $\hat{\gamma}$ with one endpoint in $\partial\mathcal{A}$ is a geodesic in $\mathcal{A}_0$. Since $x, y \in \partial\mathcal{A}$, the arc of $\hat{\gamma}$ beyond y lies in the image of map $t \mapsto \text{Flow}_{f_x}^t(y)$, where $f_x = \frac{1}{2} \cdot \text{dist}_x^2$. By (c), this arc lies in $\partial\mathcal{A}$.

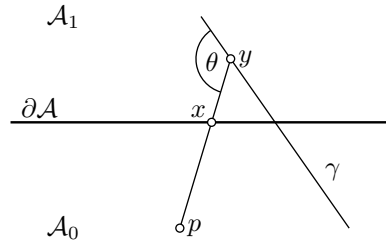
Now choose a point z on this arc, so $z \in \partial\mathcal{A}$. Applying the same argument, we get that the arc of $\hat{\gamma}$ up to y lies in $\partial\mathcal{A}$. Hence the claim follows. $\triangle$

Choose a point p in $\mathcal{W}$; let $f := \frac{1}{2} \cdot \text{dist}_p^2$. It is sufficient to show that $(f \circ \gamma)'' \leq 1$. If $p \in \partial\mathcal{A}$, then the statement follows from function comparison in $\mathcal{A}_0$ and $\mathcal{A}_1$. So, we can assume that $p \in \mathcal{A}_0 \setminus \partial\mathcal{A}$. Also, we can assume that γ does not lie in $\partial\mathcal{A}$; otherwise, the inequality again follows from the comparison in $\mathcal{A}_0$.

Choose $y = \gamma(t_0)$; without loss of generality we can assume that $t_0 = 0$.

Assume $y \in \mathcal{A}_0 \setminus \partial\mathcal{A}$. Then $(f \circ \gamma)''(0) \leq 1$ in the barrier sense; it follows from the comparison in $\mathcal{A}_0$.

Assume $y \in \mathcal{A}_1 \setminus \partial\mathcal{A}$. Suppose $[py]$ crosses $\partial\mathcal{A}$ at x ; by ❷, this point is unique. Let Σ_x be the space of directions of $\mathcal{A}$ at x ,



and let Ξ be its doubling. As before, we denote by Σ_0 and Σ_1 the two copies of Σ_x in Ξ and keep the notation $\partial\Sigma_x$ for their common boundary. By $(d)_{m-1}$, Ξ is ALEX(1).

The directions $\uparrow_{[xy]}$ and $\uparrow_{[xp]}$ lie on opposite sides of $\partial\Sigma_x$ and

$$|\uparrow_{[xy]} - \uparrow_{[xp]}|_{\Xi} \geq \pi.$$

Otherwise, we could choose a direction $\xi \in \partial\Sigma_x$ such that

$$|\uparrow_{[xy]} - \xi|_{\Xi} + |\xi - \uparrow_{[xp]}|_{\Xi} < \pi.$$

Furthermore, we could consider the curve $\alpha(t) = \text{gexp}_x(t \cdot \xi)$. By $(c)_m$, α lies in $\partial\mathcal{A}$. By 5.12

$$|p - \alpha(s)|_{\mathcal{A}_0} + |y - \alpha(s)|_{\mathcal{A}_1} < |p - y|_{\mathcal{W}}$$

for small values $s > 0$ — a contradiction.

Cone Ξ contains a line with directions $\uparrow_{[xy]}$ and $\uparrow_{[xp]}$. By the splitting theorem, Cone Ξ splits in these directions; in particular,

$$|\uparrow_{[xy]} - \xi| + |\xi - \uparrow_{[xp]}| = \pi$$

for any $\xi \in \Xi$. It follows that for any $\xi \in \Xi$ there is $\xi' \in \partial\Sigma_x$ such that ξ and ξ' lie on some geodesic $[\uparrow_{[xy]} \uparrow_{[xp]}]_{\Xi}$.

Fix $t \approx 0$ such that $t \neq 0$. Let $z = \gamma(t)$ and $\xi = \uparrow_{[xz]}$; choose a direction $\xi' \in \partial\Sigma_x$ as above for ξ . Consider the curve $\alpha(s) := \text{gexp}_x(s \cdot \xi')$. Let us show that

$$\begin{aligned} \textcircled{3} \quad |p - z|_{\mathcal{W}} &\leq |p - \alpha(s)|_{\mathcal{A}_0} + |\alpha(s) - z|_{\mathcal{A}_1} \leq \\ &\leq \tilde{\gamma}[y \overset{p}{z}]. \end{aligned}$$

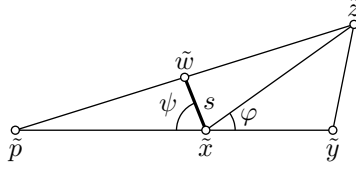
for a suitable value s .

The first inequality in $\textcircled{3}$ is evident. Set $\varphi = \angle[x \overset{y}{z}]$ and $\psi = \angle[\uparrow_{[xp]}, \xi']$. The choice of s comes from the model configuration $\tilde{p}$, $\tilde{x}$, $\tilde{y}$, $\tilde{w}$, $\tilde{z} \in \mathbb{E}^2$ such that

$$\begin{aligned} \tilde{x} \in [\tilde{p}\tilde{y}], \quad |\tilde{p} - \tilde{x}| &= |p - x|, \quad |\tilde{p} - \tilde{y}| = |p - y|, \quad |\tilde{x} - \tilde{z}| = |x - z|, \\ \tilde{w} \in [\tilde{p}\tilde{z}], \quad \angle[\tilde{x} \overset{\tilde{y}}{\tilde{z}}] &= \varphi, \quad \angle[\tilde{x} \overset{\tilde{p}}{\tilde{w}}] = \psi, \quad s = |\tilde{x} - \tilde{w}|. \end{aligned}$$

By 5.12, we get

$$\begin{aligned} |p - \alpha(s)|_{\mathcal{A}_0} &\leq |\tilde{p} - \tilde{w}|, \\ |\alpha(s) - z|_{\mathcal{A}_1} &\leq |\tilde{w} - \tilde{z}|; \end{aligned}$$



by the comparison,

$$|\tilde{p} - \tilde{z}| \leq \tilde{\gamma}[y_z^p].$$

9.11. Exercise. *Prove the last inequality.*

Hence we get $(f \circ \gamma)''(0) \leq 1$ in the barrier sense.

Finally, if $\gamma(0) \in \partial\mathcal{A}$, then a splitting argument shows that

$$(f \circ \gamma)^+(0) - (f \circ \gamma)^-(0) \leq 0.$$

Summarizing, we get that $(f \circ \gamma)'' \leq 1$ on every arc of γ that lies entirely in $\mathcal{A}_0$ or $\mathcal{A}_1$. If γ crosses $\partial\mathcal{A}$, then we know that it happens only once and at the crossing moment t_0 we have $(f \circ \gamma)^+(t_0) - (f \circ \gamma)^-(t_0) \leq 0$. All this implies that $(f \circ \gamma)'' \leq 1$. $\square$

9.12. Exercise! *Let $\mathcal{A}$ be an m -dimensional ALEX(1) space. Assume that $\partial\mathcal{A} \neq \emptyset$ and $2 \leq m < \infty$. Show that $\partial\mathcal{A}$ is connected.*

9.13. Exercise. *Let $\mathcal{A}$ be a finite-dimensional ALEX(0) space with nonempty boundary $\partial\mathcal{A}$.*

- (a) *Suppose a geodesic γ in $\mathcal{A}$ has its interior point in $\partial\mathcal{A}$. Show that γ lies in $\partial\mathcal{A}$.*
- (b) *Show that the distance function to the boundary is concave.*

9.14. Exercise! *Let $\mathcal{A}$ be a finite-dimensional ALEX(0) space with nonempty boundary $\partial\mathcal{A}$. Suppose γ is a geodesic in $\partial\mathcal{A}$ with the induced length metric. Show that the function $t \mapsto \frac{1}{2} \cdot \text{dist}_p^2 \circ \gamma(t)$ is 1-concave for any point p .*

9.15. Exercise. *Let $\text{proj}: \mathcal{W} \rightarrow \mathcal{A}$ be the natural projection to a finite-dimensional Alexandrov space $\mathcal{A}$ from its doubling $\mathcal{W}$ across the boundary. Suppose $f: \mathcal{A} \rightarrow \mathbb{R}$ is a λ -concave function. Show that $f \circ \text{proj}: \mathcal{W} \rightarrow \mathbb{R}$ is λ -concave if and only if $\nabla_x f \in \partial T_x$ for any $x \in \partial\mathcal{A}$.*

F Remarks

The doubling theorem was generalized by the third author [97] allowing gluing nonidentical spaces and generalized further by Jian Ge and Nan Li [33] allowing for gluing more than two spaces.

It easily follows by induction on dimension that the doubling of a finite-dimensional Alexandrov space across its boundary results in an Alexandrov space without boundary. This observation often helps to reduce a statement about general finite-dimensional Alexandrov spaces to Alexandrov spaces without boundary.

Let us discuss several additional tools that are available for spaces with empty boundary.

9.16. Fundamental-class lemma. *Any compact finite-dimensional Alexandrov space $\mathcal{A}$ without boundary has a fundamental class with $\mathbb{Z}/2$ coefficients; that is, if $\mathcal{A}$ is m -dimensional, then*

$$H^m(\mathcal{A}, \mathbb{Z}/2) = \mathbb{Z}/2.$$

This lemma was proved by Karsten Grove and Peter Petersen [43]. Originally it was stated for Alexander–Spanier cohomology. We do not make this distinction because for compact Alexandrov spaces it is the same as singular cohomology. Indeed, both cohomology theories are homotopy invariant [115, Chapter 6], compact Alexandrov spaces are homotopy equivalent to finite simplicial complexes (see 8.17), and for paracompact CW complexes, Alexander–Spanier cohomology is isomorphic to Čech and singular cohomology [115, Chapter 6].

The lemma implies, for example, that on finite-dimensional Alexandrov spaces without boundary the gradient flow for a λ -concave function is an onto map; in other words, gradient curves can be extended into the past. It is also used in the proof of the following version of the domain invariance theorem [56, Theorem 3.2].

9.17. Domain invariance. *Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be two m -dimensional Alexandrov spaces with empty boundary; m is finite. Suppose Ω_1 is an open subset in $\mathcal{A}_1$ and $f: \Omega_1 \rightarrow \mathcal{A}_2$ is an injective continuous map. Then $f(\Omega_1)$ is open in $\mathcal{A}_2$.*

Theorem 9.2 can be used to prove the following.

9.18. Topological stratification. *Any m -dimensional Alexandrov space with $m < \infty$ can be subdivided into topological manifolds $S_0, \dots, S_m$ such that for every i we have $\dim S_i = i$ or $S_i = \emptyset$. Moreover,*

- (a) *the closure of S_{m-1} is the boundary of the space, and*
- (b) *$S_{m-2} = \emptyset$.*

This statement implies that a compact finite-dimensional Alexandrov space has the homotopy type of a finite CW complex, but it seems to be unknown if it has to be homeomorphic to a CW complex.

The stratification theorem 9.18 can be sharpened as follows.

9.19. Boundary characterization. *Let $\mathcal{A}$ be an m -dimensional Alexandrov space with $m < \infty$. Then the following statements are equivalent.*

- (a) $p \in \partial\mathcal{A}$;
- (b) Σ_p is contractible;
- (c) $\tilde{H}_{m-1}(\Sigma_p, \mathbb{Z}/2) = 0$;
- (d) $H_m(\mathcal{A}, \mathcal{A} \setminus \{p\}, \mathbb{Z}/2) = 0$.

Let f be a semiconcave function. A point $p \in \text{Dom } f$ is called a critical point of f if $d_p f \leq 0$; otherwise it is called regular.

The following statement plays a role in the proof of the stability theorem, but it is also a useful technical tool on its own.

9.20. Morse lemma. *Let f be a semiconcave function on a finite-dimensional Alexandrov space without boundary. Suppose K is a compact set of regular points of f in its level set $f = a$. Then an open neighborhood Ω of K admits a homeomorphism $x \mapsto (h(x), f(x))$ to a product space $\Lambda \times (a - \varepsilon, a + \varepsilon)$.*

The Morse lemma was originally proved by Grigory Perelman [88] for a special kind of semiconcave functions made out of distance functions. For such functions it holds for all Alexandrov spaces (possibly with nonempty boundary). For general semiconcave functions on spaces without boundary it follows from [89] where the technical tools from [88] (the notion of inner product between differentials of semiconcave functions and the notion of admissible maps) were generalized to include general semiconcave functions. With those definitions all the main Morse theory results of [88] follow by the same proof by using [89, 2.2] instead of [88, Lemma 1]. On Alexandrov spaces with boundary all the Morse theory results hold if one requires in addition that the semiconcave functions remain semiconcave when canonically extended to the doubling.¹ This is automatic for the special type of semiconcave functions considered in [88]; therefore this distinction is not made there.

Subsets in Alexandrov spaces that satisfy the condition in 9.10c are called extremal. More precisely, a subset E is extremal if any f -gradient curve that starts in E remains in E ; here f is an arbitrary function of the form $\frac{1}{2} \cdot \text{dist}_p^2$.

¹Semiconcave functions are often defined to meet this property, but we do not follow this convention.

Extremal subsets were introduced by Grigory Perelman and the third author [85]. They will pop up in the next lecture.

The following conjecture is one of the oldest questions in Alexandrov geometry that remains open.

9.21. Conjecture. *Let S be a component of the boundary of a finite-dimensional Alexandrov space. Then S equipped with the induced length metric is an Alexandrov space with the same curvature bound.*

Lecture 10

Quotients

This lecture discusses applications of Alexandrov geometry to isometric group actions.

A Quotient space

Suppose that a group G acts isometrically on a metric space $\mathcal{X}$. Note that

$$|G \cdot x - G \cdot y|_{\mathcal{X}/G} := \inf \{ |x - g \cdot y|_{\mathcal{X}} : g \in G \}$$

defines a semimetric on the orbit space $\mathcal{X}/G$. Moreover, if the orbits of the action are closed, then it is a genuine metric.

10.1. Theorem. *Suppose that a group G acts isometrically on a proper ALEX(0) space $\mathcal{A}$, and G has closed orbits. Then the quotient space $\mathcal{A}/G$ is ALEX(0).*

A more general statement will be given in 10.5.

Proof. Denote by $\sigma: \mathcal{A} \rightarrow \mathcal{A}/G$ the quotient map.

Fix a quadruple of points $p, x_1, x_2, x_3 \in \mathcal{A}/G$. Choose $\hat{p} \in \mathcal{A}$ such that $\sigma(\hat{p}) = p$. Since $\mathcal{A}$ is proper and the G -orbits are closed, we can choose points $\hat{x}_i \in \mathcal{A}$ such that $\sigma(\hat{x}_i) = x_i$ and

$$|p - x_i|_{\mathcal{A}/G} = |\hat{p} - \hat{x}_i|_{\mathcal{A}}$$

for all i .

Note that

$$|x_i - x_j|_{\mathcal{A}/G} \leq |\hat{x}_i - \hat{x}_j|_{\mathcal{A}}$$

for all i and j . By 2.6,

$$\textcircled{1} \quad \tilde{\mathcal{L}}(p_{x_j}^{x_i}) \leq \tilde{\mathcal{L}}(\hat{p}_{\hat{x}_j}^{\hat{x}_i})$$

for all i and j .

By the $\mathbb{E}^2$ -comparison in $\mathcal{A}$, we have

$$\tilde{\mathcal{L}}(\hat{p}_{\hat{x}_2}) + \tilde{\mathcal{L}}(\hat{p}_{\hat{x}_3}) + \tilde{\mathcal{L}}(\hat{p}_{\hat{x}_1}) \leq 2 \cdot \pi.$$

Applying **1**, we get

$$\tilde{\mathcal{L}}(p_{x_2}) + \tilde{\mathcal{L}}(p_{x_3}) + \tilde{\mathcal{L}}(p_{x_1}) \leq 2 \cdot \pi;$$

that is, the $\mathbb{E}^2$ -comparison holds for any quadruple in $\mathcal{A}/G$.

Since $\mathcal{A}$ is proper and geodesic and the orbits are closed, we get that $\mathcal{A}/G$ is geodesic, and the statement follows. $\square$

10.2. Very advanced exercise. *Let G be a compact Lie group with a bi-invariant Riemannian metric. Show that G is isometric to a quotient of a Hilbert space by an isometric group action.*

Conclude that G is ALEX(0).

B Submetries

A map $\sigma: \mathcal{X} \rightarrow \mathcal{Y}$ between metric spaces is called a submetry if

$$\sigma(B(p, r)_{\mathcal{X}}) = B(\sigma(p), r)_{\mathcal{Y}}$$

for any $p \in \mathcal{X}$ and $r \geq 0$.

Suppose G and $\mathcal{A}$ are as in 10.1. Observe that the quotient map $\sigma: \mathcal{A} \rightarrow \mathcal{A}/G$ is a submetry. The following two exercises show that this is not the only source of submetries.

10.3. Exercise. *Construct submetries*

(a) $\sigma_1: \mathbb{S}^2 \rightarrow [0, \pi],$

(b) $\sigma_2: \mathbb{S}^2 \rightarrow [0, \frac{\pi}{2}],$

(c) $\sigma_n: \mathbb{S}^2 \rightarrow [0, \frac{\pi}{n}]$ (for integer $n \geq 1$)

such that the fibers $\sigma_n^{-1}\{x\}$ are connected for any x .

10.4. Exercise. *Let $\sigma: \mathbb{E}^2 \rightarrow [0, \infty)$ be a submetry. Show that $K = \sigma^{-1}\{0\}$ is a closed convex set without interior points and $\sigma(x) = \text{dist}_K x$.*

The proof of 10.1 works for submetries; that is, if $\sigma: \mathcal{A} \rightarrow \mathcal{B}$ is a submetry and $\mathcal{A}$ is a proper ALEX(0) space, then so is $\mathcal{B}$. Theorem 10.1 admits a straightforward generalization to the ALEX(−1) case.

In the ALEX(1) case, the proof produces a weaker statement: if $\sigma: \mathcal{A} \rightarrow \mathcal{B}$ is a submetry and $\mathcal{A}$ is ALEX(1) then $\mathbb{S}^2$ -comparison holds for any quadruple p, x_1, x_2, x_3 in $\mathcal{B}$ if $|p - x_i| < \frac{\pi}{2}$ for each i . In

particular, $\mathcal{B}$ is *locally* ALEX(1). As above, assuming orbits are closed, the quotient of a proper geodesic space is geodesic. Therefore, the globalization theorem implies that it is globally ALEX(1). The same holds for the targets of submetries from an ALEX(1) space. With a bit of extra work, one can extend the statement to nonproper spaces [7, 8.34]. Thus, we have the following.

10.5. Theorem. *Let $\sigma: \mathcal{A} \rightarrow \mathcal{B}$ be a submetry. If $\mathcal{A}$ is ALEX(κ), then so is $\mathcal{B}$.*

In particular, if G acts isometrically on an ALEX(κ) space $\mathcal{A}$, and G has closed orbits, then the quotient space $\mathcal{A}/G$ is ALEX(κ).

C Hopf's conjecture

Does $\mathbb{S}^2 \times \mathbb{S}^2$ admit a Riemannian metric with positive sectional curvature? Hopf's conjecture says that the answer should be negative. Let us take a close look at the following partial result obtained by Wu-Yi Hsiang and Bruce Kleiner [49].

10.6. Theorem. *There is no Riemannian metric on $\mathbb{S}^2 \times \mathbb{S}^2$ with sectional curvature ≥ 1 and a nontrivial isometric $\mathbb{S}^1$ -action.*

Recall that a group action $G \curvearrowright \mathcal{X}$ is called effective if for any $g \in G$ there is $x \in \mathcal{X}$ such that $g \cdot x \neq x$.

10.7. Key lemma. *Suppose $\mathbb{S}^1 \curvearrowright \mathbb{S}^3$ is an effective isometric action without fixed points and $\Sigma = \mathbb{S}^3/\mathbb{S}^1$ is its quotient space. Then there is a distance noncontracting map $\Sigma \rightarrow \frac{1}{2} \cdot \mathbb{S}^2$, where $\frac{1}{2} \cdot \mathbb{S}^2$ denotes the standard 2-sphere rescaled with a factor $\frac{1}{2}$.*

The proof of the lemma is guided by the following exercise.

10.8. Exercise! *Suppose $\mathbb{S}^1 \curvearrowright \mathbb{S}^3$ is an effective isometric action without fixed points. Let us think of $\mathbb{S}^3$ as the unit sphere in $\mathbb{R}^4$.*

(a) *Show that one can identify $\mathbb{R}^4$ with $\mathbb{C}^2$ so that the action is given by matrix multiplication*

$$\begin{pmatrix} u^p & 0 \\ 0 & u^q \end{pmatrix},$$

where (p, q) is a pair of relatively prime positive integers and $u \in \mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$. In particular, our $\mathbb{S}^1$ is a subgroup of the torus that acts by matrix multiplication

$$\begin{pmatrix} v & 0 \\ 0 & w \end{pmatrix},$$

where $v, w \in \mathbb{S}^1$.

Fix p and q as above. Let $\Sigma_{p,q} = \mathbb{S}^3/\mathbb{S}^1$ be the quotient space.

- (b) Show that $\Sigma_{p,q} = \mathbb{S}^3/\mathbb{S}^1$ is a topological sphere with $\mathbb{S}^1$ -symmetry. This symmetry has two fixed points, say the north pole and the south pole, that correspond to the orbits of $(1, 0)$ and $(0, 1)$ in $\mathbb{S}^3$.

Denote by $S(r)$ the circle of radius r with the center at the north pole of $\Sigma_{p,q}$.

- (c) Denote by $T(r)$ the inverse image of $S(r)$ in $\mathbb{S}^3$, and let $a(r)$ be its area. Show that $T(r)$ is an orbit of the torus action and

$$a(r) = 4 \cdot \pi^2 \cdot \sin r \cdot \cos r.$$

- (d) Let $b_{p,q}(r)$ be the length of the $\mathbb{S}^1$ -orbit in $\mathbb{S}^3$ that corresponds to a point on $S(r)$. Show that

$$b_{p,q}(r) = 2 \cdot \pi \cdot \sqrt{(p \cdot \sin r)^2 + (q \cdot \cos r)^2}.$$

- (e) Let $c_{p,q}(r)$ be the length of $S(r)$. Show that

$$a(r) = c_{p,q}(r) \cdot b_{p,q}(r).$$

- (f) Show that $c_{p,q}(r) \leq c_{1,1}(r)$ for any pair (p, q) of relatively prime positive integers. Use it to construct a distance noncontracting map $\Sigma_{p,q} \rightarrow \frac{1}{2} \cdot \mathbb{S}^2 \stackrel{\text{iso}}{=} \Sigma_{1,1}$.

Proof of 10.6. Assume $\mathcal{B} = (\mathbb{S}^2 \times \mathbb{S}^2, g)$ is a counterexample. By the Toponogov theorem, $\mathcal{B}$ is ALEX(1). By 10.1, the quotient space $\mathcal{A} = \mathcal{B}/\mathbb{S}^1$ is ALEX(1); evidently, $\mathcal{A}$ is 3-dimensional.

Denote by $F \subset \mathcal{B}$ the fixed point set of the $\mathbb{S}^1$ -action. Then $\chi(\mathcal{B}) = \chi(F)$. Each connected component of F is either an isolated point or a 2-dimensional geodesic submanifold in $\mathcal{B}$; the latter has to have positive curvature, and therefore it is homeomorphic to $\mathbb{S}^2$ or $\mathbb{RP}^2$. Notice that

- ◇ each isolated point contributes 1 to the Euler characteristic of $\mathcal{B}$,
- ◇ each sphere contributes 2 to the Euler characteristic of $\mathcal{B}$, and
- ◇ each projective plane contributes 1 to the Euler characteristic of $\mathcal{B}$.

Since $\chi(\mathcal{B}) = 4$, we are in one of the following three cases:

1. F has exactly 4 isolated points,
2. F has one 2-dimensional submanifold and at least 2 isolated points,
3. F has at least two 2-dimensional submanifolds.

In each case we will arrive at a contradiction.

Case 1. Suppose F has exactly 4 isolated points x_1, x_2, x_3 , and x_4 . Denote by y_1, y_2, y_3 , and y_4 the corresponding points in $\mathcal{A}$. Note that $\Sigma_{y_i}\mathcal{A}$ is isometric to a quotient of $\mathbb{S}^3$ by an isometric $\mathbb{S}^1$ -action without fixed points.

By 10.7, each angle $\angle[y_i^{y_j} y_k] \leq \frac{\pi}{2}$ for any three distinct points y_i, y_j, y_k . In particular, all four triangles $[y_1 y_2 y_3]$, $[y_1 y_2 y_4]$, $[y_1 y_3 y_4]$, and $[y_2 y_3 y_4]$ are nondegenerate. By the comparison, the sum of angles in each triangle is strictly greater than π .

Denote by ω the sum of all 12 angles in the 4 triangles $[y_1 y_2 y_3]$, $[y_1 y_2 y_4]$, $[y_1 y_3 y_4]$, and $[y_2 y_3 y_4]$. From above,

$$\omega > 4 \cdot \pi.$$

On the other hand, by 10.7 any triangle in $\Sigma_{y_1}\mathcal{A}$ has perimeter at most π . In particular,

$$\angle[y_1^{y_2} y_3] + \angle[y_1^{y_3} y_4] + \angle[y_1^{y_4} y_2] \leq \pi.$$

Apply the same argument in $\Sigma_{y_2}\mathcal{A}$, $\Sigma_{y_3}\mathcal{A}$, and $\Sigma_{y_4}\mathcal{A}$; adding the results, we get

$$\omega \leq 4 \cdot \pi$$

— a contradiction.

Case 2. Suppose F contains one surface S . Then the projection of S to $\mathcal{A}$ forms its boundary $\partial\mathcal{A}$. The doubling $\mathcal{W}$ of $\mathcal{A}$ across its boundary has at least 4 singular points — each singular point of $\mathcal{A}$ corresponds to two singular points of $\mathcal{W}$.

By the doubling theorem, $\mathcal{W}$ is an ALEX(1) space. Therefore we arrive at a contradiction in the same way as in the first case.

Case 3. Impossible by 9.12. □

D Erdős' problem revisited

A point p in an Alexandrov space is called extremal if $\angle[p_y^x] \leq \frac{\pi}{2}$ for any hinge $[p_y^x]$ with the vertex at p ; equivalently, $\text{diam } \Sigma_p \leq \pi/2$.

10.9. Theorem. *Let $\mathcal{A}$ be a compact m -dimensional ALEX(0) space. Then it has at most 2^m extremal points.*

Proof of 10.9. Let $\{p_1, \dots, p_N\}$ be extremal points in $\mathcal{A}$. For each p_i consider its open Voronoi domain V_i ; that is,

$$V_i = \{x \in \mathcal{A} : |p_i - x| < |p_j - x| \text{ for any } j \neq i\}.$$

Clearly $V_i \cap V_j = \emptyset$ if $i \neq j$.

Choose $0 < \alpha \leq 1$. Given a point $x \in \mathcal{A}$, choose a geodesic $[p_i x]$ and denote by x_i the point on $[p_i x]$ such that $|p_i - x_i| = \alpha \cdot |p_i - x|$; let $\Phi_i: x \mapsto x_i$ be the corresponding map. By the comparison,

$$|x_i - y_i| \geq \alpha \cdot |x - y|$$

for any x, y , and i . Therefore

$$\text{vol}(\Phi_i \mathcal{A}) \geq \alpha^m \cdot \text{vol } \mathcal{A}.$$

Now, suppose $\alpha < \frac{1}{2}$. Then $x_i \in V_i$ for any $x \in \mathcal{A}$. Indeed, assume $x_i \notin V_i$; then there is p_j such that $|p_i - x_i| \geq |p_j - x_i|$. By comparison, we have $\tilde{Z}(p_j \frac{p_i}{x})_{\mathbb{E}^2} > \frac{\pi}{2}$; that is, p_j is not an extremal point.

It follows that $\text{vol } V_i \geq \alpha^m \cdot \text{vol } \mathcal{A}$ for any $0 < \alpha < \frac{1}{2}$; hence

$$\text{vol } V_i \geq \frac{1}{2^m} \cdot \text{vol } \mathcal{A}.$$

Since $V_1, \dots, V_N$ are disjoint subsets of $\mathcal{A}$, we have $N \leq 2^m$. $\square$

10.10. Exercise. Find a bound on the number of extremal points in an $\text{ALEX}(-1)$ space in terms of its dimension m and diameter D .

E Crystallographic actions

An isometric action $\Gamma \curvearrowright \mathbb{E}^m$ is called crystallographic if it is properly discontinuous (that is, for any compact set $K \subset \mathbb{E}^m$ and $x \in \mathbb{E}^m$ there are only finitely many elements $g \in \Gamma$ such that $g \cdot x \in K$) and cocompact (that is, the quotient space $\mathcal{A} = \mathbb{E}^m / \Gamma$ is compact).

Let F be a maximal finite subgroup of Γ ; that is, if $F < H < \Gamma$ for a finite group H , then $F = H$. Denote by $\mathfrak{M}(\Gamma)$ the number of maximal finite subgroups of Γ up to conjugation.

10.11. Open question. Let $\Gamma \curvearrowright \mathbb{E}^m$ be a crystallographic action. Is it true that $\mathfrak{M}(\Gamma) \leq 2^m$?

Note that any finite subgroup F of Γ fixes an affine subspace A_F in $\mathbb{E}^m$. If F is maximal, then A_F completely describes F . Indeed, since the action is properly discontinuous, the subgroup of Γ that fixes A_F pointwise has to be finite. This subgroup must contain F , but since F is maximal, it must coincide with F .

Denote by $\mathfrak{M}_k(\Gamma)$ the number of maximal finite subgroups $F < \Gamma$ (up to conjugation) such that $\dim A_F = k$.

Choose a finite subgroup $F < \Gamma$; consider a conjugate subgroup $F' = g \cdot F \cdot g^{-1}$. Note that $A_{F'} = g \cdot A_F$. In particular, the subspaces

A_F and $A_{F'}$ have the same image in the quotient space $\mathcal{A} = \mathbb{E}^m/\Gamma$. Therefore, to count subgroups up to conjugation, we need to count the images of their fixed sets. By the lemma below (10.13), $\mathfrak{M}_0(\Gamma)$ cannot exceed the number of extremal points in $\mathcal{A} = \mathbb{E}^m/\Gamma$. Combining this observation with 10.9, we get the following.

10.12. Proposition. *Let $\Gamma \curvearrowright \mathbb{E}^m$ be a crystallographic action. Then $\mathfrak{M}_0(\Gamma) \leq 2^m$.*

10.13. Lemma. *Let $\Gamma \curvearrowright \mathbb{E}^m$ be a crystallographic action, and let F be a maximal finite subgroup of Γ that fixes an isolated point p . Then the image of p in the quotient space $\mathcal{A} = \mathbb{E}^m/\Gamma$ is an extremal point.*

Proof. Let q be the image of p . Suppose q is not extremal; that is, $\angle[q \begin{smallmatrix} y_1 \\ y_2 \end{smallmatrix}] > \frac{\pi}{2}$ for some hinge $[q \begin{smallmatrix} y_1 \\ y_2 \end{smallmatrix}]$ in $\mathcal{A}$.

Choose the inverse images $x_1, x_2 \in \mathbb{E}^m$ of $y_1, y_2 \in \mathcal{A}$ such that $|p - x_i|_{\mathbb{E}^m} = |q - y_i|_{\mathcal{A}}$. Note that $\angle[p \begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}] \geq \angle[q \begin{smallmatrix} y_1 \\ y_2 \end{smallmatrix}] > \frac{\pi}{2}$. Moreover, since p is fixed by F , we have

$$\bullet \quad \angle[p \begin{smallmatrix} x_1 \\ g \cdot x_2 \end{smallmatrix}] > \frac{\pi}{2}$$

for any $g \in F$.

Denote by $z \in \mathbb{E}^m$ the barycenter of the orbit $F \cdot x_2$. Note that z is a fixed point of F . By $\bullet$, $z \neq p$; so F must fix the line pz . But p is an isolated fixed point of F — a contradiction. $\square$

10.14. Exercise. *Let $\Gamma \curvearrowright \mathbb{E}^m$ be a crystallographic action. Show that*

- (a) $\mathfrak{M}_{m-1}(\Gamma) \leq 2$, and
- (b) if $\mathfrak{M}_{m-1}(\Gamma) = 1$, then $\mathfrak{M}_0(\Gamma) \leq 2^{m-1}$.

Construct crystallographic actions with equalities in (a) and (b).

F Remarks

Submetries were introduced by Valerii Berestovskii [14] and have attracted attention in various contexts of differential and metric geometry.

A more general form of Theorem 10.6 was found by Karsten Grove and Burkhard Wilking [44]; it classifies isometric $\mathbb{S}^1$ actions on 4-dimensional manifolds with nonnegative sectional curvature. Their proof is as beautiful as the original work of Wu-Yi Hsiang and Bruce Kleiner.

It is expected that no ALEX(1) space with a nontrivial isometric $\mathbb{S}^1$ -action can be homeomorphic to $\mathbb{S}^2 \times \mathbb{S}^2$; so 10.6 holds for a general

ALEX(1) space. The proof of 10.6 would work if we had the following generalization of 10.7; see [47].

10.15. Open question. *Let Σ be an ALEX(1) space homeomorphic to $\mathbb{S}^3$. Suppose $\mathbb{S}^1$ acts on Σ isometrically and without fixed points. Is it true that any triangle in $\Sigma/\mathbb{S}^1$ has perimeter at most π ? (And if the answer is yes, is there a distance-noncontracting map $\Sigma/\mathbb{S}^1 \rightarrow \frac{1}{2}\mathbb{S}^2$?)*

10.16. Advanced exercise. *Suppose $\mathbb{S}^1$ acts isometrically on an ALEX(1) space $\mathcal{A}$ that is homeomorphic to $\mathbb{S}^3$. Assume its fixed-point set is a closed local geodesic γ . Show that*

$$\text{length } \gamma \leq 2\pi.$$

An analogous question for a $\mathbb{Z}_2$ -action is open [91].

Theorem 10.9 is a translation of the following classical problem in discrete geometry to Alexandrov's language.

10.17. Problem. *Let F be a set of points in $\mathbb{E}^m$ such that any triangle formed by three distinct points in F has no obtuse angles. Then $|F| \leq 2^m$. Moreover, if $|F| = 2^m$, then F consists of the vertices of an m -dimensional rectangle.*

This problem was posed by Paul Erdős [30] and solved by Ludwig Danzer and Branko Grünbaum [27]. Grigory Perelman noticed that, after proper definitions, the same proof works in Alexandrov spaces [84]; thus, it proves 10.9. Applying our argument to the convex hull of F in 10.17 proves that $|F| \leq 2^m$; the case of equality requires more work.

Compact m -dimensional ALEX(0) spaces with the maximal number of extremal points include m -dimensional rectangles and the quotients of flat tori by reflections across a point (this action has 2^m isolated fixed points; each corresponds to an extremal point in the quotient space $\mathcal{A} = \mathbb{T}^m/\mathbb{Z}_2$). The second author has proved [71] that *every m -dimensional ALEX(0) space with 2^m extremal points is a quotient of Euclidean space by a crystallographic action.*

The extremal subsets of Alexandrov spaces were briefly discussed in 9F. The following definition is more relevant to isometric group actions.

A closed subset E in a finite-dimensional Alexandrov space is called extremal if $\angle[p_y^x] \leq \frac{\pi}{2}$ for any hinge $[p_y^x]$ such that $x \notin E$ and $p \in E$ minimizes $|x - p|$. A nonempty extremal set is called minimal if it contains no proper extremal subsets.

For example, the whole space and the empty set are extremal. The vertices, edges, and faces of a cube, as well as their unions, are extremal subsets. Vertices of the cube are its only minimal extremal subsets.

Counting maximal finite subgroups in a crystallographic group Γ (up to conjugation) is equivalent to counting the minimal extremal subsets in the quotient space $\mathcal{A} = \mathbb{E}^m/\Gamma$. So, 10.12 would follow from the next conjecture.

10.18. Conjecture. *Any m -dimensional compact $\text{ALEX}(0)$ space has at most 2^m minimal extremal subsets.*

Let us mention another related conjecture. A nonempty extremal set is called primitive if it contains no proper extremal subsets with nonempty relative interior. (Given an Alexandrov space and a point p in it, the set of points reachable from p by a recursive application of the gradient flow for squares of distance functions forms a primitive extremal set.) For example, each face of an m -dimensional cube is a primitive extremal subset; therefore the cube has exactly 3^m primitive extremal subsets, including the whole cube.

10.19. Conjecture. *Any m -dimensional compact $\text{ALEX}(0)$ space has at most 3^m primitive extremal subsets.*

Some crude estimates on the number of extremal subsets follow from the idea in Gromov's Betti number theorem 8.18.

Lecture 11

Surfaces

This lecture is less rigorous; it offers a glimpse into the geometry of convex surfaces, a major precursor of modern Alexandrov geometry. For a deeper dive into this theory, we recommend the classic and brilliantly written books by Alexandr Alexandrov [8, 10]. Additionally, the book by Alexey Pogorelov [107] is highly recommended, despite being a challenging read.

A Polyhedral surfaces

A polyhedral surface is defined as a 2-dimensional manifold (possibly with boundary) equipped with a length metric such that there is a finite triangulation in which each triangle is isometric to a Euclidean triangle. A triangulation of a polyhedral surface is always assumed to satisfy this condition.

Note that, according to our definition, every polyhedral surface is compact.

Consider a point p on a polyhedral surface $\mathcal{P}$. We can assume that p is a vertex of a triangulation of $\mathcal{P}$, which can be achieved by subdividing the triangulation. Let θ_p denote the total angle around p , which is the sum of all angles at p in the triangles that have p as a vertex.

Note that θ_p is independent of the choice of triangulation. If p is an interior point, then the value $2\pi - \theta_p$ is called the curvature at p . If p lies on the boundary of $\mathcal{P}$, then the value $\pi - \theta_p$ is called the inner turn at p .

A point with nonzero curvature or inner turn will be called an essential vertex of the surface. Note that an essential vertex is a vertex in any triangulation.

11.1. Exercise! Show that geodesics on a polyhedral surface with nonnegative curvature and nonnegative inner turns can have essential vertices only at their endpoints.

The following statement is an analog of the Gauss–Bonnet formula.

11.2. Exercise. Let $\mathcal{P}$ be a polyhedral surface. Denote by $K(\mathcal{P})$ and $T(\mathcal{P})$ the sum of the curvatures at all interior points and the sum of the inner turns at all boundary points of $\mathcal{P}$, respectively. Show that

$$K(\mathcal{P}) + T(\mathcal{P}) = 2 \cdot \pi \cdot \chi(\mathcal{P}),$$

where $\chi(\mathcal{P})$ denotes the Euler characteristic of $\mathcal{P}$.

The following proposition states that the new definition of curvature is consistent with the ALEX(0) comparison.

11.3. Proposition. A polyhedral surface is ALEX(0) if and only if it has nonnegative curvature at every interior point and nonnegative inner turn at each boundary point.

Proof. By 2.22, it is sufficient to check that $f = \frac{1}{2} \cdot \text{dist}_p^2 \circ \gamma$ is 1-concave for any geodesic γ and any point p .

We can assume that p is not a vertex and that the endpoints of γ are not vertices; the remaining cases can be handled by approximation. By 11.1, γ does not contain any vertices.

Given a point $x = \gamma(t_0)$, choose a geodesic $[px]$. Again, by 11.1, $[px]$ does not contain any vertices. Therefore, a neighborhood $U \supset [px]$ can be unfolded onto a plane; that is, there exists an injective length-preserving map $z \mapsto \tilde{z}$ of U into the Euclidean plane. It maps a neighborhood of x in γ to a line segment $\tilde{\gamma}$. Let

$$\tilde{f}(t) := \frac{1}{2} \cdot \text{dist}_{\tilde{p}}^2 \circ \tilde{\gamma}(t).$$

Since the geodesic $[px]$ maps to a line segment, we have $\tilde{f}(t_0) = f(t_0)$. Furthermore, since the unfolding map $z \mapsto \tilde{z}$ preserves the lengths of curves, we obtain $\tilde{f}(t) \geq f(t)$ if t is close to t_0 . That is, $\tilde{f}$ is a local upper barrier for f at t_0 ; see 2E. Evidently, $\tilde{f}'' = 1$. Therefore, f is 1-concave.

11.4. Exercise. The converse is left to the reader. □

B Approximation

The following theorem is a key additional tool in the Alexandrov geometry of surfaces. We will use this result in the proof of 11.20 to

reduce questions about ALEX(0) surfaces to polyhedral surfaces with nonnegative curvature.

11.5. Theorem. *Any closed ALEX(0) surface is a Gromov–Hausdorff limit of homeomorphic polyhedral surfaces with nonnegative curvature.*

We give only a sketch of the proof, but the sketch is not short. First we apply the following exercise to construct an approximation $\tilde{\mathcal{P}}_\delta$ of a given surface $\mathcal{P}$. It reduces the theorem to saying that this is indeed an approximation (11.7). Further, we use a couple of deep results (11.8 and 11.9) to prove the claim.

11.6. Exercise. *Let $\mathcal{P}$ be a closed ALEX(0) surface.*

- (a) *Show that any point p admits an arbitrarily small closed convex polygonal neighborhood N ; that is, N is convex and bounded by a broken geodesic.*
- (b) *Given $\delta > 0$, show that $\mathcal{P}$ admits a triangulation τ by convex triangles with positive inner turn at each vertex and diameter smaller than δ .*
- (c) *Suppose that v is a vertex of a triangulation τ of $\mathcal{P}$ by convex triangles. Let θ_v be the sum of angles at v in all the triangles of τ . Show that $\theta_v \leq 2\pi$.*

Construction. Let $\mathcal{P}$ be a closed ALEX(0) surface. By part (b), we can triangulate $\mathcal{P}$ by small convex triangles, say, diameter of each triangle is less than a given $\delta > 0$. Now, replace each triangle in the triangulation with its corresponding model solid triangle, and denote the resulting polyhedral surface by $\tilde{\mathcal{P}}_\delta$.

Note that $\tilde{\mathcal{P}}_\delta$ is homeomorphic to $\mathcal{P}$; moreover, there exists a homeomorphism $\mathcal{P} \rightarrow \tilde{\mathcal{P}}_\delta$ (it will be denoted by $x \mapsto \tilde{x}$) that maps each edge in the triangulation of $\mathcal{P}$ isometrically to the corresponding edge of $\tilde{\mathcal{P}}_\delta$, and each triangle in $\mathcal{P}$ is mapped to the corresponding model triangle in $\tilde{\mathcal{P}}_\delta$.

By the hinge comparison (2.17d) and part (c) of the exercise, the total angle around each vertex in $\tilde{\mathcal{P}}_\delta$ does not exceed 2π . Thus, the resulting polyhedral surface $\tilde{\mathcal{P}}_\delta$ has nonnegative curvature.

Observe that Theorem 11.5 follows from the following statement.

11.7. Claim. *If $\tilde{\mathcal{P}}_\delta$ is provided by the construction, then $\tilde{\mathcal{P}}_\delta \rightarrow \mathcal{P}$ as $\delta \rightarrow 0$ in the sense of Gromov–Hausdorff.*

This claim looks self-evident, but it is not. A remarkably elegant proof was given in [10, VII § 6]. We will outline an alternative proof based on the following exercise and two theorems, which will be stated without proof.

The first theorem is due to Yuri Burago, Mikhael Gromov, and Grigori Perelman [20, 10.8]; it generalizes Alexandrov's theorem for surfaces [10, X § 2]. The second theorem is due to Nan Li [74].

11.8. Theorem. *Let $\mathcal{X}_1, \mathcal{X}_2, \dots$ be m -dimensional $\text{ALEX}(\kappa)$ spaces that converge to $\mathcal{X}_\infty$ in the sense of Gromov–Hausdorff. Then m -volumes on $\mathcal{X}_i$ weakly converge to the m -volume on $\mathcal{X}_\infty$.*

11.9. Theorem. *Let $\mathcal{X}$ be an Alexandrov space without boundary, and let $\mathcal{Y}$ be an arbitrary Alexandrov space of the same dimension. Then any short volume-preserving map $\mathcal{X} \rightarrow \mathcal{Y}$ is an isometry.*

Suppose a convex solid triangle Δ in an $\text{ALEX}(0)$ surface has angles α , β , and γ . Let us define its excess by

$$\text{excess } \Delta = \alpha + \beta + \gamma - \pi.$$

Since the angles of a model triangle sum up to π , by the hinge comparison (2.17d), the excess is nonnegative.

11.10. Exercise. *Let τ be a triangulation of a closed $\text{ALEX}(0)$ surface $\mathcal{P}$ by convex triangles $\Delta_1, \dots, \Delta_n$.*

(a) *Show that*

$$\text{excess } \Delta_1 + \dots + \text{excess } \Delta_n \leq 2 \cdot \pi \cdot \chi(\mathcal{P}),$$

where $\chi(\mathcal{P})$ denotes the Euler characteristic of $\mathcal{P}$.

(b) *Let x and y be points on the sides of a triangle Δ_i , and let $\tilde{x}$ and $\tilde{y}$ be the corresponding points in the corresponding triangle $\tilde{\Delta}$ in $\tilde{\mathcal{P}}$. Show that*

$$|\tilde{x} - \tilde{y}|_{\tilde{\Delta}} \leq |x - y|_{\Delta} \leq |\tilde{x} - \tilde{y}|_{\tilde{\Delta}} + \text{excess } \Delta \cdot \text{diam } \Delta.$$

(c) *Let Δ be a solid triangle in the triangulation τ of $\mathcal{P}$, and let $\tilde{\Delta}$ be the corresponding triangle in $\tilde{\mathcal{P}}$. Show that*

$$\text{area } \tilde{\Delta} \leq \text{area } \Delta \leq \text{area } \tilde{\Delta} + \frac{1}{2} \cdot \text{excess } \Delta \cdot (\text{diam } \Delta)^2.$$

Note that part (a) implies that $\chi(\mathcal{P}) \geq 0$. Therefore, $\mathcal{P}$ is homeomorphic to a sphere, projective plane, torus, or Klein bottle. In the latter two cases, the construction produces a flat surface $\tilde{\mathcal{P}}_\delta$, which has to be isometric to $\mathcal{P}$. Therefore the cases of the sphere and the projective plane are more interesting.

Proof of 11.7. Choose a sequence of positive numbers $\delta_n \rightarrow 0$; let $\tilde{\mathcal{P}}_{\delta_n}$ be polyhedral surfaces provided by the construction and let τ_n be the corresponding triangulation.

According to part (b) of the exercise, the spaces $\tilde{\mathcal{P}}_{\delta_n}$ have bounded diameter. Therefore by Gromov's selection theorem, we can pass to a converging sequence of $\tilde{\mathcal{P}}_{\delta_n}$; denote its Gromov–Hausdorff limit by $\tilde{\mathcal{P}}$. If $\tilde{\mathcal{P}}_{\delta_n}$ does not converge to $\mathcal{P}$, then we can assume that $\tilde{\mathcal{P}}$ is not isometric to $\mathcal{P}$.

Choose two points $x, y \in \mathcal{P}$, and connect them by a geodesic. Denote by $s_1, \dots, s_m$ the points of the geodesic on the sides of the triangulation τ_n ; we assume that these points appear in the same order on the geodesic. Denote by $\tilde{x}$, $\tilde{y}$, and $\tilde{s}_1, \dots, \tilde{s}_m$ the corresponding points in $\tilde{\mathcal{P}}_{\delta_n}$. By part (b) of the exercise,

$$|\tilde{s}_{i-1} - \tilde{s}_i|_{\tilde{\mathcal{P}}_{\delta_n}} \leq |s_{i-1} - s_i|_{\mathcal{P}},$$

Note also that

$$|\tilde{x} - \tilde{s}_1|_{\tilde{\mathcal{P}}_{\delta_n}} \leq \delta_n \quad \text{and} \quad |\tilde{s}_m - \tilde{y}|_{\tilde{\mathcal{P}}_{\delta_n}} \leq \delta_n$$

Therefore

$$|\tilde{x} - \tilde{y}|_{\tilde{\mathcal{P}}_{\delta_n}} \leq |x - y|_{\mathcal{P}} + 2 \cdot \delta_n.$$

Passing to the limit, we get a short onto map $\mathcal{P} \rightarrow \tilde{\mathcal{P}}$. On the other hand, applying parts (a) and (c), we get that

$$\text{area } \mathcal{P} - \pi \cdot \chi(\mathcal{P}) \cdot \delta_n^2 \leq \text{area } \tilde{\mathcal{P}}_{\delta_n} \leq \text{area } \mathcal{P}$$

By Theorem 11.8, $\text{area } \mathcal{P} = \text{area } \tilde{\mathcal{P}}$. Applying Theorem 11.9, we get that the short map $\mathcal{P} \rightarrow \tilde{\mathcal{P}}$ is an isometry — a contradiction. $\square$

Remark. The main difficulty in the proof comes from the nonconvexity of triangles in the triangulation of $\tilde{\mathcal{P}}_{\delta}$. If these triangles were convex, then the first estimate in parts (b) and (a) would imply that $\tilde{\mathcal{P}}_{\delta}$ is close to $\mathcal{P}$ in the sense of Gromov–Hausdorff.

C Surface of polyhedrons and bodies

Let us define a convex body as a compact convex subset of $\mathbb{E}^3$ with a nonempty interior. The surface of a convex body is defined as its boundary equipped with the induced length metric.

11.11. Exercise! *Show that the surface of a convex body is homeomorphic to $\mathbb{S}^2$.*

A convex polyhedron is a convex body that is the convex hull of a finite set of points; the essential points in this set are called its vertices.

Note that the surface of a convex polyhedron K is a closed polyhedral surface.

11.12. Exercise. Assume that the surface of a nondegenerate tetrahedron T has curvature π at each vertex. Show that

- (a) all faces of T are congruent;
- (b) the line containing the midpoints of opposite edges of T intersects these edges at right angles.

11.13. Claim. The surface $\mathcal{P}$ of any convex polyhedron K has non-negative curvature. Moreover, a point v is a vertex of K if and only if v is an essential vertex of $\mathcal{P}$.

A proof of this claim can be found in a school textbook [60, § 48]; one can also deduce it from 1.11.

11.14. Exercise! Let $K_1, K_2, \dots$, and K_∞ be convex bodies in $\mathbb{E}^m$. Denote by $\mathcal{P}_n$ the surface of K_n . Suppose $K_n \rightarrow K_\infty$ in the sense of Hausdorff. Show that $\mathcal{P}_n \rightarrow \mathcal{P}_\infty$ in the sense of Gromov–Hausdorff.

Since any convex body is a Hausdorff limit of a sequence of convex polyhedrons, the next proposition follows from 11.3, 11.14, and 8.1.

11.15. Proposition. The surface of a convex body in $\mathbb{E}^3$ is ALEX(0).

D Uniqueness theorem

11.16. Theorem. Any two convex polyhedrons in $\mathbb{E}^3$ with isometric surfaces are congruent.

Moreover, any isometry between the surfaces of convex polyhedrons can be extended to an isometry of the whole $\mathbb{E}^3$.

Assuming the isometry between the surfaces to be face-to-face gives a reformulation of Cauchy’s theorem; his argument, with a small addition, proves 11.16.

First, let us recall Cauchy’s proof, assuming the reader knows it. If not, then read it in one of the classical texts; see, for example, [1, 28, 116].

Sketch of Cauchy’s proof. Suppose K and K' are convex polyhedrons in $\mathbb{E}^3$; denote their surfaces by $\mathcal{P}$ and $\mathcal{P}'$. Suppose there is an isometry $\iota: \mathcal{P} \rightarrow \mathcal{P}'$ that sends each face of K to a face of K' .

Let us mark an edge of K with “+” (or “−”) if the dihedral angle at this edge in K is smaller (respectively, bigger) than the corresponding angle in K' . Further, we consider the graph Γ that is formed by all

marked edges. If Γ is empty, then Cauchy's theorem follows; assume the contrary.

The graph Γ is embedded into $\mathcal{P}$, which is homeomorphic to the sphere. In particular, the edges coming from one vertex have a natural cyclic order. Given a vertex v of Γ , we can count the *number of sign changes* around v ; that is, the number of consecutive pairs of edges with different signs.

We need two statements:

11.17. Local lemma. *The number of sign changes at each vertex of Γ is at least 4.*

11.18. Global lemma. *No (nonempty) planar graph meets the condition of the local lemma.*

Once the lemmas are proved, Cauchy's theorem follows. □

Once more, the argument above is written only to make sure we are on the same page; it will not work without reading the actual proof.

Alexandrov's addition. We need to remove the assumption that the isometry $\iota: \mathcal{P} \rightarrow \mathcal{P}'$ is face-to-face. Mark in $\mathcal{P}$ all the edges of K as we did above. In addition, if an edge in K' does not correspond to an edge of K , then mark its inverse image in K with “−”; these lines on K will be referred to as fake edges.

The marked lines divide $\mathcal{P}$ into regions, and the restriction of ι to each region is a rigid motion. A vertex of the obtained graph can be a vertex of K , or it can be a fake vertex; that is, it might be an intersection of an edge and a fake edge (as v in the picture).



For a usual vertex, the local lemma can be proved the same way. For a fake vertex v , it is easy to see that both parts of the edge coming thru v are marked with minus while both of the fake edges at v are marked with plus. Therefore, we still have at least four sign changes at v . The remaining argument works as before. □

E Existence theorem

By 11.3, 11.13, and 11.11, the surface of a convex polyhedron is ALEX(0) and homeomorphic to the sphere. Alexandrov's theorem

states that the converse holds if one includes in the consideration *twice covered polygons*. In other words, we have to consider a plane polygon as a degenerate polyhedron; in this case, its surface is defined as the doubling of the polygon across its boundary.

From now on, we assume that *a polyhedron can degenerate to a plane polygon*.

11.19. Theorem. *A polyhedral metric on the two-sphere is isometric to the surface of a convex polyhedron (possibly degenerate) if and only if it has nonnegative curvature.*

Applying this theorem with the approximation theorem (11.5) and 11.14, we get the following statement. Here we again assume that a convex body can degenerate to a convex plane figure, and, in this case, its surface is defined as the doubling of the figure across its boundary.

11.20. Corollary. *A metric on the two-sphere is $\text{ALEX}(0)$ if and only if it is isometric to the surface of a convex body (possibly degenerate).*

The proof of the existence theorem will be discussed in the following two sections. It is instructive to solve the following exercise before going further.

11.21. Exercise! *Let $\mathcal{P}$ be the 2-sphere equipped with a polyhedral metric with nonnegative curvature.*

- (a) *Prove that $\mathcal{P}$ has at least 3 essential vertices.*
- (b) *If $\mathcal{P}$ has exactly 3 essential vertices u , v , and w , then it is isometric to the doubling of the solid model triangle $\hat{\Delta}(uvw)$.*
- (c) *If $\mathcal{P}$ has exactly 4 essential vertices, then it is isometric to the surface of a tetrahedron (possibly degenerate to a quadrangle).*

F Reformulation

In this section, we introduce several notions and use them to reformulate the existence theorem (11.22).

Space of polyhedrons. Let us denote by $\mathbf{K}$ the space of all convex polyhedrons in Euclidean space, including polyhedrons that degenerate to a plane polygon. Polyhedrons in $\mathbf{K}$ will be considered up to a motion of the space; we will not distinguish between a convex polyhedron and its congruence class.

The space $\mathbf{K}$ will be considered with the topology induced by the Hausdorff metric up to a motion; that is, the distance between (congruence classes of) polyhedrons K and L is defined by

$$|K - L| := \inf_{\mu} \{ |K - \mu(L)|_{\text{Haus}} \},$$

where μ runs among all motions of $\mathbb{E}^3$ and $|K - L|_{\text{Haus}}$ denotes Hausdorff distance; see Section 1K.

We say that a polyhedron K in $\mathbf{K}$ has no symmetries if $K \neq \mu(K)$ for any nontrivial motion μ of $\mathbb{E}^3$. The set of all polyhedrons without symmetries in $\mathbf{K}$ will be denoted by $\mathbf{K}^\circ$. Observe that $\mathbf{K}^\circ$ is an open set in $\mathbf{K}$.

Further, denote by $\mathbf{K}_n$ the polyhedrons in $\mathbf{K}$ with exactly n vertices, and let $\mathbf{K}_n^\circ = \mathbf{K}_n \cap \mathbf{K}^\circ$. Since any polyhedron has at least 3 vertices, the space $\mathbf{K}$ admits a subdivision into a countable number of subsets $\mathbf{K}_3, \mathbf{K}_4, \dots$

Space of surfaces. The space of polyhedral surfaces with nonnegative curvature that are homeomorphic to the 2-sphere will be denoted by $\mathbf{P}$. The surfaces in $\mathbf{P}$ will be considered up to an isometry, and the whole space $\mathbf{P}$ will be equipped with the natural topology induced by the Gromov–Hausdorff metric.

We say that a surface $\mathcal{P}$ in $\mathbf{P}$ has no symmetries if there is no nontrivial isometry $\mu: \mathcal{P} \rightarrow \mathcal{P}$. The set of all surfaces without symmetries in $\mathbf{P}$ will be denoted by $\mathbf{P}^\circ$. Observe that $\mathbf{P}^\circ$ is an open set in $\mathbf{P}$.

The subset of $\mathbf{P}$ of all surfaces with exactly n essential vertices will be denoted by $\mathbf{P}_n$; let $\mathbf{P}_n^\circ = \mathbf{P}_n \cap \mathbf{P}^\circ$. By 11.21a, any surface in $\mathbf{P}$ has at least 3 essential vertices. Therefore $\mathbf{P}$ is subdivided into countably many subsets $\mathbf{P}_3, \mathbf{P}_4, \dots$

From a polyhedron to its surface. Recall that the surface of a convex polyhedron is a sphere with nonnegative curvature. Therefore, passing from a polyhedron to its surface, we define a map

$$\iota: \mathbf{K} \rightarrow \mathbf{P}.$$

The existence theorem (11.19) follows from the next statement.

11.22. Theorem. *For any integer $n \geq 3$, the map ι defines a bijection from $\mathbf{K}_n$ to $\mathbf{P}_n$.*

G About the proof of existence

By 11.14, the map $\iota: \mathbf{K} \rightarrow \mathbf{P}$ is continuous. Combining 11.13 with the uniqueness theorem (11.16), we get that $\iota(\mathbf{K}_n) \subset \mathbf{P}_n$ and the map $\iota: \mathbf{K}_n \rightarrow \mathbf{P}_n$ is injective. It remains to prove the following.

11.23. Claim. *For any $n \geq 3$, the map $\iota: \mathbf{K}_n \rightarrow \mathbf{P}_n$ is surjective.*

The proof is based on the construction of a one-parameter family of polyhedrons that starts at an arbitrary polyhedron and ends at a polyhedron with its surface isometric to the given surface $\mathcal{P}$. This is called the continuity method; it is often used in the theory of differential equations.

Now let us get into details. First, observe that the second part of the uniqueness theorem (11.16) implies that $\iota(K_n^\circ) \subset P_n^\circ$.

11.24. Lemma. *For any integer $n \geq 4$, the space P_n° is connected and dense in P_n .*

Note that $P_3^\circ = \emptyset$; indeed the surface of a triangle admits a reflection symmetry. The case $n = 4$ can be deduced from 11.21c; thus, we can assume that $n \geq 5$.

The density of P_n° in P_n follows from a general-position-type argument.

The proof of the first statement is not complicated, but it requires ingenuity; it can be done by the direct construction of a one-parameter family of surfaces in P_n° that connects two given surfaces. Such a family can be obtained by concatenating the following elementary deformations (direct or reversed).

Start with a surface $\mathcal{P}$ from P_n° . Suppose v and w are essential vertices in $\mathcal{P}$. Let us cut $\mathcal{P}$ along a geodesic from v to w . This way we obtain a sphere with a hole. The hole can be patched by a family of patches so that the obtained surface remains in P_n ; in other words, the surface still has exactly n essential vertices. (After the patching, the vertices v and w may become inessential. There is a three-parameter family of such patches, so we have something to choose from.) Choosing a one-parameter family of such patches, we can get an elementary deformation of $\mathcal{P}$.

Again, applying a general-position-type argument to the above construction, we get a path in P_n° , assuming that the starting and ending surfaces are in P_n° .

11.25. Lemma. *The map $\iota: K_n^\circ \rightarrow P_n^\circ$ is open; that is, it maps any open set in K_n° to an open set in P_n° .*

In particular, for any $n \geq 3$, the image $\iota(K_n^\circ)$ is open in P_n° .

This statement follows from the so-called invariance of domain theorem, which states that a continuous injective map between manifolds of the same dimension is open.

Recall that ι defines a continuous and injective map $K_n^\circ \rightarrow P_n^\circ$. It remains to check that both spaces K_n° and P_n° are $(3 \cdot n - 6)$ -dimensional manifolds.

Choose a polyhedron K in $\mathbf{K}_n$. It is uniquely determined by the $3 \cdot n$ coordinates of its n vertices. We can assume that the first vertex is at the origin, the second has a positive x -coordinate and the remaining two coordinates vanish, and the third has a vanishing z -coordinate and a positive y -coordinate. Therefore, all polyhedrons in $\mathbf{K}_n$ that lie sufficiently close to K can be described by $3 \cdot n - 6$ parameters. If K has no symmetries, then this description is one-to-one; in this case, a neighborhood of K in $\mathbf{K}_n$ admits a parametrization by an open set in $\mathbb{R}^{3 \cdot n - 6}$.

The case of surfaces is analogous. We need to construct a subdivision of the sphere into plane triangles using only essential vertices. By Euler's formula, there are exactly $3 \cdot n - 6$ edges in this subdivision. The lengths of the edges completely describe the surface $\mathcal{P}$ and any surface nearby. If the surface has no symmetries, then this description is one-to-one, and a neighborhood of $\mathcal{P}$ in $\mathbf{P}_n$ admits a parametrization by an open set in $\mathbb{R}^{3 \cdot n - 6}$.

11.26. Lemma. *The map $\iota: \mathbf{K}_n \rightarrow \mathbf{P}_n$ is closed; that is, the image of a closed set in $\mathbf{K}_n$ is closed in $\mathbf{P}_n$.*

In particular, for any $n \geq 3$, the set $\iota(\mathbf{K}_n)$ is closed in $\mathbf{P}_n$.

Choose a sequence of polyhedrons $K_1, K_2, \dots \in \mathbf{K}_n$. Assume that the sequence $\mathcal{P}_i = \iota(K_i)$ converges in $\mathbf{P}_n$ as $i \rightarrow \infty$; denote its limit by $\mathcal{P}_\infty$. We need to construct a polyhedron $K_\infty \in \mathbf{K}_n$ such that $\iota(K_\infty) = \mathcal{P}_\infty$; let us do it.

Passing to a subsequence, we can assume that K_i converges in $\mathbf{K}$ as $i \rightarrow \infty$; denote the limit polyhedron by K_∞ . Since ι is continuous, $\iota(K_i)$ converges to $\iota(K_\infty)$ in $\mathbf{P}$; so, $\iota(K_\infty) = \mathcal{P}_\infty$. Recall that $\iota(\mathbf{K}_m) \subset \subset \mathbf{P}_m$ for each m ; therefore, $K_\infty \in \mathbf{K}_n$.

Proof of 11.23. The case $n \leq 4$ is already solved in 11.21; so we assume that $n \geq 5$. By 11.26 and 11.25, $\iota(\mathbf{K}_n^\circ)$ is a nonempty closed and open set in $\mathbf{P}_n^\circ$, and $\mathbf{P}_n^\circ$ is connected. Therefore, $\iota(\mathbf{K}_n^\circ) = \mathbf{P}_n^\circ$.

By 11.26, $\iota(\mathbf{K}_n)$ is closed in $\mathbf{P}_n$. By 11.24, $\mathbf{P}_n^\circ$ is dense in $\mathbf{P}_n$. Since $\iota(\mathbf{K}_n^\circ) = \mathbf{P}_n^\circ$, we have $\mathbf{P}_n^\circ \subset \iota(\mathbf{K}_n)$; therefore, $\iota(\mathbf{K}_n) = \mathbf{P}_n$; that is, $\iota: \mathbf{K}_n \rightarrow \mathbf{P}_n$ is surjective. $\square$

H Ambient space

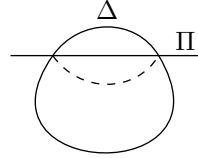
On one hand, Alexandrov's surface theory is simpler since it has extra tools. On the other hand, these tools come with extra structure, which makes the theory more complicated. The following result of Joseph Liberman [75] gives an example.

11.27. Theorem. *Any geodesic in the surface of a convex body is one-sided differentiable as a curve in $\mathbb{E}^3$.*

Proof. Let γ be a geodesic on the surface of a convex body K . Choose $p \in K$. By 9.14 (which is Liberman's lemma), the function $f_p: t \mapsto \text{dist}_p \circ \gamma(t)$ is semiconcave for any $p \in K$. In particular, one-sided derivatives $f_p^\pm(t)$ are defined for every t .

Given $x = \gamma(t)$, choose three points $p_1, p_2, p_3 \in K$ in general position; that is, the points x, p_1, p_2 , and p_3 do not lie in one plane. Observe that the distance functions dist_{p_i} give smooth coordinates in a neighborhood of x . By the above, the functions f_{p_i} have one-sided derivatives at t . Since the coordinates are smooth, we get that $\gamma^+(t)$ is defined as well. $\square$

11.28. Exercise. *Suppose a plane Π cuts from the surface of a convex body K a disk Δ , and the reflection of Δ across Π lies in K . Show that Δ is a convex subset of the surface; that is, if a geodesic has endpoints in Δ , then it lies completely in Δ .*



The following exercise gives a more exact version of comparison for convex surfaces; it is due to Anatolii Milka [80, Theorem 2].

11.29. Very advanced exercise. *Let $\mathcal{P}$ be the surface of a nondegenerate convex body $K \subset \mathbb{E}^3$, and let $\gamma_1, \gamma_2: [0, 1] \rightarrow \mathcal{P}$ be geodesic paths that start at one point $p = \gamma_1(0) = \gamma_2(0)$. Suppose $x_i = \gamma_i(1)$, and $y_i = p + \gamma_i^+(0)$. Show that*

$$|x_1 - x_2|_{\mathcal{P}} \leq |y_1 - y_2|_W,$$

where W denotes the complement to the interior of K .

I Remarks

The excess of solid triangles can be extended to a sigma-additive measure on Borel subsets of an $\text{ALEX}(\kappa)$ surface. This is the so-called curvature measure [10, V]. There have been several attempts to generalize this to higher dimensions, but so far, we only have partial results; see [36, 69] and the references therein.

A big part of the theory described in this lecture is generalized to surfaces with bounded integral curvature [11, 109].

The statement of Cauchy's theorem was conjectured by Adrien-Marie Legendre in the first edition of his geometry textbook [73]. It was motivated by a vague definition in Euclid's Elements, which could

be interpreted as *polyhedrons are equal if their faces are pairwise equal*. The local lemma was already known to Legendre. Joseph-Louis Lagrange, a colleague of Legendre, suggested this problem to Augustin-Louis Cauchy, who soon solved it [21].

The observation that the face-to-face condition can be removed was made by Alexandr Alexandrov in 1941; in the same paper he proved the uniqueness theorem [9]. A quite different proof was found by Yurii Volkov in his thesis [121]; it uses a deformation of three-dimensional polyhedral space. (Be aware that the proof of this theorem given in the book by Igor Pak contains an essential mistake [90].)

In Cauchy's proof [21], it was deduced from an analog of the following lemma. Cauchy made a small mistake in its proof that was fixed in a century [110]. Several proofs of the arm lemma can be found in the letters between Isaac Schoenberg and Stanisław Zaremba [111].

11.30. Arm lemma. *Assume that $A = [a_0 a_1 \dots a_n]$ is a convex polygon in $\mathbb{E}^2$ and $A' = [a'_0 a'_1 \dots a'_n]$ is a polygonal line in $\mathbb{E}^3$ such that $|a_i - a_{i+1}| = |a'_i - a'_{i+1}|$ for any $i \in \{0, \dots, n-1\}$ and $\angle a_i \leq \angle a'_i$ for each $i \in \{1, \dots, n-1\}$. Then*

$$|a_0 - a_n| \leq |a'_0 - a'_n|$$

and equality holds if and only if A is congruent to A' .

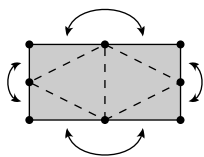
The following variation of the arm lemma is due to Viktor Zalgaller [123]; it makes sense for nonconvex spherical polygons and can be used instead of the standard arm lemma.

11.31. Another arm lemma. *Let $A = [a_1 \dots a_n]$ and $A' = [a'_1 \dots a'_n]$ be two spherical n -gons (not necessarily convex). Assume that A lies in a half-sphere, the corresponding sides of A and A' are equal, and each angle of A is at least the corresponding angle in A' . Then A is congruent to A' .*

Another close relative of the arm lemma is Reshetnyak's majorization theorem [108].

Alexandrov gave two proofs of the global lemma [8, 2.1.2 and 2.1.3]. The first is combinatorial, and the second is more visual. The argument in the second proof was reused by Anton Klyachko in his car-crash lemma [61].

Proposition 11.15 generalizes to the boundaries of convex bodies in $\mathbb{E}^m$ for any $m \geq 2$. It could be considered as a special case of the conjecture about the boundary of Alexandrov space; see 9.21. Another partial case, for convex sets in Riemannian manifolds, is proved by Stephanie Alexander and the two of us [5].



According to the uniqueness theorem, a convex polyhedron is completely defined by the intrinsic metric of its surface. In particular, knowing the metric, we could find the position of the edges. However, in practice, it is not easy to do.

For example, the surface glued from a rectangle, as shown in the picture, defines a tetrahedron. Some of the glued lines appear inside the faces of the tetrahedron, and some edges (dashed lines) do not follow the sides of the rectangle.

Let us also state the following result of Alexey Pogorelov [106, chapter III]; an alternative proof was found by Yuriy Volkov [122].

11.32. Theorem. *Any two convex bodies in $\mathbb{E}^3$ with isometric surfaces are congruent.*

Moreover, any isometry between surfaces of convex bodies can be extended to an isometry of the whole $\mathbb{E}^3$.

Notice that the statement is equivalent to the following: *Two polyhedra K and K' with close surfaces in the sense of Gromov–Hausdorff are almost congruent*; that is, there is a motion μ of $\mathbb{E}^3$ such that the Hausdorff distance from K to $\mu(K')$ is small. At first glance, this theorem might look like a small improvement of Alexandrov’s uniqueness, but this improvement is huge, and the proof is quite hard.

Semisolutions

1.1; (a). Suppose X is compact. Then for any $\varepsilon > 0$, any cover of X by open ε -balls has a finite subcover. Note that the centers of these balls form an ε -net of X .

Now suppose X has a finite ε -net. Show that any sequence x_n of points in X has a subsequence such that all of its points lie in one ε -ball. Apply this statement for $\varepsilon = \frac{1}{n}$ together with the diagonal procedure.

(b). Let Z be a compact ε -net of X . By (a), Z has a finite ε -net, say F . Note that F is a $2 \cdot \varepsilon$ -net of X . Since $\varepsilon > 0$ is arbitrary, we get the result.

1.2. If $x_1, \dots, x_n$ is not an ε -net, then there is a point y such that $|x_i - y| \geq \varepsilon$ for any i . Therefore $x_1, \dots, x_n$ is not a maximal packing — a contradiction.

1.3. Choose a sequence of positive numbers $\varepsilon_n \rightarrow 0$ and a finite ε_n -net N_n of K for each n . We can assume that $\varepsilon_0 > \text{diam } K$, and N_0 is a one-point set. If $|x - y| < \varepsilon_k$ for some $x \in N_{k+1}$ and $y \in N_k$, then connect them by a curve of length at most ε_k .

Let K' be the union of all these curves and K . Show that K' is compact and path-connected.

Source: This problem is due to Eugene Bilokopytov [15].

1.4. Choose a Cauchy sequence x_n in $(\mathcal{X}, \|* - *\|)$; it is sufficient to show that a subsequence of x_n converges.

Observe that the sequence x_n is Cauchy in $(\mathcal{X}, |* - *|)$; denote its limit by x_∞ .

Passing to a subsequence, we can assume that $\|x_n - x_{n+1}\| < \frac{1}{2^n}$ for each n . It follows that there is a 1-Lipschitz path γ in $(\mathcal{X}, \|* - *\|)$ such that $x_n = \gamma(\frac{1}{2^n})$ for each n and $x_\infty = \gamma(0)$. Therefore,

$$\|x_\infty - x_n\| \leq \text{length } \gamma|_{[0, \frac{1}{2^n}]} \leq \frac{1}{2^n}.$$

In particular, x_n converges to x_∞ in $(\mathcal{X}, \|* - *\|)$.

Source: [50, Corollary]; see also [92, Lemma 2.3].

1.5. Given a pair of points p and q , choose a sequence of paths γ_n from p to q such that

$$\text{length } \gamma_n \rightarrow |p - q| \quad \text{as } n \rightarrow \infty;$$

these paths exist since we are in a length space. We can assume that each γ_n is parametrized proportionally to the arc length; in particular, the paths γ_n are equicontinuous. Show that paths γ_n lie in a closed ball, say $\bar{B}[p, r]$ of some radius $r < \infty$. Since the space is proper, $\bar{B}[p, r]$ is compact. By the Arzelà–Ascoli theorem, we can pass to a converging subsequence of γ_n . Show that its limit is a geodesic path from p to q .

1.7. Choose a sequence $\varepsilon_n > 0$ that converges to zero very fast, say such that $\sum_n 10^n \cdot \varepsilon_n$ is small. Follow the argument in the proof of Menger’s lemma, taking ε_n -midpoints at the n -th stage.

1.8. Let us write the Riemannian metric on $\mathbb{M}^2(\kappa)$ in polar coordinates (θ, r) ; it has the form $\begin{pmatrix} h^2 & 0 \\ 0 & 1 \end{pmatrix}$, where $h = h(\kappa, r) \geq 0$. Calculate $h(\kappa, r)$. Show that for fixed r , the function $\kappa \mapsto h(\kappa, r)$ is nonincreasing in the domain of definition. Suppose $\kappa < K$, and consider the partially defined map $\mathbb{M}^2(\kappa) \rightarrow \mathbb{M}^2(K)$ that sends a point to the point with the same polar coordinates. Show that this map is short in its domain of definition. Use it to prove the statement in the exercise.

1.9. Show and use that $\tilde{\Delta}(p_y^x)_{\mathbb{S}^2} - \tilde{\Delta}(p_y^x)_{\mathbb{E}^2} = O(|p - x|^2 + |p - y|^2)$ and $\tilde{\Delta}(p_y^x)_{\mathbb{E}^2} - \tilde{\Delta}(p_y^x)_{\mathbb{H}^2} = O(|p - x|^2 + |p - y|^2)$.

1.10. Consider a hinge in the plane $\mathbb{R}^2$ with a metric defined by a norm, say by the ℓ^∞ -norm.

1.12. Show that $\angle[p_y^x] = \pi$ and apply 1.11.

1.13. Denote by α the arc-length parametrization of $[qp]$ from q to p . Choose $\varepsilon > 0$. Observe that

$$|\gamma(t) - \alpha(\tfrac{1}{\varepsilon} \cdot t)|^2 \leq t^2 \cdot (1 - \tfrac{2}{\varepsilon} \cdot \cos \varphi + \tfrac{1}{\varepsilon^2}) + o(t^2).$$

By the triangle inequality

$$|p - \gamma(t)| \leq |\gamma(t) - \alpha(\tfrac{1}{\varepsilon} \cdot t)| + |q - p| - \tfrac{1}{\varepsilon} \cdot t.$$

Conclude that

$$|p - \gamma(t)| \leq |q - p| - t \cdot \cos \varphi + \delta(\varepsilon) \cdot t + o(t),$$

where $\delta(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. The statement follows since $\varepsilon > 0$ is arbitrary.

1.15. Since the space is proper, it is separable; that is, we can choose a countable everywhere dense set $\{x_1, x_2, \dots\}$.

Let $A_1, A_2, \dots$ be a sequence of closed sets. Applying the diagonal procedure, we can pass to a subsequence such that for each i the sequence $\text{dist}_{A_n} x_i$ converges as $n \rightarrow \infty$; denote its limit by $f(x_i)$.

Since dist_{A_n} is 1-Lipschitz for any n , we have

$$|f(x_i) - f(x_j)| \leq |x_i - x_j|$$

for all i and j . Suppose $f(x_i) < \infty$ for some i . Then the same holds for any i . Therefore, the function f can be extended to a continuous function defined on the ambient space. Show that $A_\infty = f^{-1}\{0\}$ is the limit of A_n in the sense of Hausdorff.

If $f(x_i) = \infty$ for some i , then the same holds for any i . Show that in this case $A_n \rightarrow \emptyset$ in the sense of Hausdorff.

1.16. Set $|x_i - x_j| = 2$ for $0 \neq i \neq j \neq 0$.

1.19. Apply the definition of Hausdorff distance (1.17). Without assuming that all A_i are contained in a compact subset of $\mathcal{X}$ the statement is false as can be seen by the example of $X = \mathbb{R}, A_n = \{0, n\}, A_\infty = \{0\}$.

1.23. Given a pair of points $x_0, y_0 \in \mathcal{K}$, consider two sequences $x_0, x_1, \dots$ and $y_0, y_1, \dots$ such that $x_{n+1} = f(x_n)$ and $y_{n+1} = f(y_n)$ for each n .

Since $\mathcal{K}$ is compact, we can choose an increasing sequence of integers n_k such that both sequences $(x_{n_i})_{i=1}^\infty$ and $(y_{n_i})_{i=1}^\infty$ converge. In particular, both are Cauchy; that is,

$$|x_{n_i} - x_{n_j}|_{\mathcal{K}} \rightarrow 0 \quad \text{and} \quad |y_{n_i} - y_{n_j}|_{\mathcal{K}} \rightarrow 0$$

as $\min\{i, j\} \rightarrow \infty$.

Since f is distance-noncontracting,

$$|x_0 - x_{|n_i - n_j|}| \leq |x_{n_i} - x_{n_j}|$$

for any i and j . Therefore, there is a sequence $m_i \rightarrow \infty$ such that

$$(*) \quad x_{m_i} \rightarrow x_0 \quad \text{and} \quad y_{m_i} \rightarrow y_0$$

as $i \rightarrow \infty$.

Since f is distance-noncontracting, the sequence $\ell_n = |x_n - y_n|_{\mathcal{K}}$ is nondecreasing. By (*), $\ell_{m_i} \rightarrow \ell_0$ as $m_i \rightarrow \infty$. It follows that

$$\ell_0 = \ell_1 = \dots$$

In particular,

$$|x_0 - y_0|_{\mathcal{K}} = \ell_0 = \ell_1 = |f(x_0) - f(y_0)|_{\mathcal{K}}$$

for any pair of points (x_0, y_0) in $\mathcal{K}$. That is, the map f is distance-preserving and hence injective. From (*), we also get that $f(\mathcal{K})$ is everywhere dense. Since $\mathcal{K}$ is compact, f is surjective — hence the result.

Remarks. This is a basic lemma in the introduction to Gromov–Hausdorff distance [see 7.3.30 in 19]. The presented proof was suggested by Travis Morrison.

1.24. Apply 1.23 to a left inverse of the map.

1.27. The only-if part is trivial. Let us prove the if part.

Let $f_n: \mathcal{X}_n \rightarrow \mathcal{X}_\infty$ and $h_n: \mathcal{X}_\infty \rightarrow \mathcal{X}_n$ be the maps in the definition of the inequalities $\mathcal{X}_n \leq \mathcal{X}_\infty + \varepsilon_n$ and $\mathcal{X}_\infty \leq \mathcal{X}_n + \varepsilon_n$, respectively. Apply 1.23 to show that any partial limit of $f_n \circ h_n$ is an isometry of $\mathcal{X}_\infty$. Conclude that f_n is an ε'_n -isometry for some converging-to-zero sequence ε'_n and apply 1.26.

1.29; (a) Consider the graphs of the following functions with the induced metric from $\mathbb{R}^2$.

$$x \mapsto \cos x + \cos \frac{x}{\pi} \quad \text{and} \quad x \mapsto \cos x + \sin \frac{x}{\pi}.$$

(b) For every rational number $q \in [1, 2]$ consider an interval of length q . Let $\mathcal{X}$ be obtained by identifying all endpoints of the intervals.

Let $\mathcal{Y}$ be constructed in the same way but skipping the interval of length 1.5.

2.1. Note that we can assume that we have equality in all inequalities except one. In other words, we can assume that equality holds in all inequalities except $|y_1 - y_2| \leq |x_1 - x_2|$ or $|q - y_2| \geq |p - x_2|$.

In the first case,

$$\tilde{\Delta}(q \frac{y_1}{y_2}) \leq \tilde{\Delta}(p \frac{x_1}{x_2}), \quad \tilde{\Delta}(q \frac{y_2}{y_3}) = \tilde{\Delta}(p \frac{x_2}{x_3}), \quad \text{and} \quad \tilde{\Delta}(q \frac{y_3}{y_1}) = \tilde{\Delta}(p \frac{x_3}{x_1}).$$

Therefore, the comparison for p, x_1, x_2, x_3 implies the comparison for q, y_1, y_2, y_3 .

In the second case, let us argue by contradiction. Assume that the comparison does not hold for q, y_1, y_2, y_3 . Show that

$$\tilde{\Delta}(q \frac{y_1}{y_2}) + \tilde{\Delta}(q \frac{y_2}{y_3}) > \pi.$$

Then show and use that

$$\tilde{\Delta}(p \frac{x_1}{x_2}) + \tilde{\Delta}(p \frac{x_2}{x_3}) \geq \tilde{\Delta}(q \frac{y_1}{y_2}) + \tilde{\Delta}(q \frac{y_2}{y_3}).$$

2.3. The 4-point comparison (2.2) reduces our question to the following. *Any spherical triangle has perimeter at most $2\cdot\pi$.* Choose a spherical triangle $[xyz]$. Let x' be the antipode of x ; that is $x' = -x$. The spherical triangle inequality (1.11) implies that

$$|x - z|_{\mathbb{S}^2} \leq |y - x'|_{\mathbb{S}^2} + |x' - z|_{\mathbb{S}^2}.$$

Observe that

$$|x - y|_{\mathbb{S}^2} + |y - x'|_{\mathbb{S}^2} = \pi, \quad \text{and} \quad |x - z|_{\mathbb{S}^2} + |z - x'|_{\mathbb{S}^2} = \pi.$$

Hence

$$|x - y|_{\mathbb{S}^2} + |x - z|_{\mathbb{S}^2} + |y - z|_{\mathbb{S}^2} \leq 2\cdot\pi.$$

2.4. For the only-if part consider the following two cases.

If $\tilde{\angle}(p_{x_2}^{x_1}) + \tilde{\angle}(p_{x_3}^{x_2}) \geq \pi$, then choose two model triangles $[qy_1y_2] = \tilde{\Delta}(px_1x_2)$ and $[qy_2y_3] = \tilde{\Delta}(px_2x_3)$ that lie on the opposite sides of $[qy_2]$. By the comparison, $|y_1 - y_3| \geq |x_1 - x_3|$. Therefore the obtained configuration meets all the conditions.

If $\tilde{\angle}(p_{x_2}^{x_1}) + \tilde{\angle}(p_{x_3}^{x_2}) \leq \pi$, then choose a model triangle $[qy_1y_2] = \tilde{\Delta}(px_1x_2)$ and take y_3 on the extension of $[y_1q]$ behind q such that $|q - y_3| = |p - x_3|$. Then $\angle[q_{y_3}^{y_2}] \geq \tilde{\angle}(p_{x_3}^{x_2})$, therefore $|y_2 - y_3| \geq |x_2 - x_3|$. Further, $|y_2 - y_3| = |x_2 - p| + |p - x_3| \geq |x_2 - x_3|$, and again, the obtained configuration meets all the conditions.

To prove the if part, choose a configuration q, y_1, y_2, y_3 that meets all the conditions and maximize the sum

$$|y_1 - y_2| + |y_2 - y_3| + |y_3 - y_1|.$$

Show that $|q - y_i| = |p - x_i|$ for each i and q lies in the solid triangle $y_1y_2y_3$; in particular

$$\angle[q_{y_2}^{y_1}] + \angle[q_{y_3}^{y_2}] + \angle[q_{y_1}^{y_3}] = 2\cdot\pi.$$

Applying 2.6, we get

$$\tilde{\angle}(p_{x_2}^{x_1}) + \tilde{\angle}(p_{x_3}^{x_2}) + \tilde{\angle}(p_{x_1}^{x_3}) \leq 2\cdot\pi.$$

2.7. Consider model triangles $[\tilde{p}\tilde{x}\tilde{z}] = \tilde{\Delta}(pxz)$ and $[\tilde{p}\tilde{y}\tilde{z}] = \tilde{\Delta}(pyz)$ that share side $[\tilde{p}\tilde{z}]$ and lie on its opposite sides. Note that

$$\begin{aligned} |\tilde{x} - \tilde{y}|_{\mathbb{E}^2} &\geq |\tilde{x} - \tilde{z}|_{\mathbb{E}^2} + |\tilde{z} - \tilde{y}|_{\mathbb{E}^2} = \\ &= |x - z|_{\mathcal{X}} + |z - y|_{\mathcal{X}} = \\ &= |x - y|_{\mathcal{X}}, \end{aligned}$$

where $\mathcal{X}$ is our metric space. It remains to apply 2.6.

2.9. Apply 2.8.

2.11. Without loss of generality, we can assume that $|p - x| \leq |p - y|$. Choose $\bar{x} \in [px]$; let $\bar{y} \in [py]$ be such that $|p - \bar{x}| = |p - \bar{y}|$. Apply 2.8 to show that $\bar{x} = \bar{y}$. Conclude that $[px] \subset [py]$.

2.12. Assume that there are two distinct geodesics from z to x . Then we can choose distinct points p and q on these geodesics such that $|z - p| = |z - q|$. Observe that $\tilde{\angle}(z_p^p) > 0$. By comparison,

$$\tilde{\angle}(z_q^p) + \tilde{\angle}(z_y^p) + \tilde{\angle}(z_y^q) \leq 2 \cdot \pi.$$

Therefore, one of the angles $\tilde{\angle}(z_y^p)$ or $\tilde{\angle}(z_y^q)$ is strictly less than π . The latter contradicts the triangle inequality.

2.13. By 1.12, we have

$$\angle[p_z^x] + \angle[p_z^y] \geq \pi.$$

Since $p \in]xy[$ we have

$$\tilde{\angle}(p_{\bar{y}}^{\bar{x}}) = \pi$$

for any $\bar{x} \in [xz[$ and $\bar{y} \in]zy]$. By comparison, we have that

$$\tilde{\angle}(p_{\bar{z}}^{\bar{x}}) + \tilde{\angle}(p_{\bar{y}}^{\bar{z}}) \leq \pi$$

for any $\bar{z} \in]zp]$. Passing to the limit as $|p - \bar{x}| \rightarrow 0$, $|p - \bar{y}| \rightarrow 0$, and $|p - \bar{z}| \rightarrow 0$, we get

$$\angle[p_z^x] + \angle[p_z^y] \leq \pi.$$

2.14. Without loss of generality, we can assume that x , v , w , and y appear on $[xy]$ in this order. By 2.8,

$$\tilde{\angle}(x_p^y) \geq \tilde{\angle}(x_p^w) \geq \tilde{\angle}(x_p^v).$$

Hence, $\Rightarrow$ follows.

By Alexandrov's lemma,

$$\begin{aligned} \tilde{\angle}(x_p^y) = \tilde{\angle}(x_p^v) &\Leftrightarrow \tilde{\angle}(y_p^x) = \tilde{\angle}(y_p^v), \\ \tilde{\angle}(x_p^y) = \tilde{\angle}(x_p^w) &\Leftrightarrow \tilde{\angle}(y_p^x) = \tilde{\angle}(y_p^w). \end{aligned}$$

Whence, $\Leftarrow$ follows.

2.15. Suppose $\angle[x_\infty y_\infty] > \alpha$. Then we can choose $\bar{y}_\infty \in]x_\infty y_\infty]$ and $\bar{z}_\infty \in]x_\infty z_\infty]$ such that $\tilde{\angle}(x_\infty \bar{y}_\infty) > \alpha$. Now choose $\bar{y}_n \in]x_n y_n]$ and $\bar{z}_n \in]x_n z_n]$ such that $\bar{y}_n \rightarrow \bar{y}_\infty$ and $\bar{z}_n \rightarrow \bar{z}_\infty$. Observe that

$$\lim_{n \rightarrow \infty} \angle[x_n y_n] \geq \lim_{n \rightarrow \infty} \tilde{\angle}(x_n \bar{y}_n) = \tilde{\angle}(x_\infty \bar{y}_\infty),$$

hence the result.

2.18. The Urysohn space provides an example; see, for example, [102, Lecture 2].

2.19. Choose a triangle $[0vw]$. Note that $m = \frac{1}{2}(v+w)$ is the midpoint of $[vw]$.

Use comparison to show that

$$2 \cdot |\tfrac{1}{2}(v+w)|^2 + 2 \cdot |\tfrac{1}{2}(v-w)|^2 \geq |v|^2 + |w|^2.$$

This inequality implies the opposite one; it follows if we rewrite it via $x = \frac{1}{2}(v+w)$ and $y = \frac{1}{2}(v-w)$. Hence we have

$$2 \cdot |\tfrac{1}{2}(v+w)|^2 + 2 \cdot |\tfrac{1}{2}(v-w)|^2 = |v|^2 + |w|^2$$

for any v, w . That is, the norm meets the parallelogram identity. It is well known that all such norms are quadratic, and the statement follows.

2.20. Choose a λ -concave function $f: \mathbb{I} \rightarrow \mathbb{R}$. Note that

$$h: t \mapsto f(t) - \lambda \cdot \frac{t^2}{2}$$

is concave, and it is sufficient to show that h is one-sided differentiable.

Choose $t_0 \in \mathbb{I}$. Show and use that the function $t \mapsto \frac{h(t) - h(t_0)}{t - t_0}$ is nonincreasing.

2.21. We can assume that $\lambda = 0$; otherwise pass to the function $t \mapsto f(t) - \lambda \cdot \frac{t^2}{2}$.

Suppose that f is defined at t_0 . Since f is defined on an open interval, we can choose a and b such that $a < t_0 < b$ and f is defined on $[a, b]$. Show and use that

$$\frac{f(b) - f(t_0)}{b - t_0} \leq \frac{f(t) - f(t_0)}{t - t_0} \leq \frac{f(a) - f(t_0)}{a - t_0}$$

for any $t \in [a, b] \setminus \{t_0\}$.

2.23. From 2.22, this inequality follows in the sense of distributions, and hence in any other sense.

3.4. Suppose such a point does not exist; that is, for any $p \in \mathcal{X}$ there is a point p' such that $r(p') \leq (1 - \varepsilon) \cdot r(p)$ and $|p - p'| < \frac{1}{\varepsilon} \cdot r(p)$. Construct a sequence of points $p_0, p_1, \dots$ such that $p_n = p'_{n-1}$ for any n . Show that this sequence is Cauchy; denote its limit by p_∞ . Arrive at a contradiction by showing that $r(p_\infty) \leq 0$.

Comment. A slightly more general version of this statement was proved by Ivar Ekeland [29].

3.6. Suppose $\angle[m_p^x] \neq 0$ and $\angle[m_p^x] \neq \pi$; equivalently, $\angle[m_p^x] \neq 0$ and $\angle[m_q^x] \neq 0$.

We can assume that $|p - q|$ only slightly exceeds π , so $|p - m| < \pi$ and $|q - m| < \pi$. We can also assume that $|x - m| < \pi$. Use the comparison to show that

$$|p - x| + |q - x| < |p - q|,$$

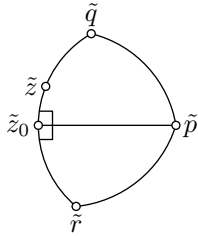
and arrive at a contradiction with the triangle inequality.

Extend $[pq]$ to a maximal local geodesic γ . Argue as above to show that any point lies on γ . If γ is closed, then the space is isometric to a circle; otherwise, it is isometric to a line segment.

3.7. Arguing by contradiction, suppose

$$\bullet \quad |p - q| + |q - r| + |r - p| > 2 \cdot \pi$$

for $p, q, r \in \mathcal{A}$. Rescaling the space slightly, we can assume that $\text{diam } \mathcal{A} < \pi$, but the inequality $\bullet$ still holds. By 3.5, after rescaling $\mathcal{A}$ is still ALEX(1).



Take $z_0 \in [qr]$ at maximal distance from p . Consider the following model configuration: two geodesics $[\tilde{p}\tilde{z}_0]$, $[\tilde{q}\tilde{r}]$ in $\mathbb{S}^2$ such that

$$\begin{aligned} |\tilde{p} - \tilde{z}_0| &= |p - z_0|, & |\tilde{q} - \tilde{r}| &= |q - r|, \\ |\tilde{z}_0 - \tilde{q}| &= |z_0 - q|, & |\tilde{z}_0 - \tilde{r}| &= |z_0 - r|, \end{aligned}$$

and

$$\angle[\tilde{z}_0 \tilde{q}] = \angle[\tilde{z}_0 \tilde{r}] = \frac{\pi}{2}.$$

Choose $\tilde{z} \in [\tilde{q}\tilde{r}]$, and let $z \in [qr]$ be the corresponding point. By comparison, $|p - z| \leq |\tilde{p} - \tilde{z}|$ if z lies near z_0 . Moreover, this inequality holds as far as

$$|\tilde{p} - \tilde{z}_0| + |\tilde{z}_0 - \tilde{z}| + |\tilde{p} - \tilde{z}| < 2 \cdot \pi.$$

But this inequality holds for all $\tilde{z}$ since $|\tilde{p} - \tilde{z}_0| < \pi$, $|\tilde{z}_0 - \tilde{q}| < \pi$, and $|\tilde{z}_0 - \tilde{r}| < \pi$. Hence $|p - q| \leq |\tilde{p} - \tilde{q}|$ and $|p - r| \leq |\tilde{p} - \tilde{r}|$. The latter contradicts $\bullet$.

4.1. Suppose $\uparrow_{[p_{x_n}]} \not\rightarrow \uparrow_{[p_{x_\infty}]}$. Since Σ_p is compact, we may pass to a converging subsequence of $\uparrow_{[p_{x_n}]}$; denote its limit by ξ . We may assume that $\angle(\uparrow_{[p_{x_\infty}]}, \xi) > 0$.

Denote by γ_n and γ_∞ the arc-length parametrization of $[px_n]$ and $[px_\infty]$ from p . For a geodesic α that starts from p and goes in a direction sufficiently close to ξ , we have

$$|\alpha(t) - \gamma_n(t)| < \varepsilon \cdot t \quad \text{and} \quad |\alpha(t) - \gamma_\infty(t)| > a \cdot t$$

for some $\varepsilon \ll a$, all large n , and all sufficiently small t . These two inequalities imply that

$$|\gamma_n(t) - \gamma_\infty(t)| > \frac{a}{2} \cdot t$$

for all small t and all large n . On the other hand, by assumption, $|\gamma_n(t) - \gamma_\infty(t)| \rightarrow 0$ as $n \rightarrow \infty$ — a contradiction.

Comments. The compactness of Σ_p is necessary. A counterexample with noncompact Σ_p can be built using an iterated warped product of line segments and applying [3, Theorem 1.2]. The space $\mathcal{A}$ can be assumed to be compact.

4.2. Any point of Cone $\mathcal{X}$ can be connected to the origin by a geodesic. Given a nonzero element $v \in \text{Cone } \mathcal{X}$, denote by v' its projection in $\mathcal{X}$; so, $v = |v| \cdot v'$.

Suppose $\mathcal{X}$ is π -geodesic. Choose two nonzero elements $v, w \in \text{Cone } \mathcal{X}$; let $\alpha = \angle(v, w) = |v' - w'|_{\mathcal{X}}$. If $\alpha \geq \pi$, then the product of geodesics $[v0]$ and $[0w]$ is a geodesic $[vw]$. If $\alpha < \pi$, there is a geodesic $\gamma: [0, \alpha] \rightarrow \mathcal{X}$ from v' to w' . Consider the hinge $[\tilde{o} \frac{\tilde{v}}{\tilde{w}}]$ in the plane such that $\angle[\tilde{o} \frac{\tilde{v}}{\tilde{w}}] = \alpha$, $|\tilde{o} - \tilde{v}| = |v|$, and $|\tilde{o} - \tilde{w}| = |w|$. Let $t \mapsto (\varphi(t), r(t))$ be the geodesic $[\tilde{v}\tilde{w}]$ written in polar coordinates with origin at $\tilde{o}$, so that $\varphi(0) = 0$. Show that $t \mapsto r(t) \cdot \gamma \circ \varphi(t)$ is a geodesic from v to w ; here we identify $\mathcal{X}$ with the unit sphere in Cone $\mathcal{X}$.

To prove the converse, reverse the steps in the argument.

4.4. Let $\mathcal{A}_n = \lambda_n \cdot \mathcal{A}$. Note that for any n the space $\Sigma_p \mathcal{A}$ is identical to $\Sigma_{\iota_n(p)} \mathcal{A}_n$. In particular, we can identify isometrically $T_p \mathcal{A}$ with $T_{\iota_n(p)} \mathcal{A}_n$. So for any geodesic γ that starts at p , the vector $\gamma^+(0)$ corresponds to $\frac{1}{\lambda_p} \cdot (\iota_n \circ \gamma)^+(0)$.

Consider the logarithm maps $f_n = \log_{\iota_n(p)}: \mathcal{A}_n \rightarrow T_p \mathcal{A}$. We claim that this sequence of maps satisfies the assumptions in 1.28.

The condition in (a) is evident, and it is sufficient to check the conditions in (b) and (c) only for $R = 1$.

Choose $\varepsilon > 0$. By compactness of Σ_p we can find a finite ε -net $\xi_1, \dots, \xi_N$ in Σ_p . Moreover, without loss of generality we can assume that these directions are geodesic; that is, there exist geodesics $\gamma_1, \dots, \gamma_N$ starting at p such that $\xi_i = \gamma_i^+(0)$ for each i .

Choose $T > 0$ such that all γ_i are defined on $[0, T]$. Show that for any $\lambda_n > \frac{1}{T}$ the image under f_n of the union $\bigcup_i \gamma_i([0, T])$ is an ε -net in $B(0, 1)_{T_p}$. This proves (c).

By comparison, we have that

$$|\xi_i - \xi_j|_{\Sigma_p} \geq \tilde{Z}(p^{\gamma_i(t_i)})$$

for all $i \neq j$ and any $t_i, t_j \in (0, T]$. By the definition of angle, we can assume that T has been chosen so that in addition

$$|\xi_i - \xi_j|_{\Sigma_p} \leq \tilde{Z}(p^{\gamma_i(t)}) + \varepsilon$$

for all $i \neq j$ and any $t \in (0, T]$.

By construction of the map f_n this implies that

$$\left| |x - x'|_{A_n} - |f_n(x) - f_n(x')|_{T_p} \right| < \varepsilon$$

for all $\lambda_n > \frac{1}{T}$ and all points x, x' in $\bigcup_i \gamma_i([0, \frac{1}{\lambda_n}]) \subset B(p, 1)_{A_n}$.

Now hinge comparison and the triangle inequality imply that the same holds for arbitrary points x, x' in $B(p, 1)_{A_n}$ with ε replaced by $3 \cdot \varepsilon$. This verifies (b).

4.5. Since angles are defined, it follows that

$$|\gamma_1(t) - \gamma_2(t)| \leq \theta \cdot t$$

for all small $t > 0$. Since f is L -Lipschitz, we get

$$|f(\gamma_1(t)) - f(\gamma_2(t))| \leq L \cdot \theta \cdot t,$$

and the statement follows.

4.6; (a) Show that we can assume there is a geodesic in the direction of v , and apply 1.13.

(b). By (a), $(\mathbf{d}_p \text{dist}_q)(v) \leq -\max_{\xi \in \mathbb{H}_p^q} \langle \xi, v \rangle$. Suppose this inequality is strict for some v . We can assume that $|v| = 1$ and there is a geodesic, say γ in the direction of v . Let $(\mathbf{d}_p \text{dist}_q)(v) = -\cos \alpha_0$ for some $\alpha_0 \in [0, \pi]$. Note that any geodesic from p to q makes angle bigger than α_0 with γ .

The function $f = \text{dist}_q \circ \gamma$ is Lipschitz. By Rademacher's theorem it is differentiable almost everywhere; moreover,

$$f(t) - f(0) = \int_0^t f'(s) \cdot ds.$$

Suppose $f'(t)$ is defined. Use (a) to show that $f'(t) = -\cos \alpha(t)$, where $\alpha(t)$ is the angle between γ and any geodesic from $\gamma(t)$ to q . We can choose a sequence $t_n \rightarrow 0$ such that

$$\lim_{n \rightarrow \infty} \alpha(t_n) \leq \alpha_0.$$

Consider a sequence of geodesics $[\gamma(t_n)q]$. Since the space is proper, we can pass to a convergent subsequence. Its limit is a geodesic from p to q , denote it by $[pq]$.

Use 2.15 to show that $[pq]$ makes an angle at most α_0 with $\gamma - a$ contradiction.

4.8. Let $\gamma: [0, \ell] \rightarrow \mathcal{A}$ be the geodesic $[xy]$ parametrized from x to y , and let $\varphi = f \circ \gamma$. Observe that

$$\varphi'(0) = \mathbf{d}_x f(\uparrow_{[xy]}) \leq \langle \uparrow_{[xy]}, \nabla_x f \rangle.$$

The same way we get $-\varphi'(\ell) \leq \langle \uparrow_{[yx]}, \nabla_y f \rangle$. Since f is λ -concave, we have

$$\begin{aligned} f(y) &\leq f(x) + \varphi'(0) \cdot \ell + \frac{\lambda}{2} \cdot \ell^2, \\ f(x) &\leq f(y) - \varphi'(\ell) \cdot \ell + \frac{\lambda}{2} \cdot \ell^2. \end{aligned}$$

Hence the statement follows.

4.11. If the space is proper, then the statement follows from 4.6b and 2.12.

To do the general case, let us argue by contradiction. By the assumption, we can choose a point z on the extension of $[pq]$ behind q . We can assume that $|v| = 1$ and it is a direction of a geodesic, say $[px]$.

Show that there is a sequence $x_n \in]px]$ such that $|p - x_n| \rightarrow 0$ and $\angle[q_p^{x_n}] > \varepsilon$ for each n and some fixed $\varepsilon > 0$. Observe that $\angle[q_z^{x_n}] < \pi - \varepsilon$; therefore

$$|z - x_n| < |x_n - q| + |q - z| - \delta$$

for each n and some fixed $\delta > 0$. Pass to the limit as $x_n \rightarrow p$ and arrive at a contradiction to the triangle inequality.

4.12. Note that $|(\mathbf{d}_p f)(v) - (\mathbf{d}_p g)(v)| \leq s \cdot |v|$ for any $v \in T_p$. From the definition of gradient (4.7) we have:

$$\begin{aligned} (\mathbf{d}_p f)(\nabla_p g) &\leq \langle \nabla_p f, \nabla_p g \rangle, & (\mathbf{d}_p g)(\nabla_p f) &\leq \langle \nabla_p f, \nabla_p g \rangle, \\ (\mathbf{d}_p f)(\nabla_p f) &= \langle \nabla_p f, \nabla_p f \rangle, & (\mathbf{d}_p g)(\nabla_p g) &= \langle \nabla_p g, \nabla_p g \rangle. \end{aligned}$$

Therefore,

$$\begin{aligned} |\nabla_p f - \nabla_p g|^2 &= \langle \nabla_p f, \nabla_p f \rangle + \langle \nabla_p g, \nabla_p g \rangle - 2 \cdot \langle \nabla_p f, \nabla_p g \rangle \leq \\ &\leq (\mathbf{d}_p f)(\nabla_p f) + (\mathbf{d}_p g)(\nabla_p g) - (\mathbf{d}_p f)(\nabla_p g) - (\mathbf{d}_p g)(\nabla_p f) \leq \\ &\leq s \cdot (|\nabla_p f| + |\nabla_p g|). \end{aligned}$$

4.13. Suppose $|\nabla_x f| > s$. Then we can choose a geodesic γ that starts at x such that $(f \circ \gamma)^+(0) > s$. In particular, there is $\varepsilon > 0$ such that

$$f \circ \gamma(t) > (s + \varepsilon) \cdot t + o(t),$$

and the only-if part follows.

Now suppose $f(y) - f(x) > s \cdot \ell + \lambda \cdot \frac{\ell^2}{2}$, where $\ell = |x - y|$. Let $\gamma: [0, \ell] \rightarrow \mathcal{A}$ be a geodesic from x to y . Since $f \circ \gamma$ is λ -concave, we have

$$f \circ \gamma(\ell) \leq f \circ \gamma(0) + (f \circ \gamma)^+(0) \cdot \ell + \lambda \cdot \frac{\ell^2}{2}.$$

It follows that

$$(\mathbf{d}_x f)(\uparrow_{[xy]}) = (f \circ \gamma)^+(0) > s,$$

and by 4.9, $|\nabla_x f| > s$.

5.7. Note that $f \circ \alpha$ is a nondecreasing function. Apply 4.6a and the definition of gradient to show that

$$\begin{aligned} -(\mathbf{d}_{\alpha(t)} \text{dist}_{\alpha(t_3)})(\nabla_{\alpha(t)} f) &\geq \langle \nabla_{\alpha(t)} f, \uparrow_{[\alpha(t)\alpha(t_3)]} \rangle \geq \\ &\geq \mathbf{d}_{\alpha(t)} f(\uparrow_{[\alpha(t)\alpha(t_3)]}) \geq \\ &\geq 0 \end{aligned}$$

for any $t < t_3$. Conclude that the function $t \mapsto \text{dist}_{\alpha(t_3)} \circ \alpha(t)$ is nonincreasing for $t \leq t_3$.

5.8. Suppose $s > s_0$. Then

$$\begin{aligned} (f \circ \hat{\alpha})^+(s_0) &= |\nabla_{\hat{\alpha}(s_0)} f| \geq \\ &\geq (d_{\hat{\alpha}(s_0)} f)(\uparrow_{[\hat{\alpha}(s_0)\hat{\alpha}(s)]}) \geq \\ &\geq \frac{f \circ \hat{\alpha}(s) - f \circ \hat{\alpha}(s_0)}{|\hat{\alpha}(s) - \hat{\alpha}(s_0)|}. \end{aligned}$$

Since $s - s_0 \geq |\hat{\alpha}(s) - \hat{\alpha}(s_0)|$,

$$(f \circ \hat{\alpha})^+(s_0) \geq \frac{f \circ \hat{\alpha}(s) - f \circ \hat{\alpha}(s_0)}{s - s_0},$$

which implies the statement.

5.9. Fix t , and let $p = \alpha(t)$ and $q = \beta(t)$. Apply 5.5 to get

$$\begin{aligned} \ell^+ &\leq -\langle \uparrow_{[pq]}, \nabla_p f \rangle - \langle \uparrow_{[qp]}, \nabla_q g \rangle \leq \\ &\leq -\left(f(q) - f(p) - \lambda \cdot \frac{\ell^2}{2}\right)/\ell - \left(g(p) - g(q) - \lambda \cdot \frac{\ell^2}{2}\right)/\ell \leq \\ &\leq \lambda \cdot \ell + \frac{2 \cdot \varepsilon}{\ell}. \end{aligned}$$

Integrating this inequality, we get the second statement.

5.13. Choose a point $p \in M$. Observe that gexp_p^1 provides a short map from the unit hemisphere $\mathbb{S}_+^m$ onto $\overline{B}[p, \frac{\pi}{2}]_{(M,g)}$. Composing this map with the quotient map $\mathbb{S}^m \rightarrow \mathbb{S}_+^m$ gives us a short map $s: \mathbb{S}^m \rightarrow (M, g)$.

Since M is not homeomorphic to the sphere, the diameter-sphere theorem implies that $\text{diam}(M, g) \leq \frac{\pi}{2}$. In particular, $\overline{B}[p, \frac{\pi}{2}]_{(M,g)} = (M, g)$. Hence, the map s is onto.

6.2. Apply 2.23.

6.3. By the triangle inequality,

$$|\gamma(-t) - x| + |\gamma(t) - x| - 2 \cdot t \geq 0$$

for any $t \geq 0$. Passing to the limit as $t \rightarrow \infty$, we get the result.

6.4; (a). Show and use that $\text{proj}_A B = \text{proj}_B A$ is a one-point set.

(b). Suppose α is a rectifiable curve with the endpoints in A . Show and use that the curve $\beta = \text{proj}_A \circ \alpha$ has the same endpoints and

$$\text{length } \beta \leq \text{length } \alpha;$$

moreover, in case of equality, $\beta = \alpha$.

6.8. Suppose $\text{Cone } \mathcal{X}$ is ALEX(0). Observe that two half-lines in $\text{Cone } \mathcal{X}$ that start from the origin and go into directions x and $y \in \mathcal{X}$ form a line if and only if $|x - y|_{\mathcal{X}} \geq \pi$. Apply the splitting theorem to show that for any $x \in \mathcal{X}$ there is at most one point y such that $|x - y|_{\mathcal{X}} \geq \pi$ and in this case we have equality. Conclude that $\text{diam } \mathcal{X} \leq \pi$.

Now choose a quadruple of points $p, x_1, x_2, x_3 \in \mathcal{X}$; we will identify $\mathcal{X}$ with the unit sphere in $\text{Cone } \mathcal{X}$. Suppose $|p - x_i| < \frac{\pi}{2}$ for any i . Consider the following points in the cone: $y_i = \frac{1}{\cos |p - x_i|_{\mathcal{X}}} \cdot x_i$, and $q = p$. Show that $\mathbb{E}^2$ -comparison for q, y_1, y_2, y_3 in $\text{Cone } \mathcal{X}$ implies $\mathbb{S}^2$ -comparison for p, x_1, x_2, x_3 in $\mathcal{X}$. Conclude that $\mathcal{X}$ is locally ALEX(1). Apply the globalization theorem (3.3).

Now assume $\mathcal{X}$ is ALEX(1) and $\text{diam } \mathcal{X} \leq \pi$. By 3.7, the perimeter of any triangle in $\mathcal{X}$ is at most $2 \cdot \pi$. We need to check $\mathbb{E}^2$ -comparison for a given quadruple of points q, y_1, y_2, y_3 in $\text{Cone } \mathcal{X}$. We can assume that none of these points is the origin; otherwise perturb them a bit.

Set $x_i = y_i / |y_i|$ for each i and $p = q / |q|$; we can assume that p, x_1, x_2, x_3 are distinct in $\mathcal{X}$, which is the unit sphere in $\text{Cone } \mathcal{X}$.

Assume the model triangles $\tilde{\Delta}(px_1x_2)$, $\tilde{\Delta}(px_2x_3)$, and $\tilde{\Delta}(px_3x_1)$ are defined; that is, the perimeters of the triangles $[px_1x_2]$, $[px_2x_3]$,

and $[px_3x_1]$ are strictly less than $2\cdot\pi$. Note that $\mathbb{E}^3 \stackrel{\text{iso}}{=} \text{Cone}\mathbb{S}^2$. Use this together with the $\mathbb{S}^2$ -comparison for p, x_1, x_2, x_3 in $\mathcal{X}$ to show that $\mathbb{E}^2$ -comparison holds for q, y_1, y_2, y_3 in $\text{Cone}\mathcal{X}$.

Finally, if one of the model triangles is undefined, consider rescaling of $\mathcal{X}$ with a coefficient λ slightly smaller than 1. Apply the argument above to show that the comparison holds for the corresponding points in $\text{Cone}(\lambda\cdot\mathcal{X})$ and pass to the limit as $\lambda \rightarrow 1$.

Comment. The last part of the proof is close to the argument in 8.1.

6.10. Observe that

$$\begin{aligned}\langle u, u \rangle + \langle v, u \rangle + \langle w, u \rangle &\geq 0, \\ \langle u, v \rangle + \langle v, v \rangle + \langle w, v \rangle &\geq 0, \\ \langle u, w \rangle + \langle v, w \rangle + \langle w, w \rangle &= 0.\end{aligned}$$

Add the first two inequalities and subtract the last identity.

6.14. Apply 6.13 to show that $\langle v, v \rangle = \langle v, w \rangle = \langle w, w \rangle$, and use it.

6.15. Show and use that

$$\langle u, x \rangle + \langle v, x \rangle + \langle w, x \rangle \geq 0$$

and

$$\langle u, -x \rangle + \langle v, -x \rangle + \langle w, -x \rangle \geq 0.$$

6.16. Part $\Rightarrow$ is evident. To prove part $\Leftarrow$, observe that

$$\langle u^*, u^* \rangle = -\langle u, u^* \rangle \leq \langle u, u \rangle$$

and since $|u| = |u^*|$, we have equality.

6.19. Apply 6.16.

7.2. Let $L \subset T_p$ be a subspace isometric to $\mathbb{E}^m$. Assume $L \neq T_p$.

Let $N = \text{pack}_\varepsilon \mathbb{S}^{m-1}$. Observe that the unit sphere in L is isometric to $\mathbb{S}^{m-1}$ and use it to show that $\text{pack}_\varepsilon \Sigma_p > N$ for some $\varepsilon > 0$.

It follows that we can choose $N+1$ directions in Σ_p at angles larger than ε from each other. Use it to construct points $x_1, \dots, x_{N+1}$ such that $\tilde{\angle}(p_{x_j}^{x_i}) > \varepsilon$ for $i \neq j$, and apply 7.1.

7.3. Argue as in 3.6.

7.4. The only-if part is trivial. Suppose the configuration $p, a_0, \dots, a_m \in \mathcal{A}$ meets the condition. By 6.19 the directions $\uparrow_{[qa_0]}, \dots, \uparrow_{[qa_m]} \in \text{Lin}_q$ for a G-delta dense set of points $q \in \mathcal{A}$. If q is sufficiently close to p , then $\tilde{\angle}(q_{a_j}^{a_i}) > \frac{\pi}{2}$, and therefore, $\angle[q_{a_j}^{a_i}] > \frac{\pi}{2}$ for

$i \neq j$. Conclude that $\dim \text{Lin}_q \geq m$ in this case. To see this, show and use that given $m+1$ vectors in $\mathbb{R}^k$ making pairwise obtuse angles with each other, any m of them must be linearly independent.

7.7. We can assume that $f(p) = 0$ and f is 1-concave.

Choose a sequence of positive numbers $\lambda_n \rightarrow \infty$. Let $\iota_n: \mathcal{A} \rightarrow \lambda_n \cdot \mathcal{A}$ be the rescaling maps. Consider functions $f_n: \lambda_n \cdot \mathcal{A} \rightarrow \mathbb{R}$ defined by

$$f_n(\iota_n(x)) = \lambda_n \cdot f(x).$$

Observe that f_n is $\frac{1}{\lambda_n}$ -concave and $f_n \rightarrow \mathbf{d}_p f$ as $n \rightarrow \infty$ for the convergence $(\lambda_n \cdot \mathcal{A}, p) \rightarrow (T_p, 0)$ from 7.6.

Choose a geodesic γ in T_p ; assume it can be extended behind its endpoints. Construct a sequence of geodesics γ_n in $\lambda_n \cdot \mathcal{A}$ that converge to γ (here you should use that γ can be extended). Since $f_n \circ \gamma_n \rightarrow \mathbf{d}_p f \circ \gamma$, we get that $\mathbf{d}_p f$ is concave on γ , which implies the statement.

7.9; (a). By 2.23, each function dist_{a_i} is semiconcave in a small neighborhood of p . Therefore we can choose λ and $r > 0$ so that $f_{\mathbf{y}}$ is λ -concave in $B(p, r)$; further we will assume that r is sufficiently small. Choose $\alpha > 0$ such that $\tilde{\angle}(x_{a_j}^{a_i}) > \frac{\pi}{2} + \alpha$ for all $i \neq j$; we may assume that $\alpha < \frac{1}{10}$; in particular,

$$\textcircled{2} \quad (\mathbf{d}_x \text{dist}_{a_j})(\uparrow_{[xa_i]}) \geq -\cos \tilde{\angle}(x_{a_j}^{a_i}) > \frac{\alpha}{2}$$

for $j \neq i$.

By the definition of gradient and 4.6a, we have

$$\begin{aligned} -(\mathbf{d}_x \text{dist}_{a_i})(\nabla_x f_{\mathbf{y}}) &\geq \langle \uparrow_{[xa_i]}, \nabla_x f_{\mathbf{y}} \rangle \geq \\ &\geq (\mathbf{d}_x f_{\mathbf{y}})(\uparrow_{[xa_i]}). \end{aligned}$$

If $|a_i - x| > y_i$, then

$$\mathbf{d}_x f_{\mathbf{y}} = \sigma + \varepsilon \cdot \mathbf{d}_x \text{dist}_{a_0},$$

where σ is a minimum of a subset of the following functions: 0, and $\mathbf{d}_x \text{dist}_{a_j}$ for $0 \neq j \neq i$. By $\textcircled{2}$,

$$(\mathbf{d}_x \text{dist}_{a_i})(\nabla_x f_{\mathbf{y}}) < -\frac{\alpha}{2} \cdot \varepsilon.$$

Hence (i) holds for all sufficiently small $\varepsilon > 0$.

Now assume that $|a_i - x| - y_i = \min_j \{|a_j - x| - y_j\} < 0$. Then

$$\begin{aligned} \mathbf{d}_x f_{\mathbf{y}} &= \min_{j \in S} \{ \mathbf{d}_x \text{dist}_{a_j} \} + \varepsilon \cdot \mathbf{d}_x \text{dist}_{a_0} \leq \\ &\leq \mathbf{d}_x \text{dist}_{a_i} + \varepsilon \cdot \mathbf{d}_x \text{dist}_{a_0}, \end{aligned}$$

where $j \in S$ if and only if $|a_i - x| - y_i = |a_j - x| - y_j$. Applying ❷, we get

$$\begin{aligned} (\mathbf{d}_x \text{dist}_{a_i})(\nabla_x f_{\mathbf{y}}) &\geq \mathbf{d}_x f_{\mathbf{y}}(\nabla_x f_{\mathbf{y}}) - \varepsilon \cdot (\mathbf{d}_x \text{dist}_{a_0})(\nabla_x f_{\mathbf{y}}) \geq \\ &\geq [(\mathbf{d}_x f_{\mathbf{y}})(\uparrow_{[x a_0]})]^2 - 2 \cdot \varepsilon \geq \\ &\geq \left[\frac{\alpha}{2} - \varepsilon\right]^2 - 2 \cdot \varepsilon. \end{aligned}$$

Thus, (ii) holds for all sufficiently small $\varepsilon > 0$.

(b) Consider the following real-to-real functions:

$$\begin{aligned} \varphi(t) &:= \max_i \{|a_i - \alpha_{\mathbf{y}}(t)| - y_i\}, \\ \psi(t) &:= \min_i \{|a_i - \alpha_{\mathbf{y}}(t)| - y_i\}. \end{aligned}$$

Use (a) to show that for $t \in [0, t_0]$, we have $\varphi^+(t) < -\frac{1}{10} \cdot \varepsilon^2$ if $\varphi(t) > 0$ and $\psi^+(t) > \frac{1}{10} \cdot \varepsilon^2$ if $\psi(t) < 0$. Conclude that $\varphi(t_0) = \psi(t_0) = 0$; hence the result.

(c) A straightforward application of 5.9 and a reformulation of (b).

Remarks. By 5.9, the constructed map Φ is bi-Hölder with the exponent $\frac{1}{2}$. In particular, *every infinite-dimensional Alexandrov space $\mathcal{A}$ contains a bi-Hölder copy of a Euclidean ball of arbitrary dimension.* It seems plausible that $\mathcal{A}$ should contain a bi-Lipschitz copy of a Euclidean ball of arbitrary dimension, but this question is open.

7.12. Apply the $(n+1)$ -comparison (7.1) to show that at least one of the inequalities

$$\angle[x_{a_0}^y] < \frac{\pi}{2} - \varepsilon, \dots, \angle[x_{a_m}^y] < \frac{\pi}{2} - \varepsilon,$$

holds. Similarly, we get that at least one of the inequalities

$$\angle[y_{a_0}^x] < \frac{\pi}{2} - \varepsilon, \dots, \angle[y_{a_m}^x] < \frac{\pi}{2} - \varepsilon,$$

holds.

Suppose our statement does not hold for x and y in a sufficiently small neighborhood of p . It follows that

$$\angle[x_{a_0}^y] < \frac{\pi}{2} - \varepsilon \quad \text{and} \quad \angle[y_{a_0}^x] < \frac{\pi}{2} - \varepsilon.$$

Note that $|x - y|$ is small compared to $|a_0 - x|$ and $|a_0 - y|$. Therefore, the comparison contradicts ❹.

By the construction, f is Lipschitz. By the above, we can choose an index $i > 0$ so that $\angle[x_{a_i}^y] < \frac{\pi}{2} - \varepsilon$ (if $\angle[y_{a_i}^x] < \frac{\pi}{2} - \varepsilon$, then swap x and

y). By comparison, there is $c > 0$ such that $|a_i - y| \leq |a_i - x| + c \cdot |x - y|$. Hence f is bi-Lipschitz, and now 7.8 implies 7.11.

7.14. Follow the proof of the Bishop–Gromov inequality, plus prove the following two inequalities

$$\begin{aligned}\sinh r_2 \cdot |\log_p x - \log_p y|_{T_p} &\geq |x - y|_{\mathcal{A}}, \\ \sinh r_2 \cdot |w(x) - w(y)|_{\mathcal{A}} &\geq \sinh r_1 \cdot |x - y|_{\mathcal{A}}\end{aligned}$$

for any $x, y \in B(p, r_2)$.

7.15. Reuse the argument from the first part of the proof of the Bishop–Gromov inequality.

7.16. Applying the right-inverse theorem, show that some ball has positive volume. Further, use the Bishop–Gromov inequality to show that any ball with positive radius has positive volume.

Alternatively, use 7.11 applied to a neighborhood of any point p with $T_p \stackrel{iso}{=} \mathbb{E}^m$.

7.17; (a). Apply the dilation map that was used in the proof of the Bishop–Gromov inequality (7.13 and 7.14).

(b). Apply the Bishop–Gromov inequality (7.13 and 7.14).

7.19. Suppose K is a compact set in $\mathcal{A}$ such that $\text{HausDim } K \geq m$. Use the map w from the proof of the Bishop–Gromov inequality (7.13 and 7.14) to show that any open ball in $\mathcal{A}$ contains a compact set K' such that $\text{HausDim } K' \geq m$.

Use this in addition to the arguments in 7.18.

7.20; (a). Apply 4.2, 4.3, and 7.5.

(b). Apply 7.4 to show that $\text{LinDim } T_p \leq \text{LinDim } \mathcal{A}$ (argue as in 4.3). Argue as in the finite-dimensional case in the proof of 7.18 to obtain that $\text{LinDim } T_p \geq \text{LinDim } \mathcal{A}$.

(c). By 7.5 for any two points $\xi, \zeta \in \Sigma_p$ such that $|\xi - \zeta|_{\Sigma_p} < \pi$ there is a geodesic $[\xi\zeta]_{\Sigma_p}$. Suppose $|\xi - \zeta|_{\Sigma_p} \geq \pi$. Then T_p contains a line thru the origin in the directions ξ and ζ . By (a) we can apply the splitting theorem (6.5) to T_p . We get that Σ_p is a spherical suspension with poles ξ and ζ . Hence, $|\xi - \zeta| = \pi$ and there is a geodesic $[\xi\zeta]$.

8.2. Apply 7.4.

8.4. Argue as in 8.3 to construct a Gromov–Hausdorff convergence of $\bar{B}(p_n, R)_{\mathcal{A}_n}$ for a given $R > 0$, then apply the diagonal procedure to construct the needed convergence.

8.5. By 8.2, $\dim \mathcal{A}_\infty \leq m$. To show that $\dim \mathcal{A}_\infty \geq m$, apply 6.20 and 7.17b.

Comment. The following stronger statement holds

$$\operatorname{vol}_m \mathcal{A}_\infty = \lim_{n \rightarrow \infty} \operatorname{vol}_m \mathcal{A}_n.$$

In other words, if $\mathbf{K}_m \subset \text{GH}$ denotes the set of isometry classes of all compact ALEX(κ) spaces with dimension $\leq m$, then the function $\operatorname{vol}_m: \mathbf{K}_m \rightarrow \mathbb{R}$ is continuous [20, 10.8].

8.8. Choose a smooth upper barrier $\bar{f}$ of f at t_0 with $\bar{f}'' \leq \lambda_1$, provided by the lemma. Observe that the composition $\varphi \circ \bar{f}$ is an upper barrier of $\varphi \circ f$ at t_0 . It remains to apply the chain rule to $\varphi \circ \bar{f}$ and use the lemma again.

8.9. Choose a ball $B = \bar{B}[x, R]$ such that $\bar{B}[x, 2 \cdot R] \subset \Omega$. By 7.17b, we can choose a constant $c_1 > 0$ such that for every sufficiently small $\varepsilon > 0$ the ball B contains at least c_1/ε^m points $q_1, \dots, q_N$ such that $|q_i - q_j| > 10 \cdot \varepsilon$. Slice B into sets

$$B_k = \{z \in B : k \cdot \varepsilon \leq |p - z| < k + 1 \cdot \varepsilon\}.$$

There are at most $1 + 2 \cdot R/\varepsilon$ nonempty sets B_k . Therefore, for a fixed constant $c_2 > 0$ we can choose B_k that contains at least c_2/ε^{m-1} points from $\{q_1, \dots, q_N\}$. Let $r = k \cdot \varepsilon$, for each $q_i \in B_k$ choose $q'_i \in [pq_i]$ such that $|p - q'_i| = r$. It remains to observe that (1) $q'_i \in \Omega$ for each i and (2) $|q'_i - q'_j| > 8 \cdot \varepsilon$ and $\angle[pq'_i] > 8 \cdot \varepsilon/r$ if $i \neq j$.

8.11. Consider the infinite product $\mathbb{S}^1 \times (\frac{1}{2} \cdot \mathbb{S}^1) \times (\frac{1}{4} \cdot \mathbb{S}^1) \times \dots$

8.12. Combine 7.12 and the proof of 8.6.

8.13. Consider the canonical metric g on the round unit sphere $\mathbb{S}^3$ and the Hopf bundle $\mathbb{S}^3 \rightarrow \mathbb{S}^2$. Let g_n be obtained from g by rescaling the Hopf fibers by $\frac{1}{n}$. Then all $(\mathbb{S}^3, g_n)$ have nonnegative curvature and this sequence converges to the round metric on $\mathbb{S}^2$ of radius $\frac{1}{2}$. Verify that this convergence provides the desired counterexample.

Comments. The spaces $(\mathbb{S}^3, g_n)$ are called Berger spheres. Checking the curvature bounds is easy once you realize that $(\mathbb{S}^3, g)$ is isometric to a quotient space $\mathbb{S}^3 \times (\varepsilon \cdot \mathbb{S}^1)/\mathbb{S}^1$, where $\mathbb{S}^1$ acts diagonally by the Hopf action on $\mathbb{S}^3$ and the standard action on $\mathbb{S}^1$.

9.4. Let V and W be two conic neighborhoods of a point p . Without loss of generality, we may assume that $V \subseteq W$; that is, the closure of V lies in W .

Construct a sequence of embeddings $f_n: V \rightarrow W$ such that

- ◇ For any compact set $K \subset V$ there is a positive integer $n = n_K$ such that $f_n(k) = f_m(k)$ for any $k \in K$ and $m, n \geq n_K$.
- ◇ For any point $w \in W$ there is a point $v \in V$ such that $f_n(v) = w$ for all large n .

Once such a sequence is constructed, $f: V \rightarrow W$ can be defined by $f(v) = f_n(v)$ for all large values of n ; this gives the needed homeomorphism.

The sequence f_n can be constructed recursively

$$f_{n+1} = \Psi_n \circ f_n \circ \Phi_n,$$

where $\Phi_n: V \rightarrow V$ and $\Psi_n: W \rightarrow W$ are homeomorphisms of the form

$$\Phi_n(x) = \varphi_n(x) * x \quad \text{and} \quad \Psi_n(x) = \psi_n(x) \star x,$$

where $\varphi_n: V \rightarrow \mathbb{R}_{\geq 0}$, $\psi_n: W \rightarrow \mathbb{R}_{\geq 0}$ are suitable continuous functions; “ $*$ ” and “ $\star$ ” denote the multiplications in the cone structures of V and W respectively.

Comment. One may also read the original proof by Kyung Whan Kwun [62].

9.5; (a). Apply 9.2 and 9.3.

(b). Apply (a).

(c). Recall that the Poincaré homology sphere can be obtained as a quotient space $\Sigma = \mathbb{S}^3/\Gamma$ by an isometric action of a finite group Γ — the so-called binary icosahedral group. By the double suspension theorem, $\text{Susp}^2 \Sigma \cong \mathbb{S}^5$. Note that $\text{Susp}^2 \Sigma$ is an Alexandrov space and it has a point with space of directions isometric to $\text{Susp} \Sigma$. Observe that $\text{Susp} \Sigma$ is not a manifold; in particular $\text{Susp} \Sigma \not\cong \mathbb{S}^4$. Therefore the pair $\text{Susp}^2 \Sigma$ and $\mathbb{S}^5$ provides the needed example.

9.7. Apply 9.2, 9.3, and 9.6.

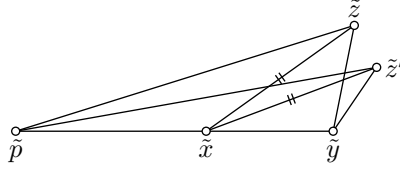
9.8. Let $\mathcal{A}$ be a finite-dimensional Alexandrov space. Choose $x \in \mathcal{A}$. By 9.2, a neighborhood $U \ni x$ is homeomorphic to T_x . Therefore, 9.7 implies that $U \cap \partial \mathcal{A} = \emptyset \Leftrightarrow x \notin \partial \mathcal{A}$; that is, the complement $\mathcal{A} \setminus \partial \mathcal{A}$ is open, and therefore, $\partial \mathcal{A}$ is closed.

9.11. Consider the model triangle $[\tilde{x}\tilde{y}\tilde{z}'] = \tilde{\Delta}(xyz)$.

Show that

$$|\tilde{p} - \tilde{z}| \leq |\tilde{p} - \tilde{z}'| \leq \tilde{\Upsilon}[y_z^p].$$

9.12. Assume that $\partial \mathcal{A}$ has at least two connected components, say A and B . Let γ be a geodesic that minimizes the distance from A to B .



Consider a two-sided infinite sequence of copies of $\mathcal{A}$

$$\dots, \mathcal{A}_{-1}, \mathcal{A}_0, \mathcal{A}_1, \dots$$

Let us glue $\mathcal{A}_i$ to $\mathcal{A}_{i+1}$ along A if i is even and along B if i is odd.

By the doubling theorem, every point in the obtained space $\mathcal{N}$ has a neighborhood that is isometric to a neighborhood of the corresponding point in $\mathcal{A}$ or its doubling. By the globalization theorem, $\mathcal{N}$ is ALEX(1).

The copies of γ in $\mathcal{A}_i$ form a line in $\mathcal{N}$. By the splitting theorem, $\mathcal{N}$ is isometric to a product $\mathcal{N}' \oplus \mathbb{R}$. Since $\dim \mathcal{N} > 1$, Exercise 7.3 implies that $\text{diam} \mathcal{N} \leq \pi$ — a contradiction.

9.13; (a) Apply the gradient flow as in the proof of 9.10 (part (c)_m + (d)_{m-1} $\Rightarrow$ (d)_m).

(b). We can assume that $\dim \mathcal{A} \geq 2$; otherwise the statement is trivial.

Choose an interior point x on γ ; we can assume that $x = \gamma(0)$. Let $y \in \partial \mathcal{A}$ be a closest point to x , and let $\alpha = \angle(\uparrow_{[xy]}, \gamma^+(0))$.

By (a), we can assume that $x \notin \partial \mathcal{A}$. Show that $T_y = [0, \infty) \times T_y \partial \mathcal{A}$ and $\uparrow_{[yx]} \perp T_y \partial \mathcal{A}$.

Given a vector $v \in T_y$, denote by $\bar{v}$ its projection to $T_y \partial \mathcal{A}$. Apply the comparison and 5.12 to show that

$$|\gamma(t) - \text{gexp}_y(\overline{\log_y \gamma(t)})| \leq |x - y| + t \cdot \cos \alpha.$$

Conclude that $(\text{dist}_{\partial \mathcal{A}} \circ \gamma)''(0) \leq 0$ in the barrier sense.

9.14. Suppose γ is defined on the interval $[0, \ell]$. Assume that the function $\rho: t \mapsto \frac{1}{2} \cdot \text{dist}_p^2 \circ \gamma(t)$ is not 1-concave. Let $\bar{\rho}: [0, \ell] \rightarrow \mathbb{R}$ be the minimal 1-concave function such that $\bar{\rho} \geq \rho$. Note that $\bar{\rho} = \rho$ at the ends of $[0, \ell]$.

Consider the curve $\bar{\gamma}(t) := \text{Flow}_f^{s(t)} \gamma(t)$, where $f = \frac{1}{2} \cdot \text{dist}_p^2$ and $s(t) = \ln \bar{\rho}(t) - \ln \rho(t)$. Use the first distance estimate to show that $\text{length } \bar{\gamma} < \text{length } \gamma$ and arrive at a contradiction.

Comment. This is the so-called Liberman's lemma; it was proved by Grigory Perelman and the third author [85] and generalizes a theorem of Joseph Liberman [75] about geodesics on convex surfaces.

9.15. Choose a geodesic γ in $\mathcal{W}$. Arguing as in the proof of 9.10*d*, we get that γ can cross the common boundary of two halves $\mathcal{A}_0$ and $\mathcal{A}_1$ of $\mathcal{W}$ at most once, or it lies in the common boundary.

In the latter case λ -concavity of $f \circ \text{proj} \circ \gamma$ follows from λ -concavity of f . In the former case the concavity has to be checked only at the point of crossing; we may assume that it happens at $x = \gamma(0)$. Since $\nabla_x f \in \partial T_x$ for any $x \in \partial \mathcal{A}$, the f -gradient flows on $\mathcal{A}_0$ and $\mathcal{A}_1$ agree on the common boundary; so they induce a continuous flow on $\mathcal{W}$.

Assume $f \circ \text{proj} \circ \gamma$ is not λ -concave at 0. Apply the constructed flow on $\mathcal{W}$ to shorten γ keeping its ends as in the proof of 9.14, and arrive at a contradiction.

10.2. Read [117, Section 4] and/or the solution for “Quotient of the Hilbert space” in [101].

10.3; (a). Choose an isometric $\mathbb{S}^1$ -action on $\mathbb{S}^2$ that fixes the poles of the sphere. Consider the projection to the quotient space $\sigma_1: \mathbb{S}^2 \rightarrow \mathbb{S}^2/\mathbb{S}^1 = [0, \pi]$.

(b). Take a half-circle γ on $\mathbb{S}^2$ and define $\sigma_2(x) := \text{dist}_\gamma(x)_{\mathbb{S}^2}$.

(c). Subdivide $\mathbb{S}^2$ into $\mathbb{S}^1$ -orbits of the action from (a). Cut $\mathbb{S}^2$ into two hemispheres by meridians, rotate one hemisphere by an angle $\alpha = \pi/n$ and glue it back. Observe that there is a submetry σ_n such that the inverse image $\sigma_n^{-1}\{y\}$ is a union of the arcs from the original $\mathbb{S}^1$ -orbits.

Note that for $n = 2$ we get the solution in (b).

10.4. Show that for any $x \in \mathbb{E}^2$ there is a half-line $H \ni x$ such that the restriction $\sigma|_H$ is an isometry. Suppose such a half-line H starts at p and passes thru q . Show that $\langle x - p, q - p \rangle \leq 0$ for any $x \in \sigma^{-1}\{0\}$. Conclude that $\sigma^{-1}\{0\}$ is a convex closed set. Use the definition of submetry to show that $\sigma^{-1}\{0\}$ has no interior points. Make a conclusion.

10.8; (a). Our $\mathbb{S}^1$ is a commutative subgroup of $\text{SO}(4)$. Therefore it is a subgroup of a maximal torus in $\text{SO}(4)$. Show that the described torus action is induced by a maximal torus in $\text{SO}(4)$. Use that maximal tori in $\text{SO}(4)$ are conjugate.

(b). Cut $\mathbb{S}^3$ along the Clifford torus $\frac{1}{\sqrt{2}} \cdot (\mathbb{S}^1 \times \mathbb{S}^1)$. Observe that the quotient of each half is a disk; conclude that $\Sigma_{p,q}$ is a sphere. The torus action on $\mathbb{S}^3$ induces the needed $\mathbb{S}^1$ -action on $\Sigma_{p,q}$.

(c)+(d)+(e). Calculations.

(f). Consider the map $\Sigma_{p,q} \rightarrow \Sigma_{1,1}$ that sends poles to poles, preserves the distance to the poles and respects the $\mathbb{S}^1$ -actions.

10.10. Choose an m -dimensional ALEX(-1) space $\mathcal{A}$. Repeat the proof of 10.9 for fixed $\alpha = \frac{1}{10}$ showing that any unit ball in $\mathcal{A}$ has at most 1000^m extremal points. Then apply 7.17b+1.2 for $\varepsilon = 1$, $\kappa = -1$, and $r = \text{diam } \mathcal{A}$.

10.14; (a). Suppose $\mathfrak{M}_{m-1}(\Gamma) \geq 3$; that is, $\mathcal{A} = \mathbb{E}^m/\Gamma$ has at least 3 boundary components. Follow Case 3 in the proof of 10.6 to glue a train-like space from copies of $\mathcal{A}$ using two of these components. Show that the obtained space splits and arrive at a contradiction.

(Alternatively, apply a similar construction to all components of the boundary. Show that the obtained space has exponential volume growth; that is, there is $a > 1$ such that $\text{vol } B(p, r) > a^r$ for all large r . Arrive at a contradiction with the Bishop–Gromov inequality.)

(b). Apply the doubling theorem as in Case 2 in the proof of 10.6.

10.16. Show that the quotient space $\Delta = \mathcal{A}/\mathbb{S}^1$ is an ALEX(1) disk and γ projects isometrically to its boundary $\partial\Delta$. It remains to show that the perimeter of Δ cannot exceed $2\cdot\pi$. The latter follows from Lytchak's problem [96, 3.3.5]; it states that if Δ is an m -dimensional ALEX(1) space, then $\text{vol}_{m-1} \partial\Delta \leq \text{vol}_{m-1} \partial\mathbb{S}^{m-1}$.

11.1. Suppose a geodesic γ passes thru a vertex v . Denote by α and β the angles that γ cuts at v . Since v is essential, $\alpha + \beta < 2\cdot\pi$. Therefore $\alpha < \pi$ or $\beta < \pi$. Arrive at a contradiction by showing that γ is not length-minimizing.

11.2. Assume $\mathcal{P}$ has no boundary. Denote by k , l , and m the number of vertices, edges, and triangles, respectively, in a chosen triangulation of $\mathcal{P}$. Note that

$$2\cdot l = 3\cdot m \quad \text{and} \quad k - l + m = \chi(\mathcal{P}).$$

The first identity follows since each edge appears in two triangles and each triangle has 3 edges; the second identity is Euler's formula.

Since each triangle contributes π to the total sum of angles, we get that total curvature is $2\cdot\pi\cdot k - \pi\cdot m$. It remains to apply straightforward algebraic manipulations.

If $\mathcal{P}$ has nonempty boundary, then pass to its doubling, apply the formula and rewrite the result using inner turns.

11.4. We need to show that if a polyhedral surface is ALEX(0), then the total angle θ at every vertex p is at most $2\cdot\pi$.

Assume that $\theta > 2\cdot\pi$, let $\varphi = \min\{\pi, \frac{1}{3}\cdot\theta\}$. We can choose three points x_1 , x_2 , and x_3 close to p such that $\angle[p_{x_j}^{x_i}] = \varphi$ for $i \neq j$. Since the points x_i are close to p , we have $\angle[p_{x_j}^{x_i}] = \tilde{\angle}(p_{x_j}^{x_i})$. The latter contradicts $\mathbb{E}^2$ -comparison.

11.6. (a). Apply 8.10.

(b). Observe that any chord divides a convex figure on $\mathcal{P}$ into two convex figures. Use it to show that the union of two convex polygons can be triangulated by convex triangles. Apply the last statement recursively to a finite cover of $\mathcal{P}$ by convex polygons.

(c). By comparison, $\tilde{\mathcal{P}}$ has nonnegative curvature. It remains to apply 11.3.

11.10; (a). Show that the total angle around each vertex of the triangulation is at most $2 \cdot \pi$ and argue as in 11.2.

(b). Apply the hinge comparison (2.17d).

(c). The first inequality follows from the generalized Kiszbraun theorem proved by Urs Lang and Viktor Schroeder [4, 7, 64]; we will not discuss its proof since this inequality is not used in the proof. In the proof of the second inequality, we use Alexandrov's idea [10, X § 1].

Suppose that a convex triangle Δ has vertices x, y , and z , with opposite side lengths a, b and c and angles α, β , and γ , respectively. Let $\tilde{\Delta}$ be its solid model triangle and let $\tilde{\alpha}, \tilde{\beta}$, and $\tilde{\gamma}$ be the corresponding angles.

Use that $\log_x$ is a noncontracting map to show that

$$\text{area } \Delta - \text{area } \tilde{\Delta} \leq \frac{1}{2} \cdot (\alpha - \tilde{\alpha}) \cdot (\text{diam } \Delta)^2 + o(a).$$

Now let us subdivide Δ into triangles $\Delta_1, \dots, \Delta_n$ with common vertex x and small opposite sides $a_1, \dots, a_n$; so, $a = a_1 + \dots + a_n$. The model triangles $\tilde{\Delta}_1, \dots, \tilde{\Delta}_n$ can be arranged on the plane with common corresponding sides. This way we get a convex triangle $\tilde{\Lambda}$ with two straight sides b and c and a polygonal side of total length a . Summing up the above inequalities for $\Delta_1, \dots, \Delta_n$, we get

$$\text{area } \Delta - \text{area } \tilde{\Lambda} \leq \frac{1}{2} \cdot (\alpha - \tilde{\alpha}) \cdot (\text{diam } \Delta)^2 + o(a_1) + \dots + o(a_n).$$

It follows that given $\varepsilon > 0$, we can choose the subdivision so that

$$\text{area } \Delta - \text{area } \tilde{\Lambda} \leq \frac{1}{2} \cdot (\alpha - \tilde{\alpha}) \cdot (\text{diam } \Delta)^2 + \varepsilon.$$

By the hinge comparison (2.17d), the angles adjacent to polygonal sides of $\tilde{\Lambda}$ do not exceed β and γ . It follows that

$$\text{area } \tilde{\Lambda} - \text{area } \tilde{\Delta} \leq \frac{1}{2} \cdot (\beta + \gamma - \tilde{\beta} - \tilde{\gamma}) \cdot (\text{diam } \Delta)^2.$$

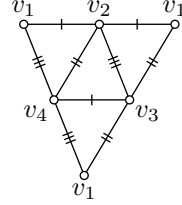
Therefore

$$\text{area } \Delta - \text{area } \tilde{\Delta} \leq \frac{1}{2} \cdot \text{excess } \Delta \cdot (\text{diam } \Delta)^2 + \varepsilon$$

for arbitrary $\varepsilon > 0$, hence the result.

11.11. We can assume that the origin lies in the interior of the convex body. Consider the central projection from its surface, say Σ , to the sphere $\mathbb{S}^2$ centered at the origin. Show that this projection $\Sigma \rightarrow \mathbb{S}^2$ is a homeomorphism.

11.12; (a). Cut the surface of T along three edges coming from one vertex v_1 and unfold the obtained surface onto the plane. Show that this way we get a triangle; the three vertices correspond to v_1 and the midpoints of sides correspond to the remaining three vertices. Make a conclusion.



(b). Let $v_1, v_2, v_3, v_4 \in \mathbb{R}^3$ be the vertices of T . From (a), we have that

$$|v_1 - v_2| = |v_3 - v_4|, \quad |v_1 - v_3| = |v_2 - v_4|, \quad |v_1 - v_4| = |v_2 - v_3|.$$

Use it to show that $\langle v_1 - v_2, v_1 + v_2 - v_3 - v_4 \rangle = 0$. Make a conclusion.

11.14. We will use that the nearest-point projection from the Euclidean space to a convex body is short; that is, distance-nonexpanding [93, 13.3].

Assume K_∞ is nondegenerate. Without loss of generality, we may assume that

$$\overline{B}(0, r) \subset K_\infty \subset \overline{B}(0, 1)$$

for some $r > 0$. Then there is a sequence $\varepsilon_n \rightarrow 0$ such that

$$K_n \subset (1 + \varepsilon_n) \cdot K_\infty \quad \text{and} \quad K_\infty \subset (1 + \varepsilon_n) \cdot K_n$$

for each large n .

Given $x \in K_n$, denote by $g_n(x)$ the nearest-point projection of $(1 + \varepsilon_n) \cdot x$ to K_∞ . Similarly, given $x \in K_\infty$, denote by $h_n(x)$ the nearest-point projection of $(1 + \varepsilon_n) \cdot x$ to K_n . Note that

$$|g_n(x) - g_n(y)| \leq (1 + \varepsilon_n) \cdot |x - y|$$

and

$$|h_n(x) - h_n(y)| \leq (1 + \varepsilon_n) \cdot |x - y|.$$

Denote by Σ_∞ and Σ_n the surface of K_∞ and K_n respectively. The above inequalities imply

$$|g_n(x) - g_n(y)|_{\Sigma_\infty} \leq (1 + \varepsilon_n) \cdot |x - y|_{\Sigma_n}$$

for any $x, y \in \Sigma_n$, and

$$|h_n(x) - h_n(y)|_{\Sigma_n} \leq (1 + \varepsilon_n) \cdot |x - y|_{\Sigma_\infty}$$

for any $x, y \in \Sigma_\infty$.

By a degree argument the maps g_n and h_n are onto. Apply 1.27 and finish the proof.

Alternatively, since the nearest-point projection cannot increase the length of a curve, we also get

$$\begin{aligned} |x - h_n \circ g_n(x)|_{\Sigma_n} &\leq 10 \cdot \varepsilon_n \\ |y - g_n \circ h_n(y)|_{\Sigma_\infty} &\leq 10 \cdot \varepsilon_n \end{aligned}$$

for all large n . Therefore, g_n is a δ_n -isometry $\Sigma_n \rightarrow \Sigma_\infty$ for a sequence $\delta_n \rightarrow 0$.

Comments. More generally, if a sequence of m -dimensional $\text{ALEX}(\kappa)$ spaces $\mathcal{A}_1, \mathcal{A}_2, \dots$ converges to $\mathcal{A}_\infty$ and $\dim \mathcal{A}_\infty = m < \infty$, then $\partial \mathcal{A}_n$ equipped with the induced length metrics converge to $\partial \mathcal{A}_\infty$. This statement is a special case of the theorem about extremal subsets proved by the third author [97, 1.2].

11.21. (a). Observe that the curvature of any essential vertex is less than $2 \cdot \pi$ and apply the Gauss–Bonnet formula 11.2.

(b). Show that geodesics between essential vertices divide the surface into two flat triangles, which have to be isometric since their sides are equal. Make a conclusion.

(c). Show that geodesics between the essential vertices can be chosen so that they do not cross each other; that is, every pair of geodesics intersects only at the common vertex. In this case they divide the surface into plane triangles.

Since the curvature is nonnegative, the sum of three angles of the triangles at each vertex is at most $2 \cdot \pi$. Show that the three angles at one (and therefore any) vertex satisfy the triangle inequality. Conclude that these triangles form faces of a tetrahedron (possibly degenerate to a quadrangle).

11.28. Assume the contrary. Then there is a minimizing geodesic $\gamma \not\subset \Delta$ with ends p and q in Δ .

Without loss of generality, we may assume that only one arc of γ lies outside of Δ . Reflection of this arc with respect to Π together with the remaining part of γ forms another curve $\hat{\gamma}$ from p to q ; it runs partly along the surface and partly outside K , but does not get in the interior of K . Note that

$$\text{length } \hat{\gamma} = \text{length } \gamma.$$

Denote by $\bar{\gamma}$ the nearest-point projection of $\hat{\gamma}$ to K . Since K is convex, the projection decreases the length. Therefore the curve $\bar{\gamma}$ lies on the surface of K , has the same ends as γ , and satisfies

$$\text{length } \bar{\gamma} < \text{length } \gamma.$$

This means that γ is not length-minimizing — a contradiction.

11.29. If the plane py_1y_2 supports K , then $\angle[p_{y_2}^{y_1}]_{\mathbb{E}^3} = \angle[p_{x_2}^{x_1}]_{\mathcal{P}}$. In this case, the statement follows from 11.15.

Now suppose that the line segment $[y_1y_2]_{\mathbb{E}^3}$ intersects K . Choose a geodesic $[y_1y_2]_W$; note that it contains a point of K , say z . Now consider the one-parameter family of points

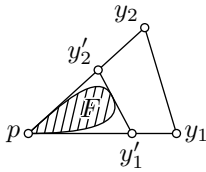
$$y_i(t) := \gamma_i(t) + \gamma_i^+(t) \cdot (1 - t).$$

This family is not necessarily continuous; note that $y_i(0) = y_i$ and $y_i(1) = x_i$.

Show that for any point $q \in K$, the function $t \mapsto |q - y_i(t)|_{\mathbb{E}^3}$ is nonincreasing. Conclude that the function $t \mapsto |q - y_i(t)|_W$ is nonincreasing for any $q \in \mathcal{P}$. Therefore,

$$\begin{aligned} |y_1 - y_2|_W &= |y_1(0) - y_2(0)|_W = \\ &= |y_1(0) - z|_W + |y_2(0) - z|_W \geq \\ &\geq |y_1(1) - z|_W + |y_2(1) - z|_W \geq \\ &\geq |x_1 - x_2|_{\mathcal{P}}. \end{aligned}$$

The last inequality follows since the nearest-point projection $W \rightarrow \mathcal{P}$ is short.



It remains to consider the case when the plane py_1y_2 does not support K , and $[y_1y_2]_{\mathbb{E}^3}$ does not intersect K . In this case the plane py_1y_2 intersects K along a convex figure F that lies in the solid triangle py_1y_2 and contains its vertex p .

Choose points $y'_1 \in [py_1]_{\mathbb{E}^3}$ and $y'_2 \in [py_2]_{\mathbb{E}^3}$ such that $[y'_1y'_2]$ touches F . Denote by $x'_1 \in [px_1]_{\mathcal{P}}$ and $x'_2 \in [px_2]_{\mathcal{P}}$ the corresponding points; that is, $|p - y'_1|_{\mathbb{E}^3} = |p - x'_1|_{\mathcal{P}}$ and $|p - y'_2|_{\mathbb{E}^3} = |p - x'_2|_{\mathcal{P}}$. From above, we have that $|y'_1 - y'_2|_{\mathbb{E}^3} \geq |x'_1 - x'_2|_{\mathcal{P}}$; in other words,

$$\tilde{\mathcal{Z}}(p_{y'_2}^{y'_1}) \geq \tilde{\mathcal{Z}}(p_{x'_2}^{x'_1});$$

here we think of $[py'_1y'_2]$ as a triangle in $\mathbb{E}^3$, and $[px'_1x'_2]$ as a triangle in $\mathcal{P}$. Note that

$$\tilde{\mathcal{Z}}(p_{y'_2}^{y'_1}) = \tilde{\mathcal{Z}}(p_{y_2}^{y_1}) \quad \text{and} \quad \tilde{\mathcal{Z}}(p_{x_2}^{x_1}) \leq \tilde{\mathcal{Z}}(p_{x'_2}^{x'_1});$$

the second inequality follows from 2.9. Hence the remaining case follows.

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